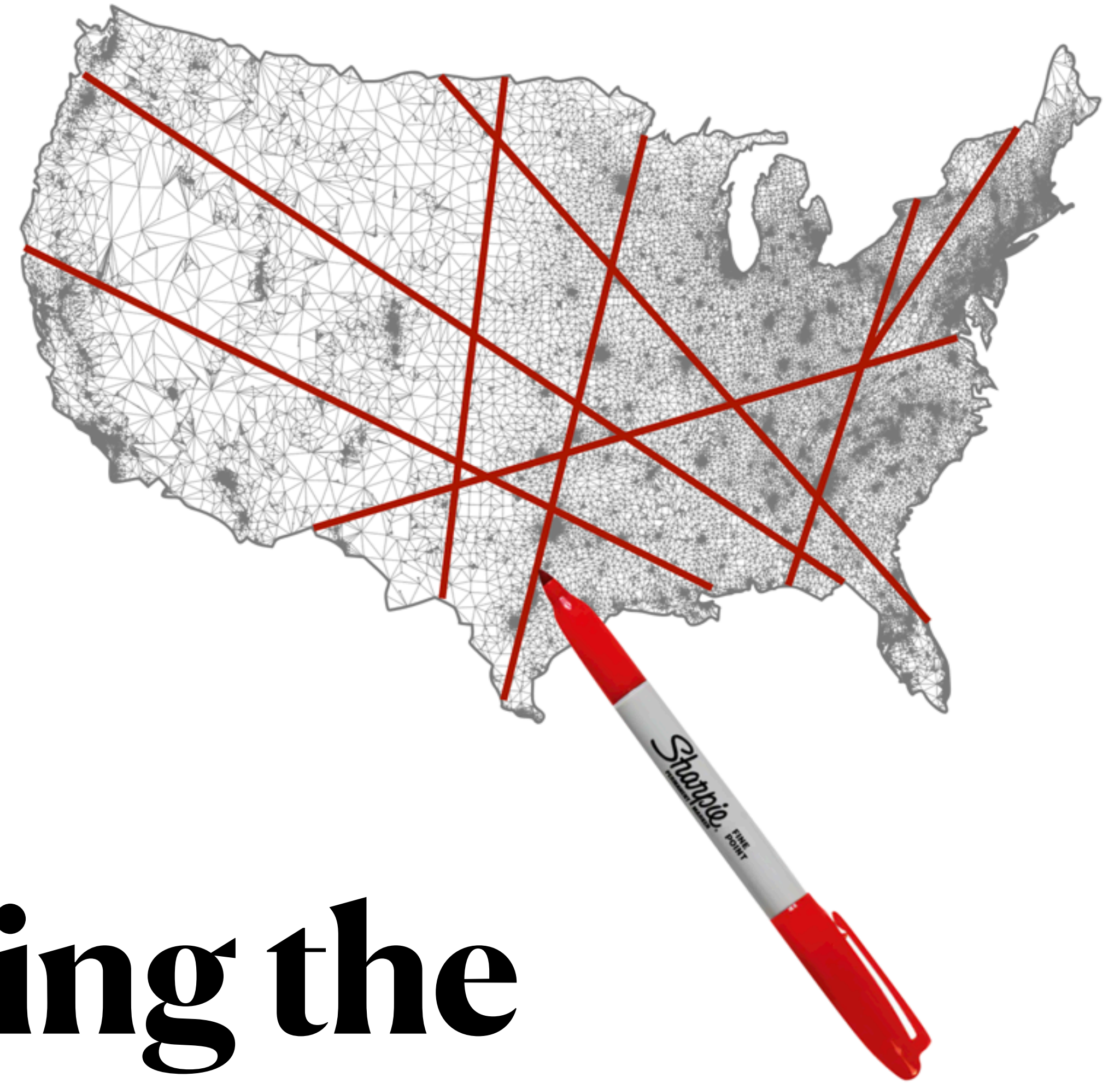




understanding the gerrymandering wars

Moon Duchin

Winner-take-all elections

“first past the post”

Winner-take-all elections

“first past the post”

- **Whoever has the most votes in a district wins!**

Winner-take-all elections

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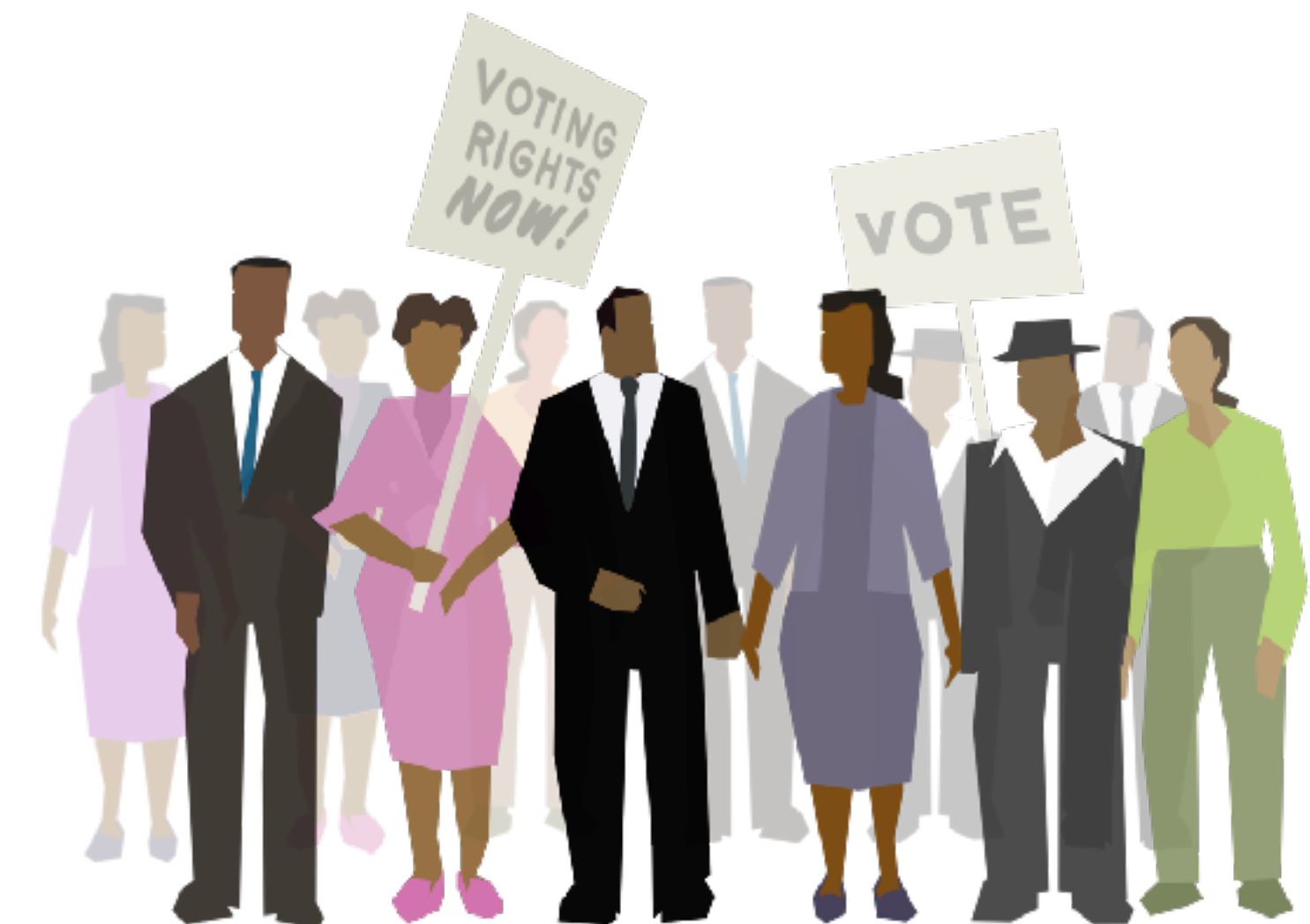
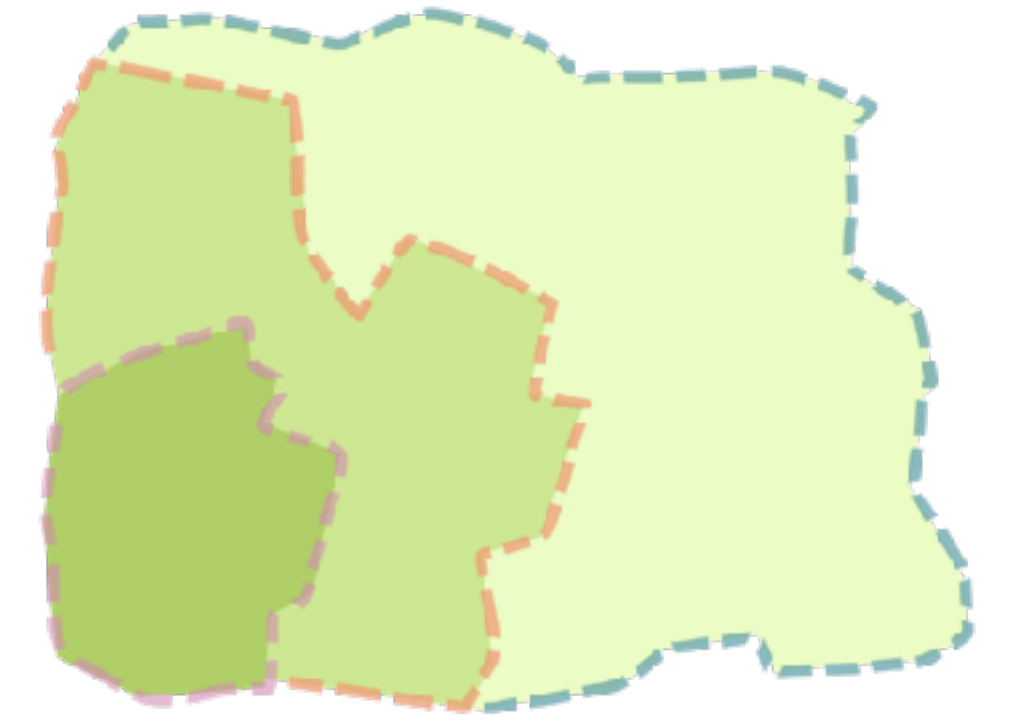
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Winner-take-all elections

“first past the post”

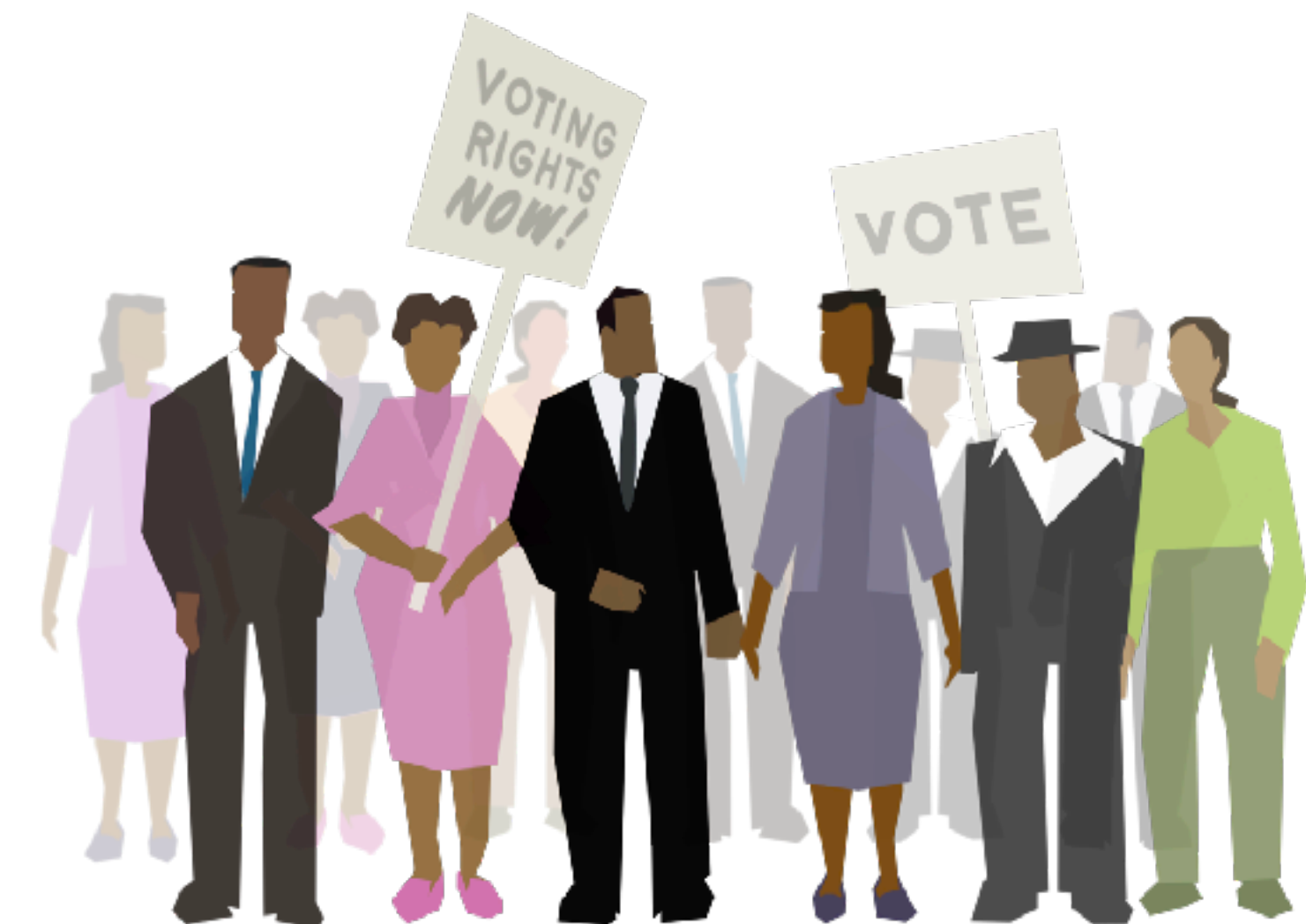
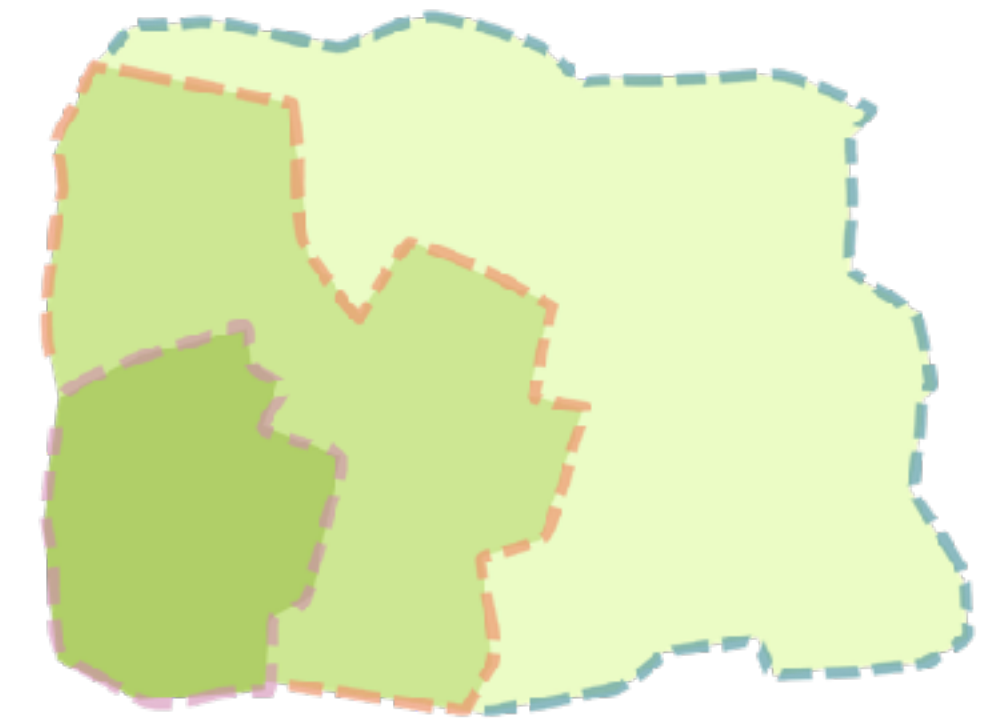
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 - Now we cut up the state into 17 pieces.

How to draw districts



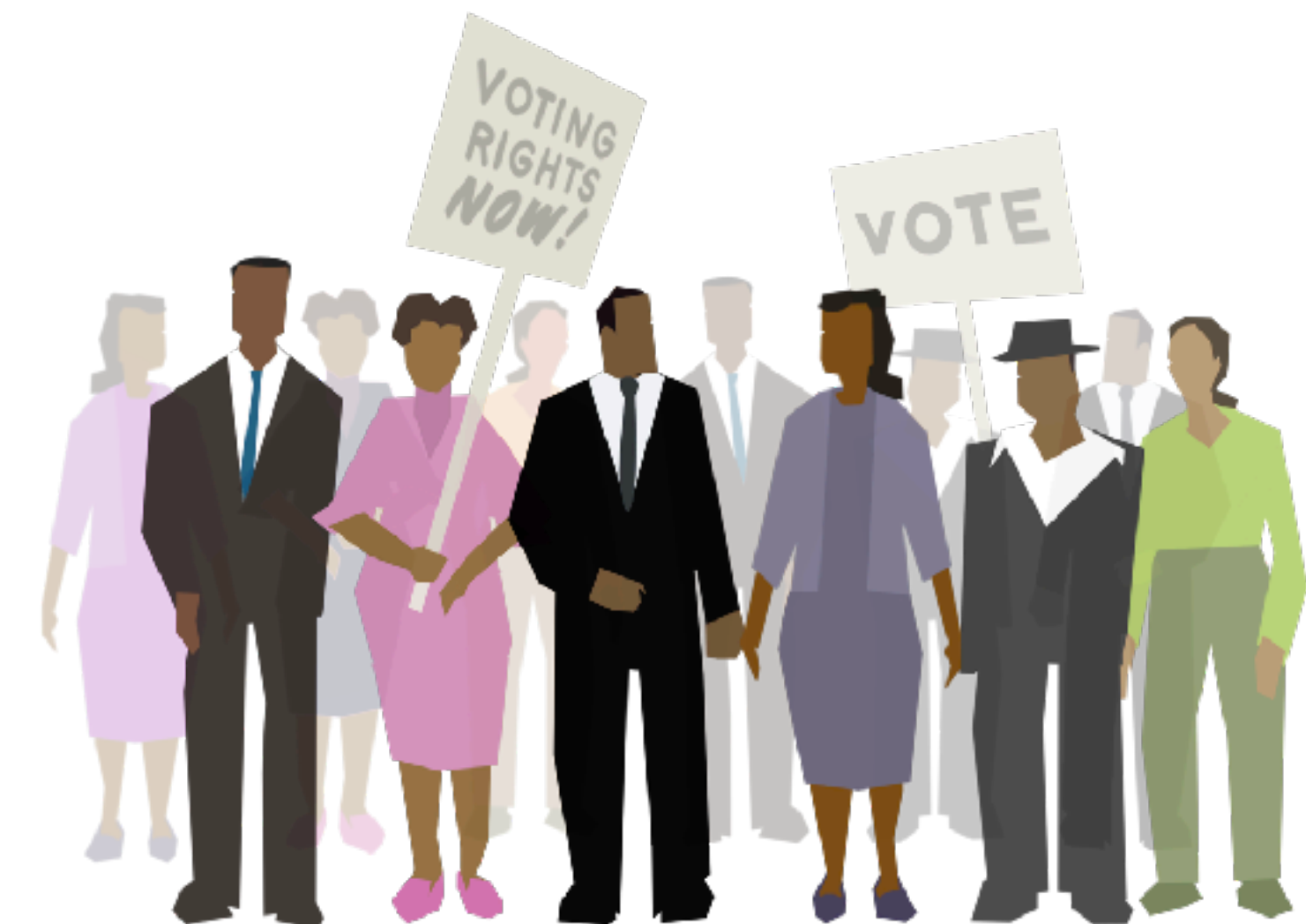
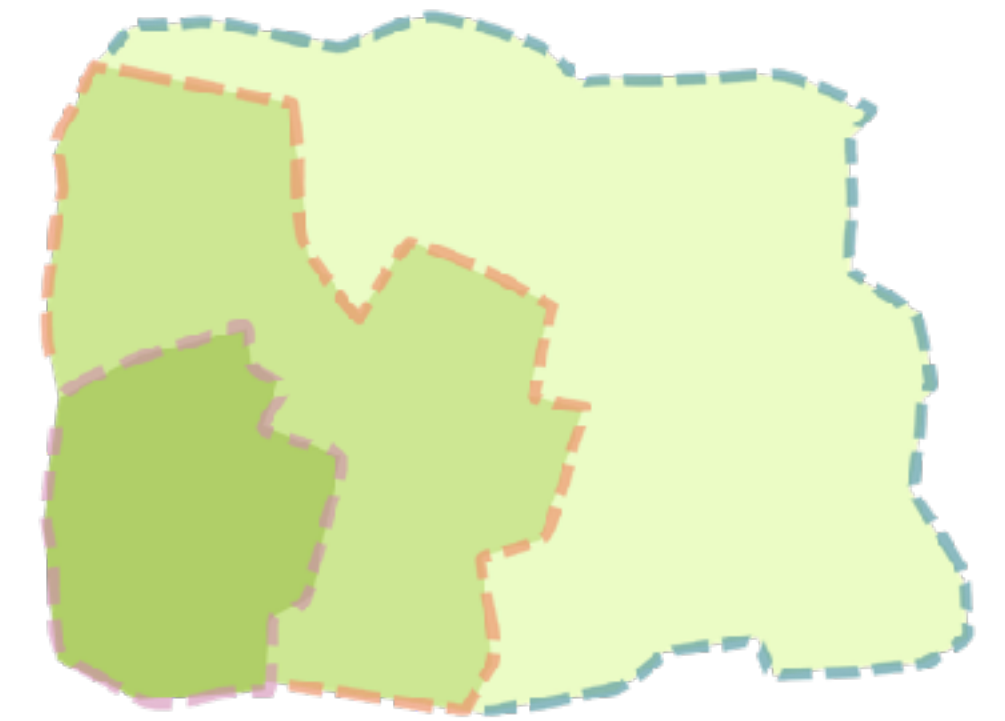
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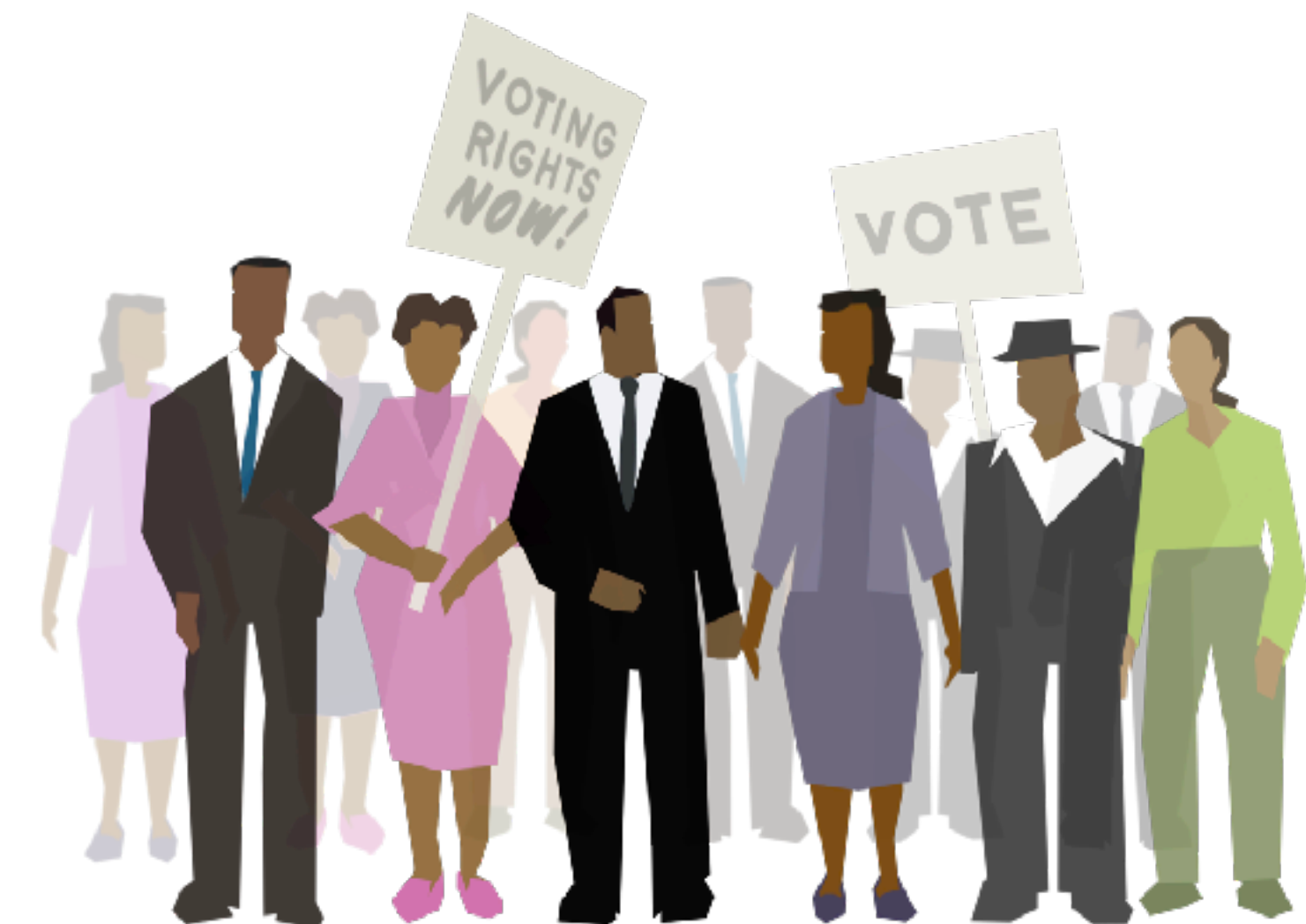
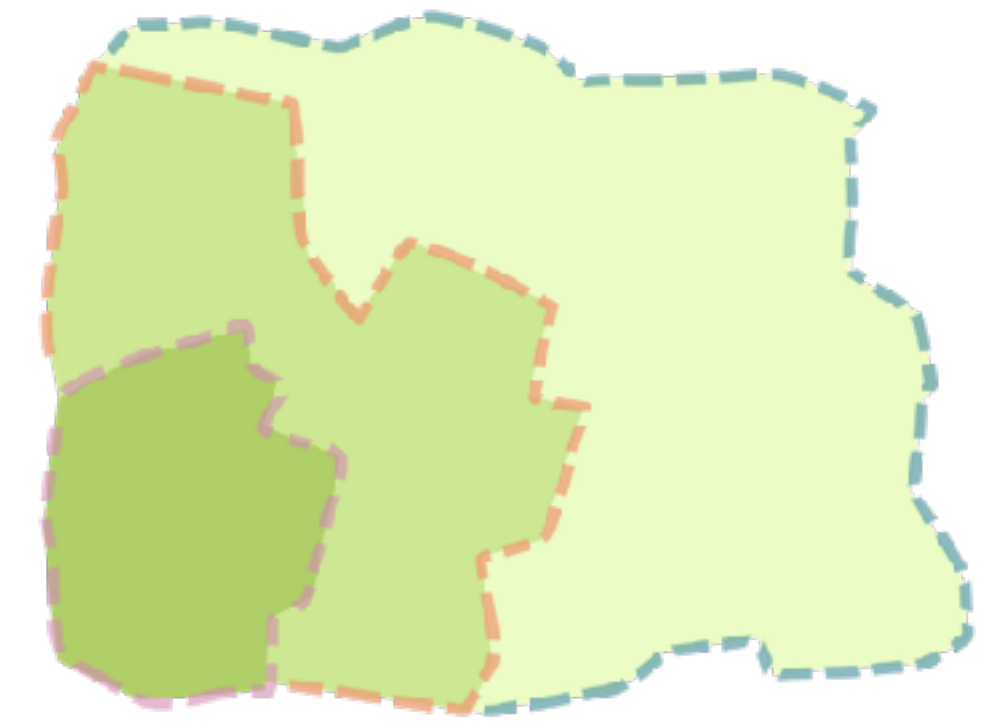
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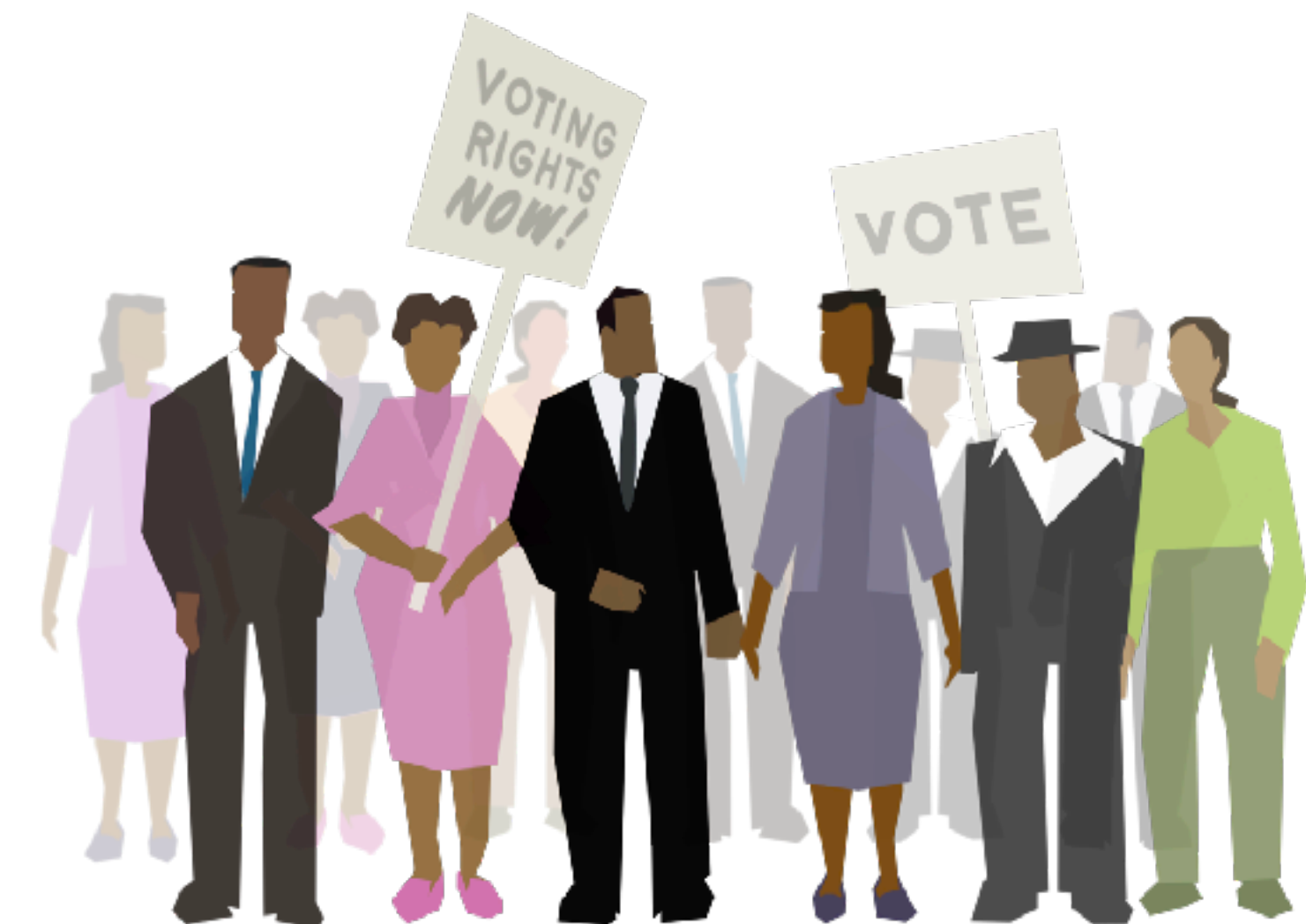
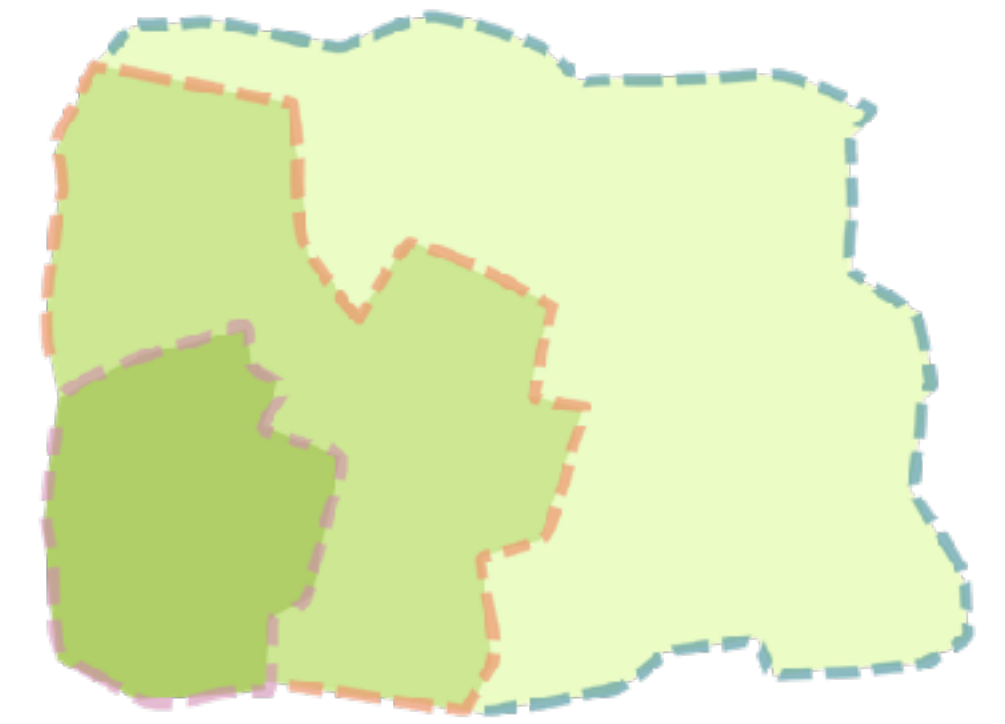
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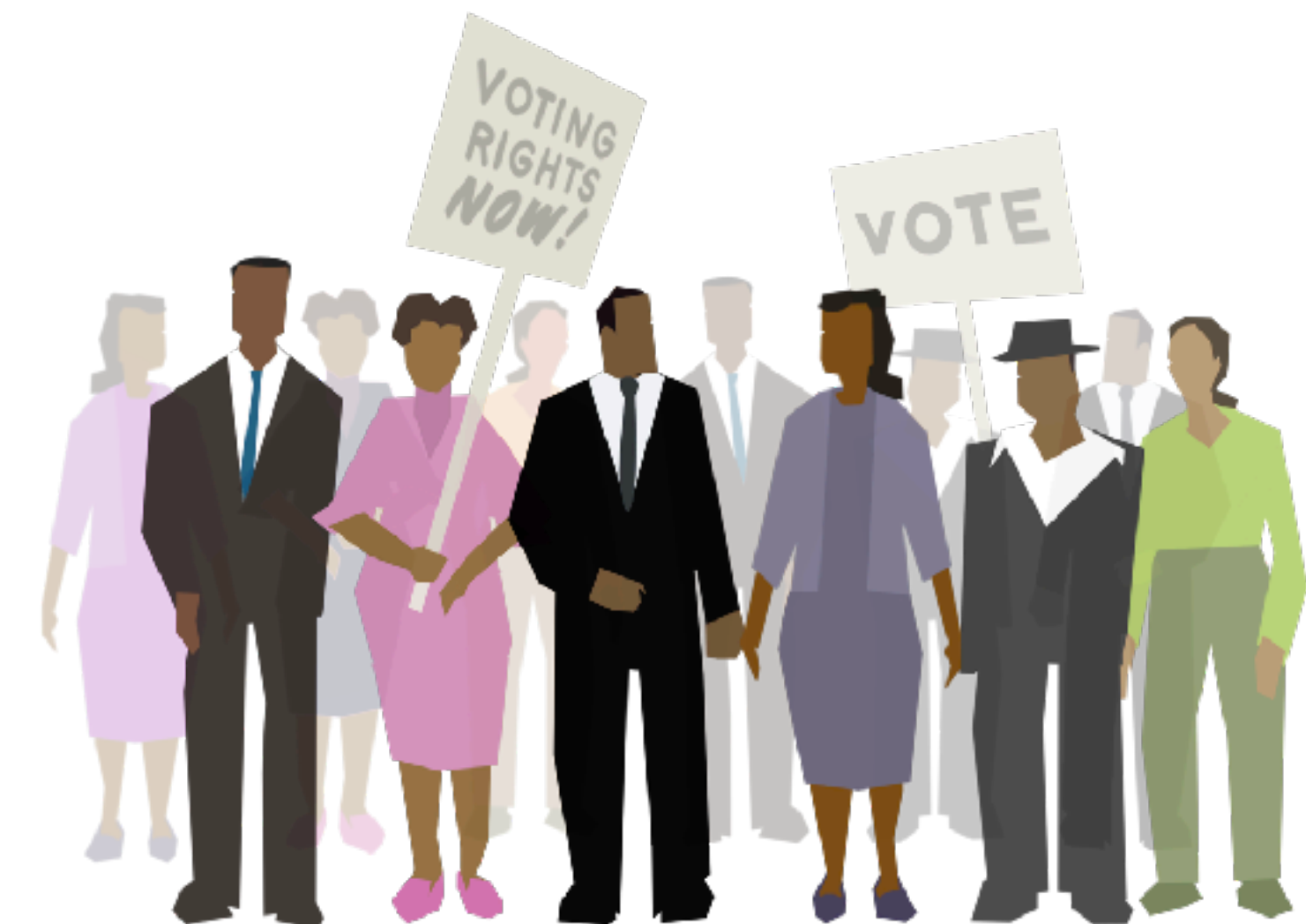
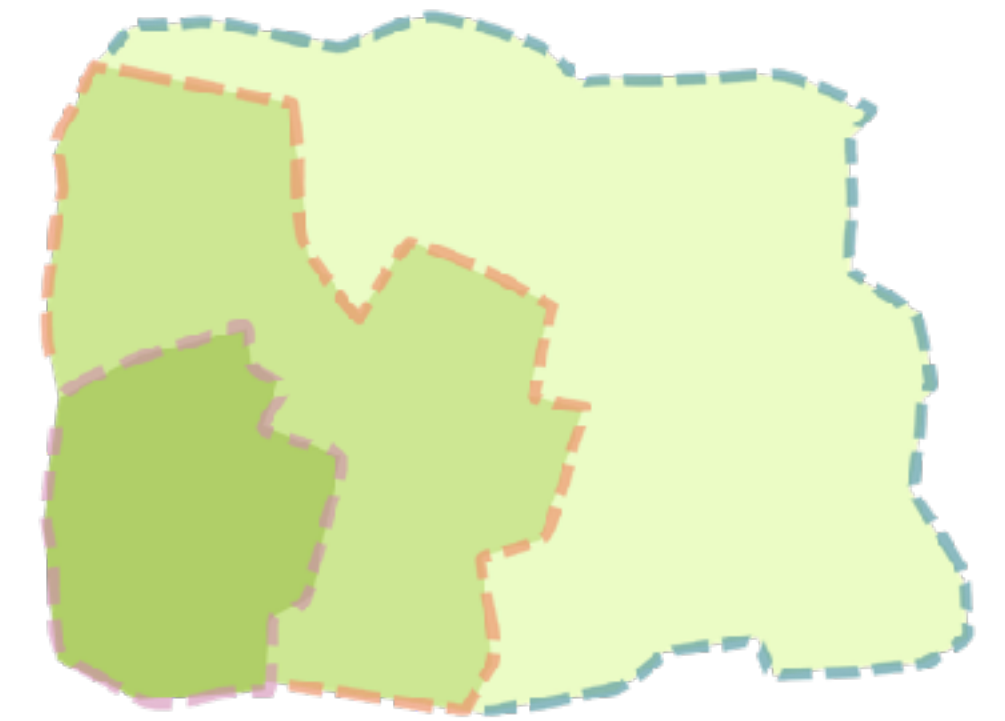
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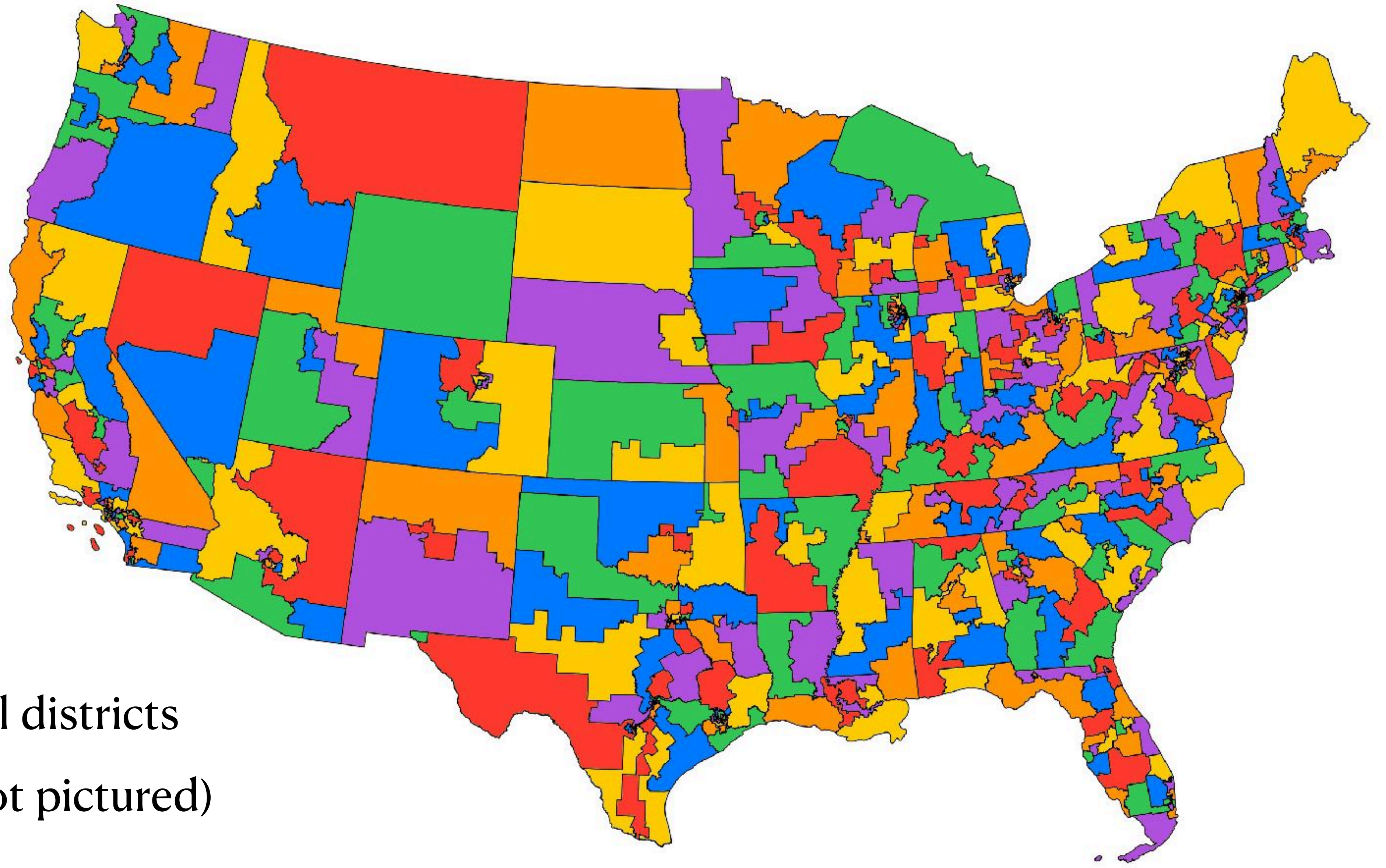
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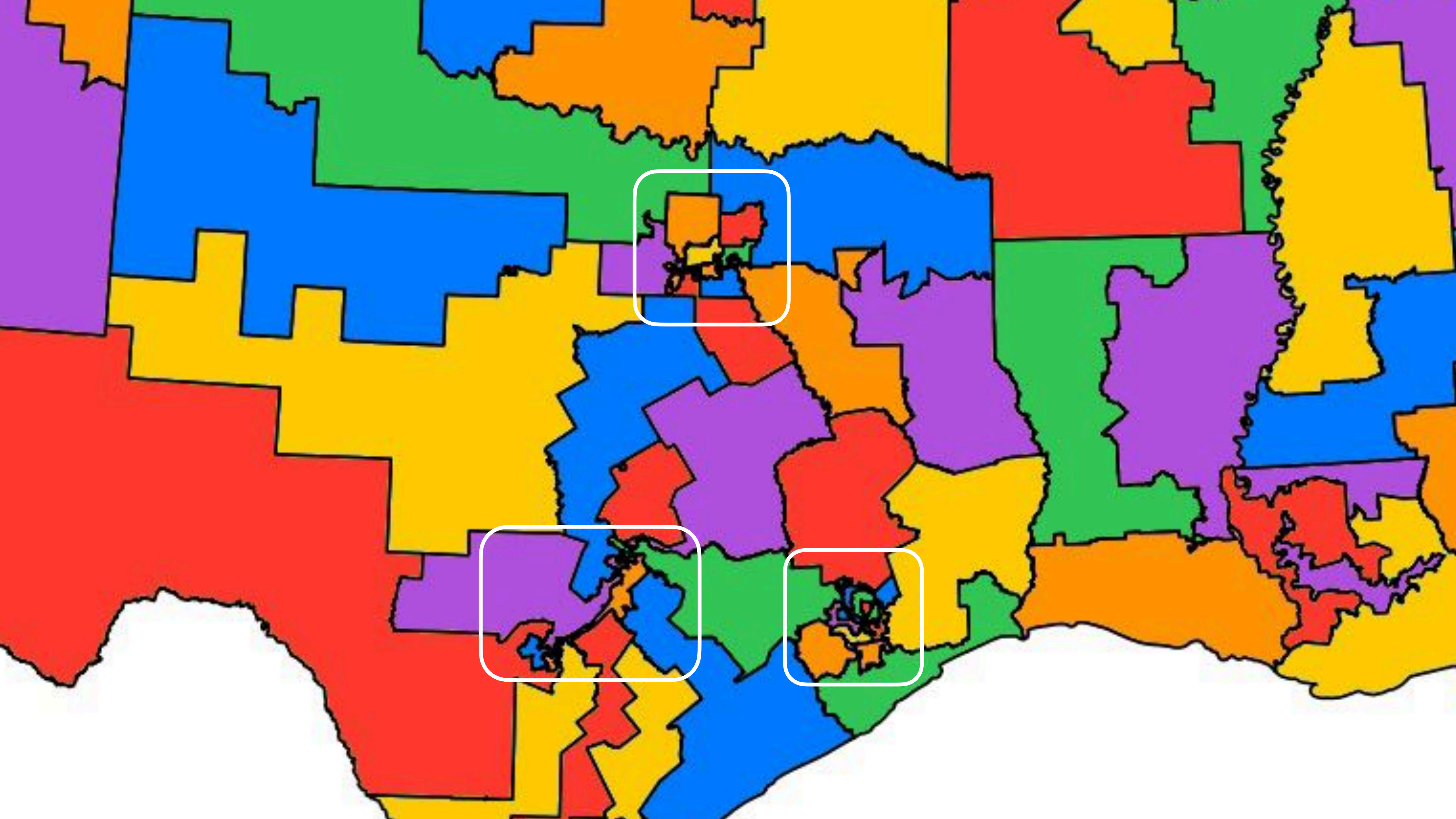
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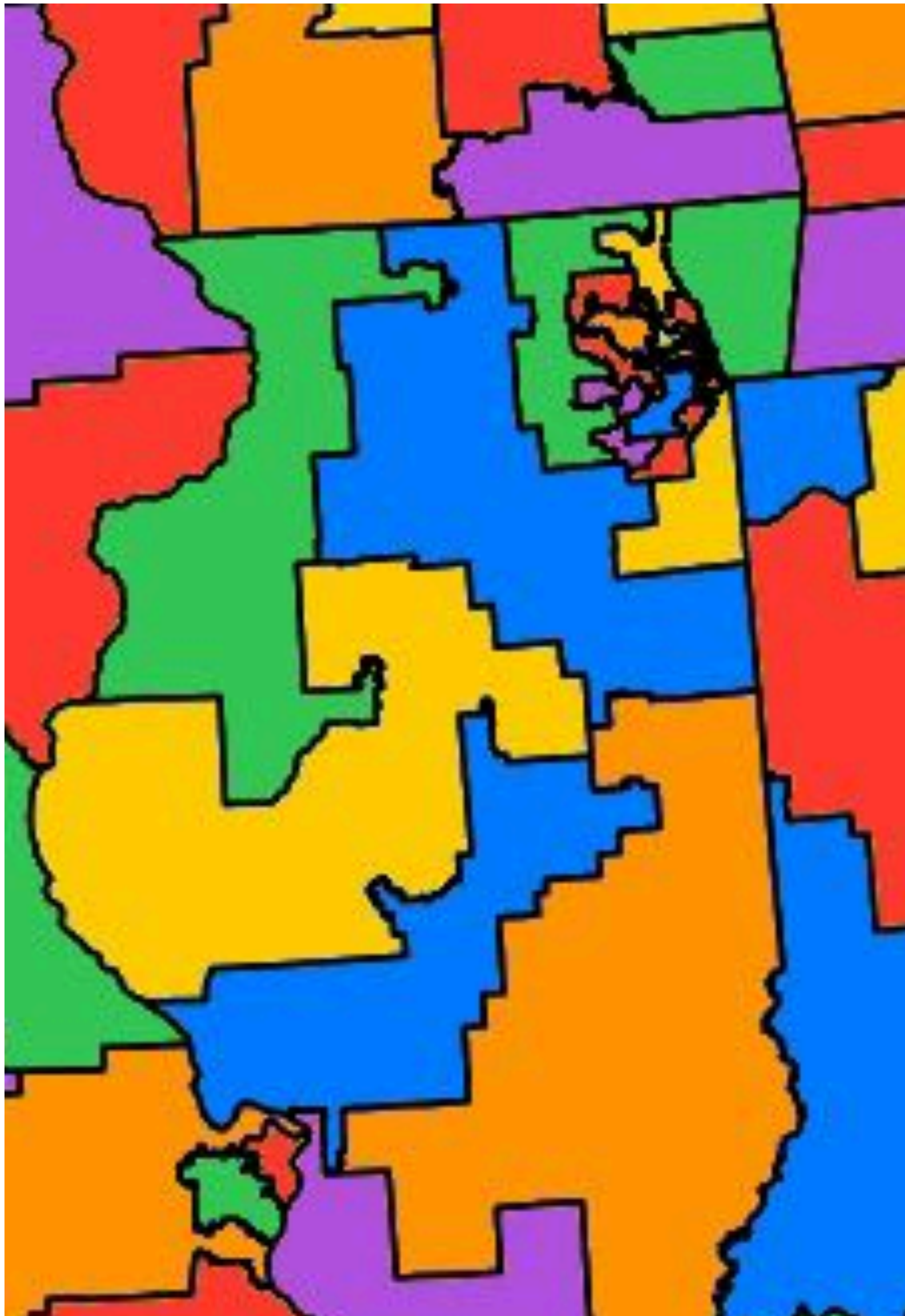
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- **RESPECT COMMUNITIES:** Pay attention to “communities of interest.”
- **RACIAL FAIRNESS:** Don’t deny or dilute votes of minorities. But also don’t pay too much attention to race.





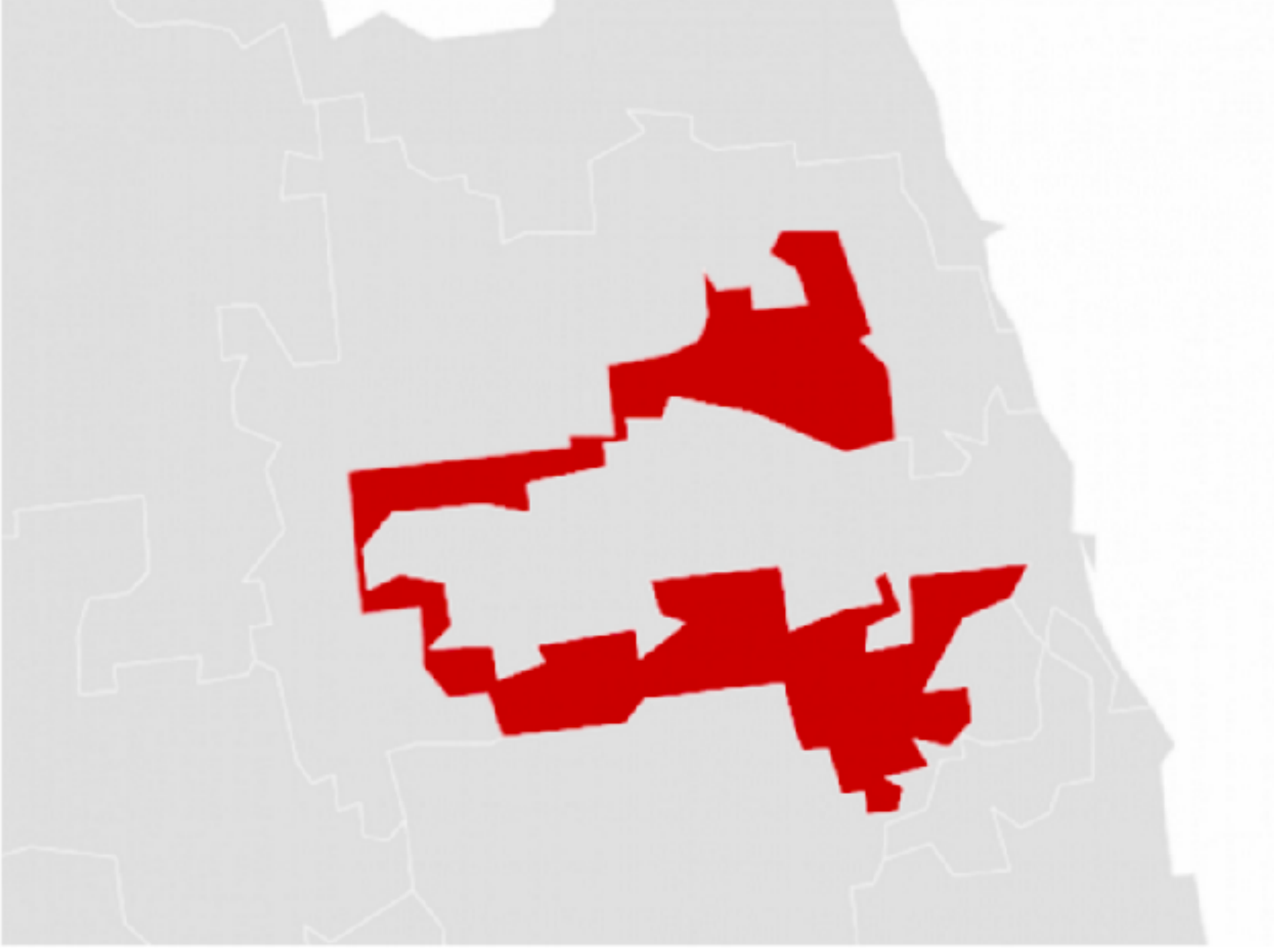
432 Congressional districts
(Alaska, Hawaii not pictured)
as of November 1, 2019



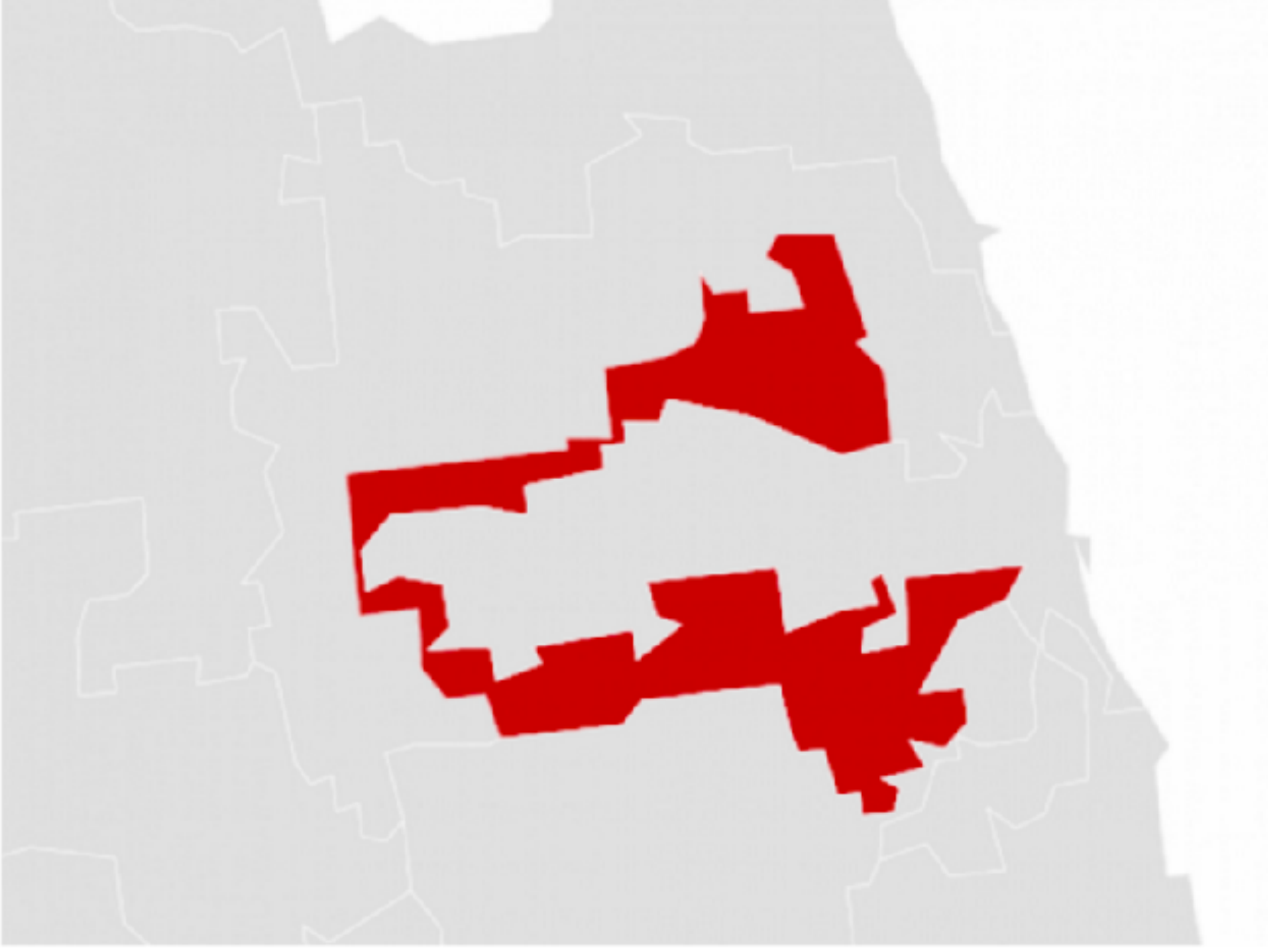


how about Chicago?

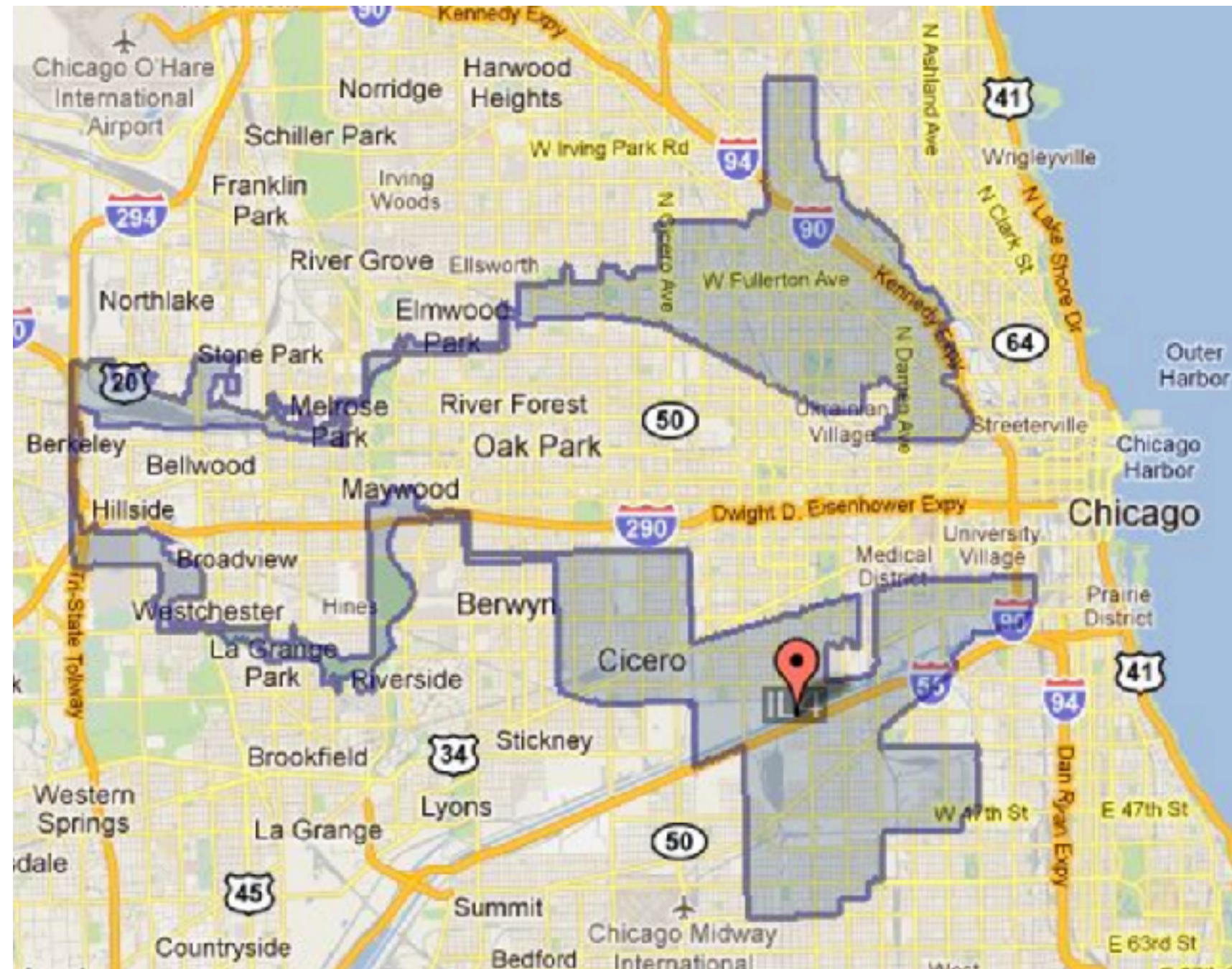
Illinois 4th district



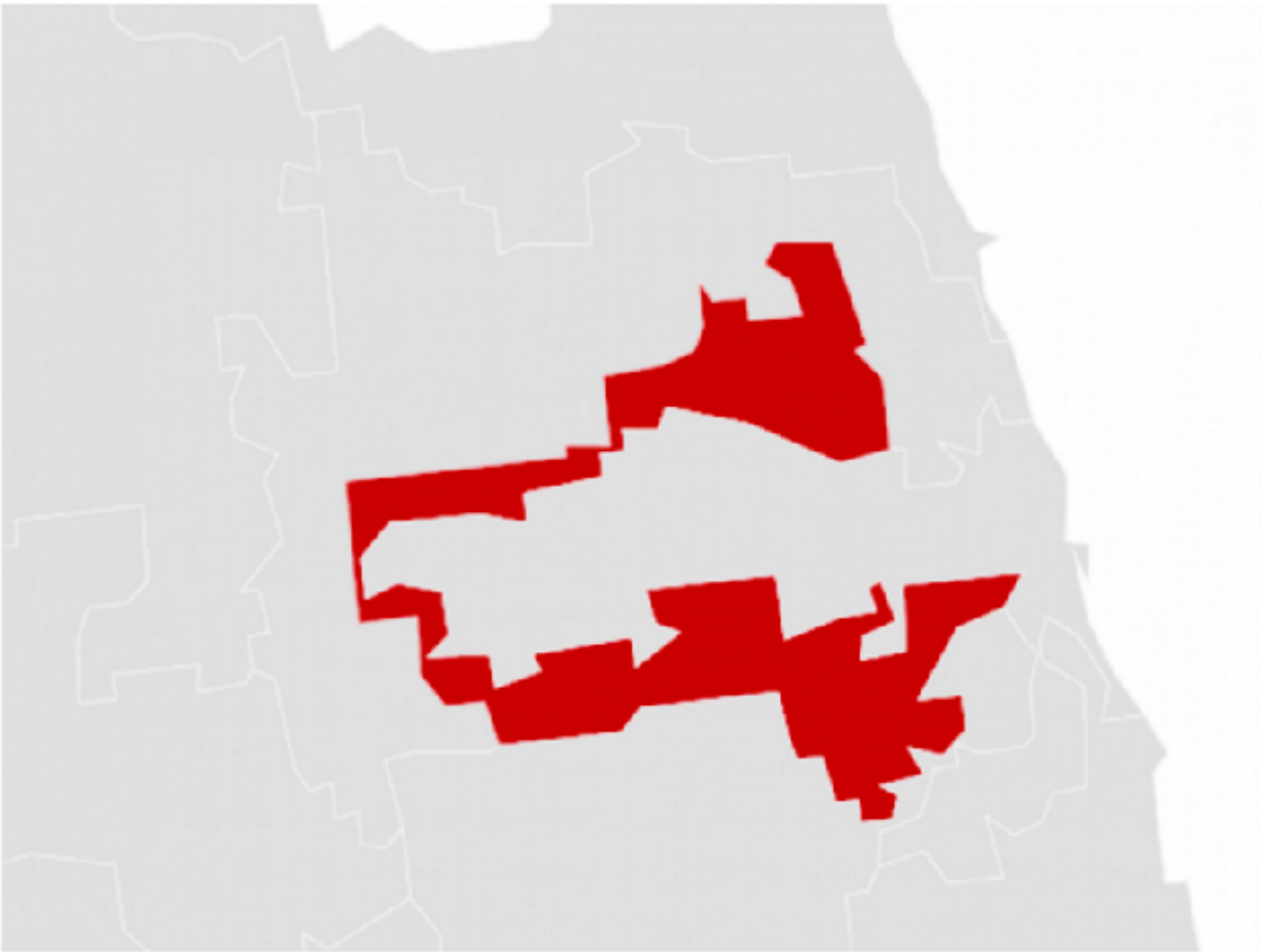
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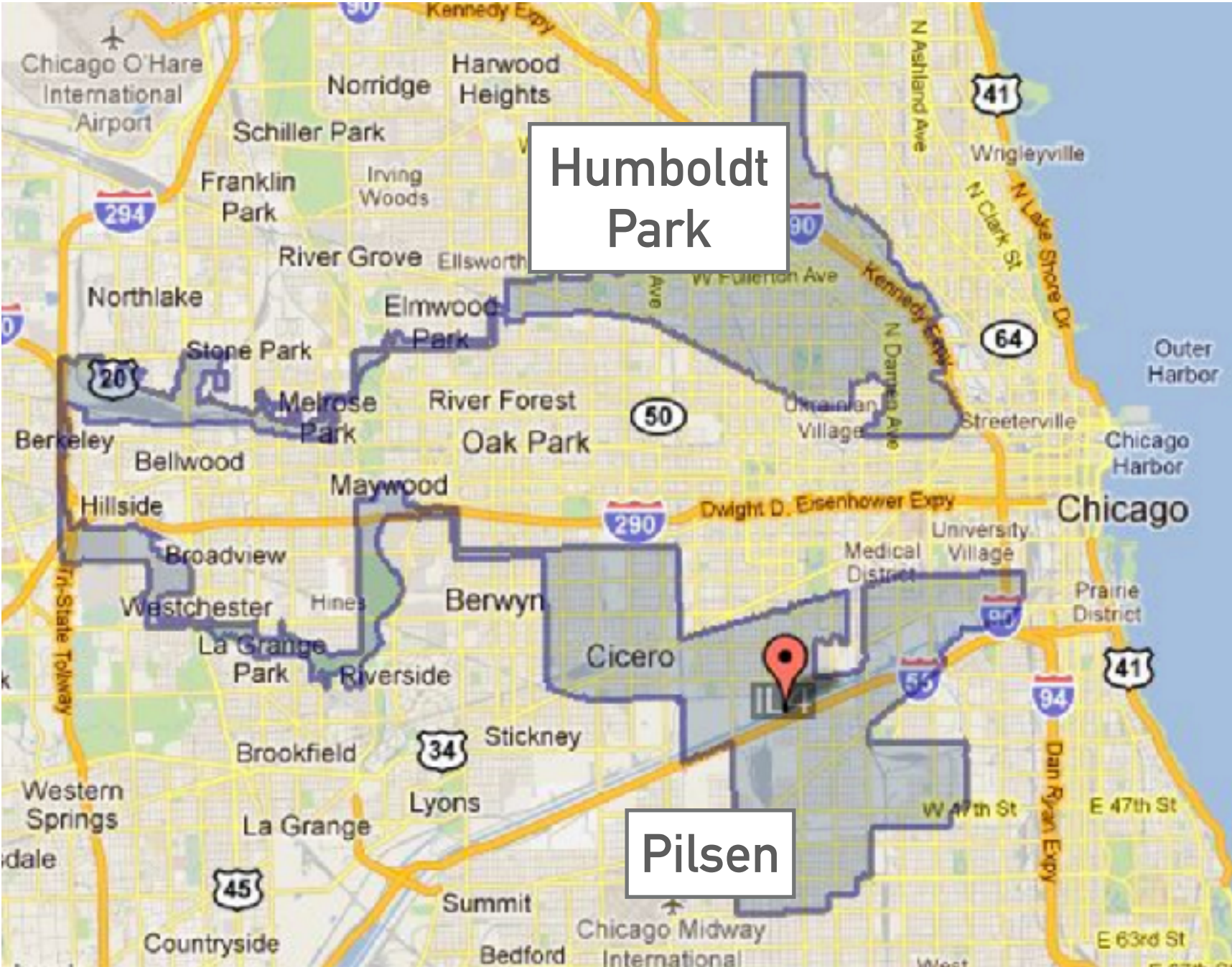
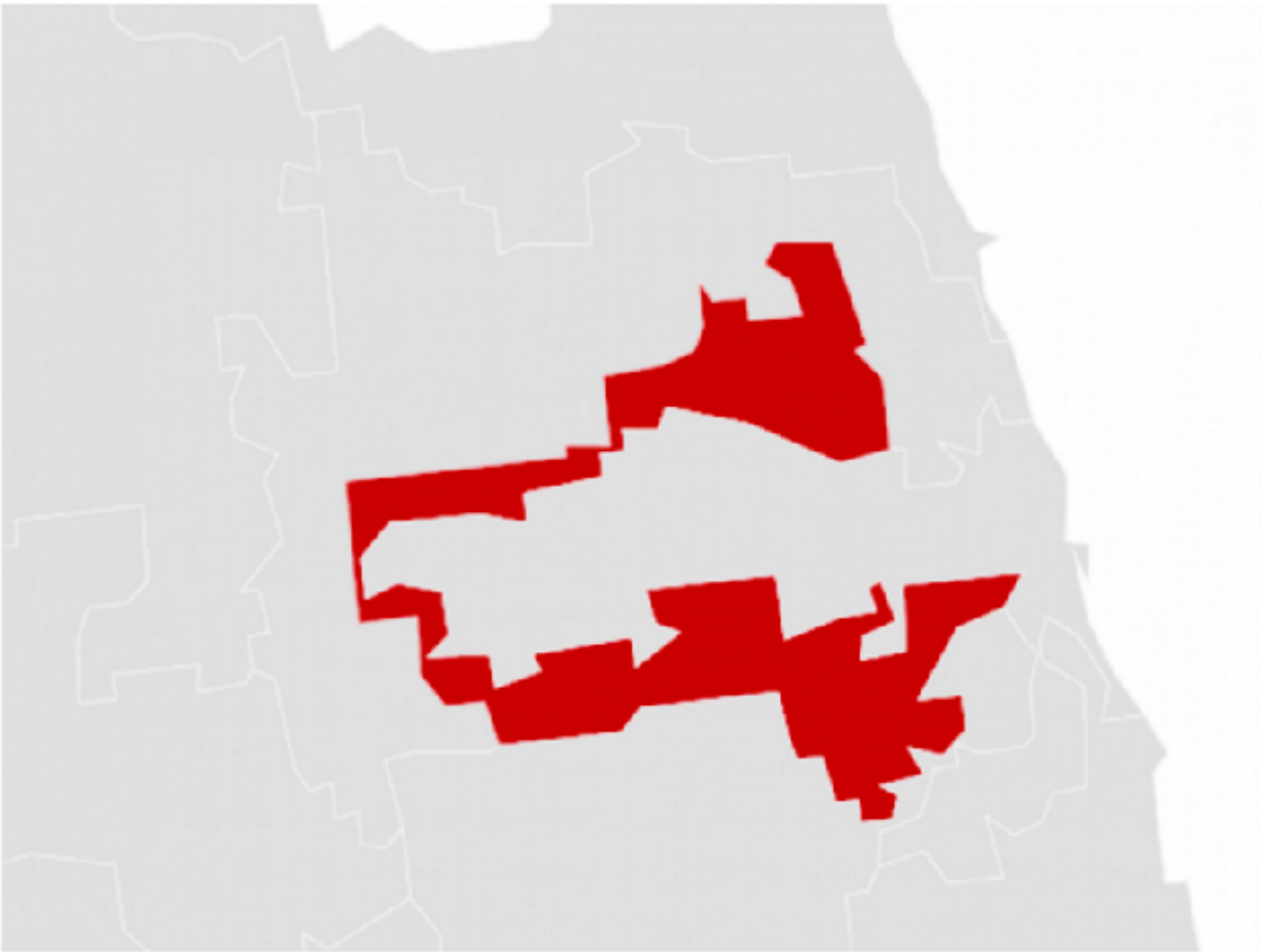
A map of the United Kingdom with the study area highlighted in red. The red area is located in the central part of England, covering parts of Northamptonshire, Leicestershire, and Lincolnshire. The surrounding areas are shown in light grey, and the coastlines are indicated by a thin white line.



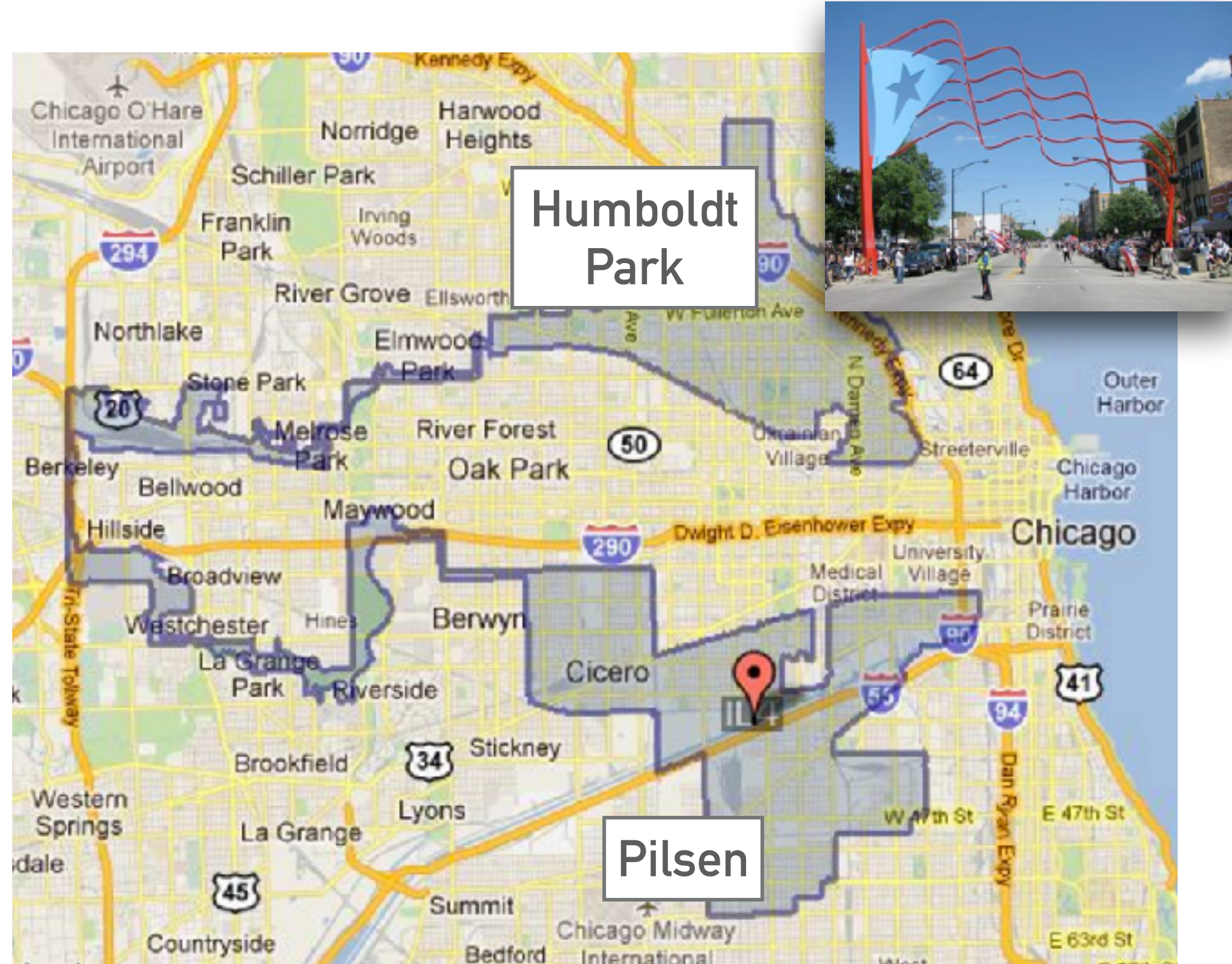
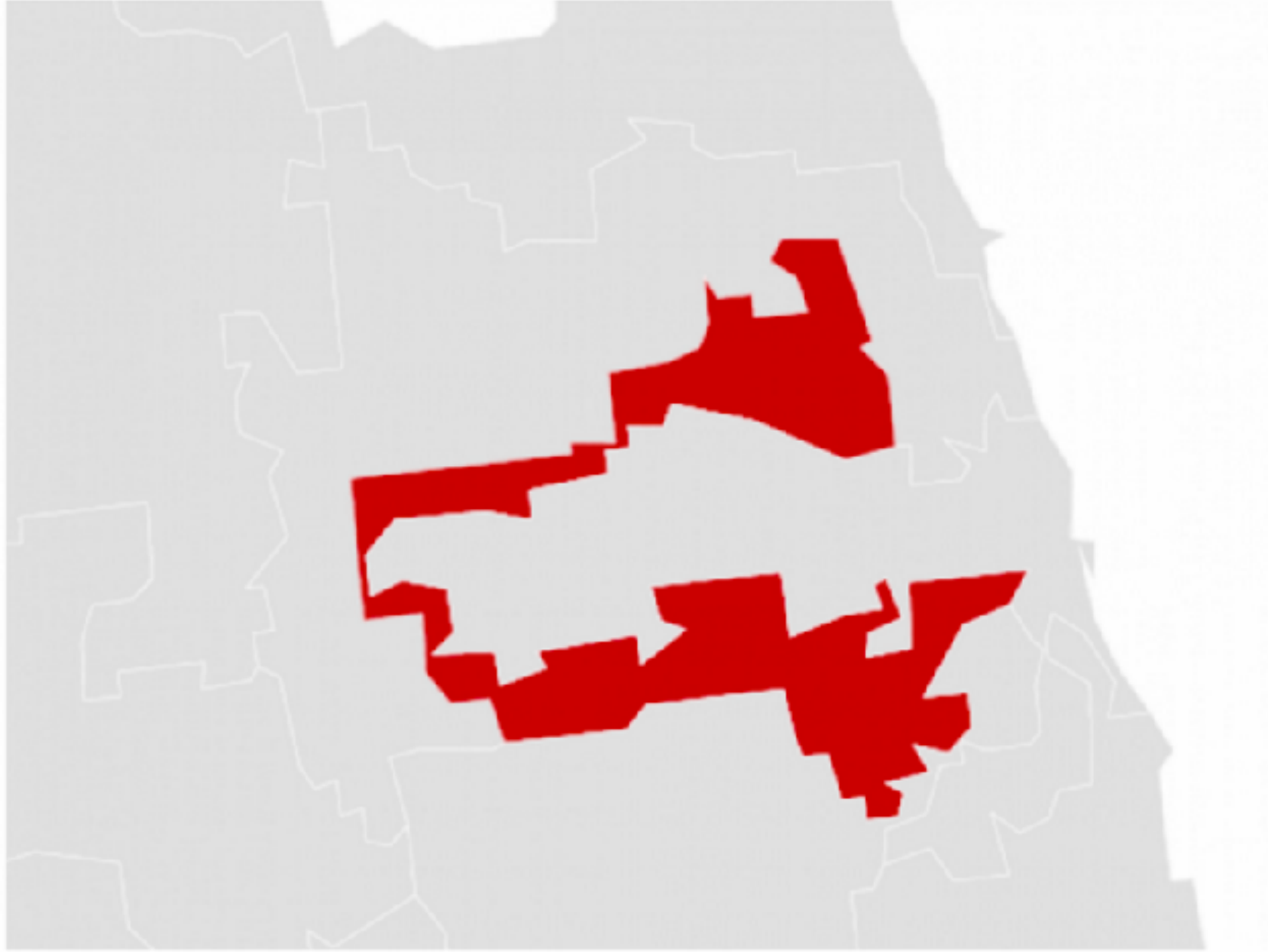
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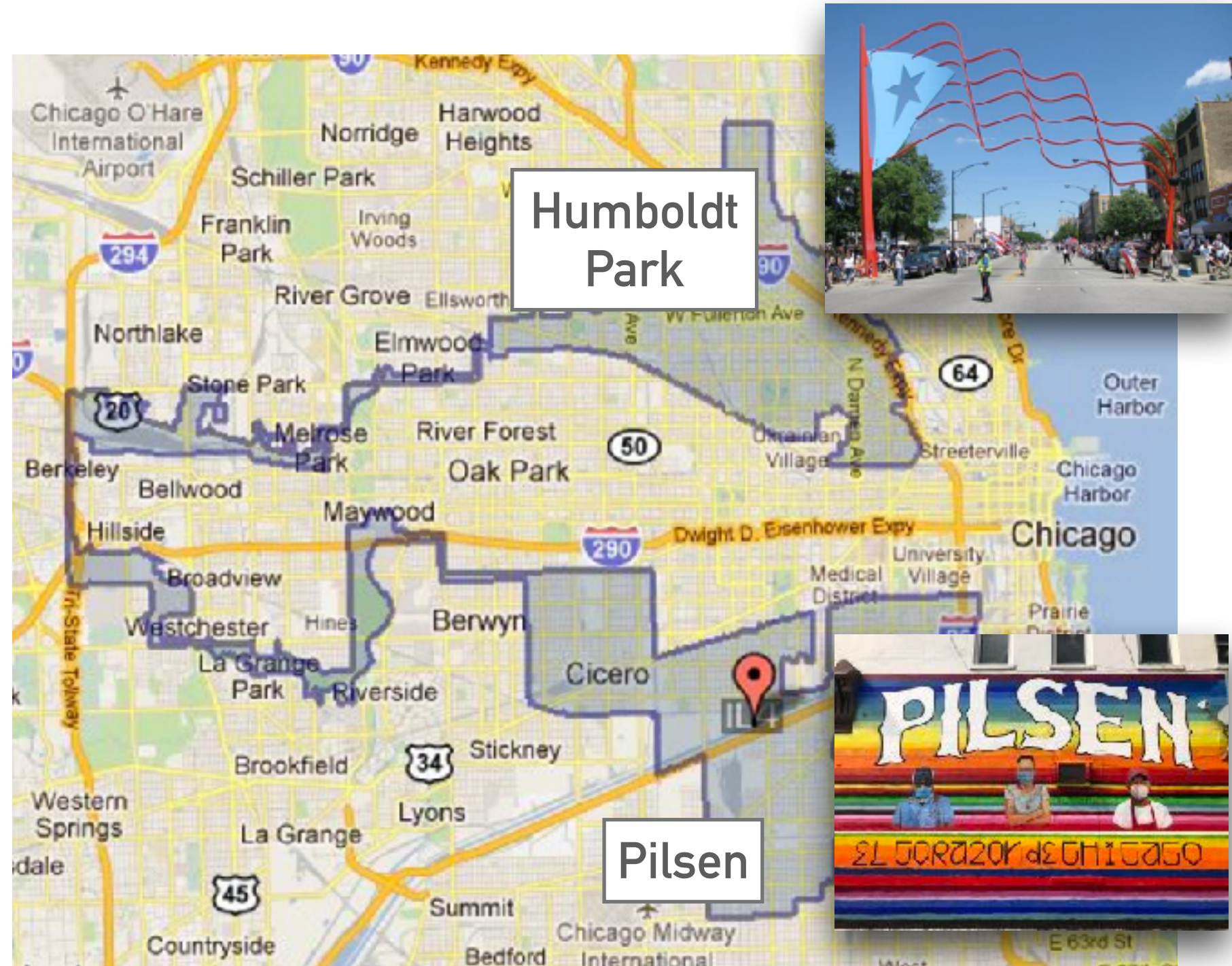
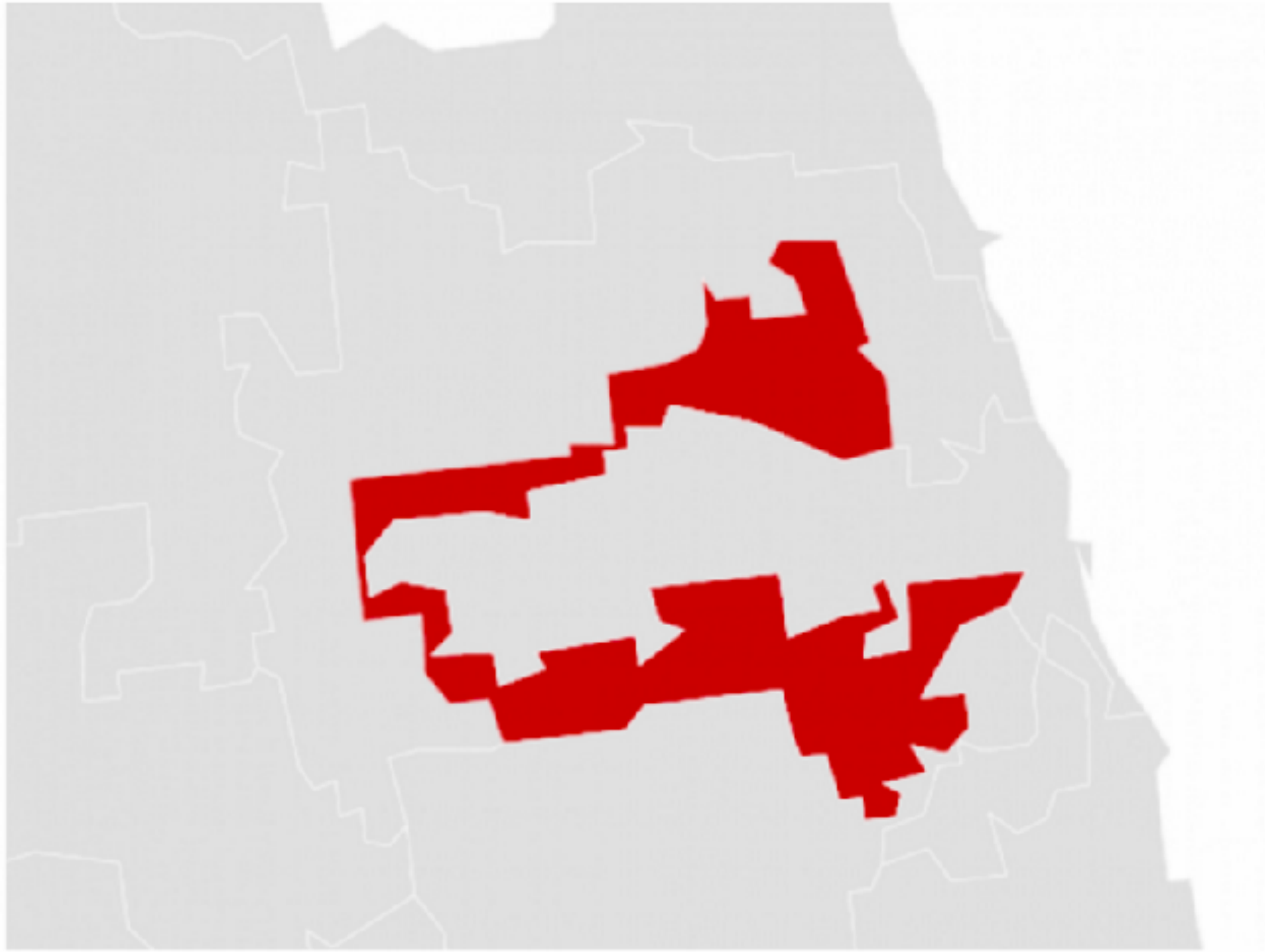
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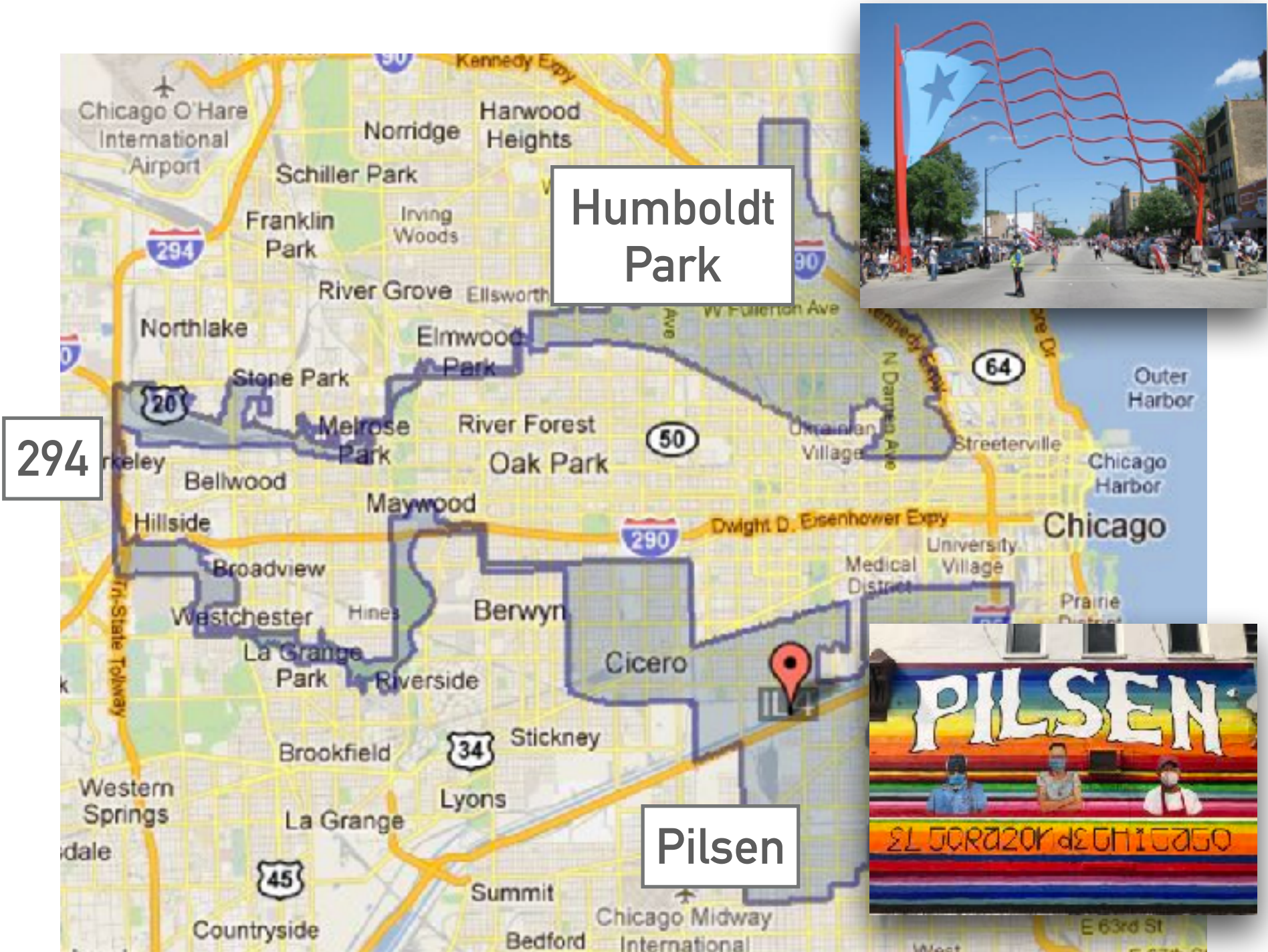
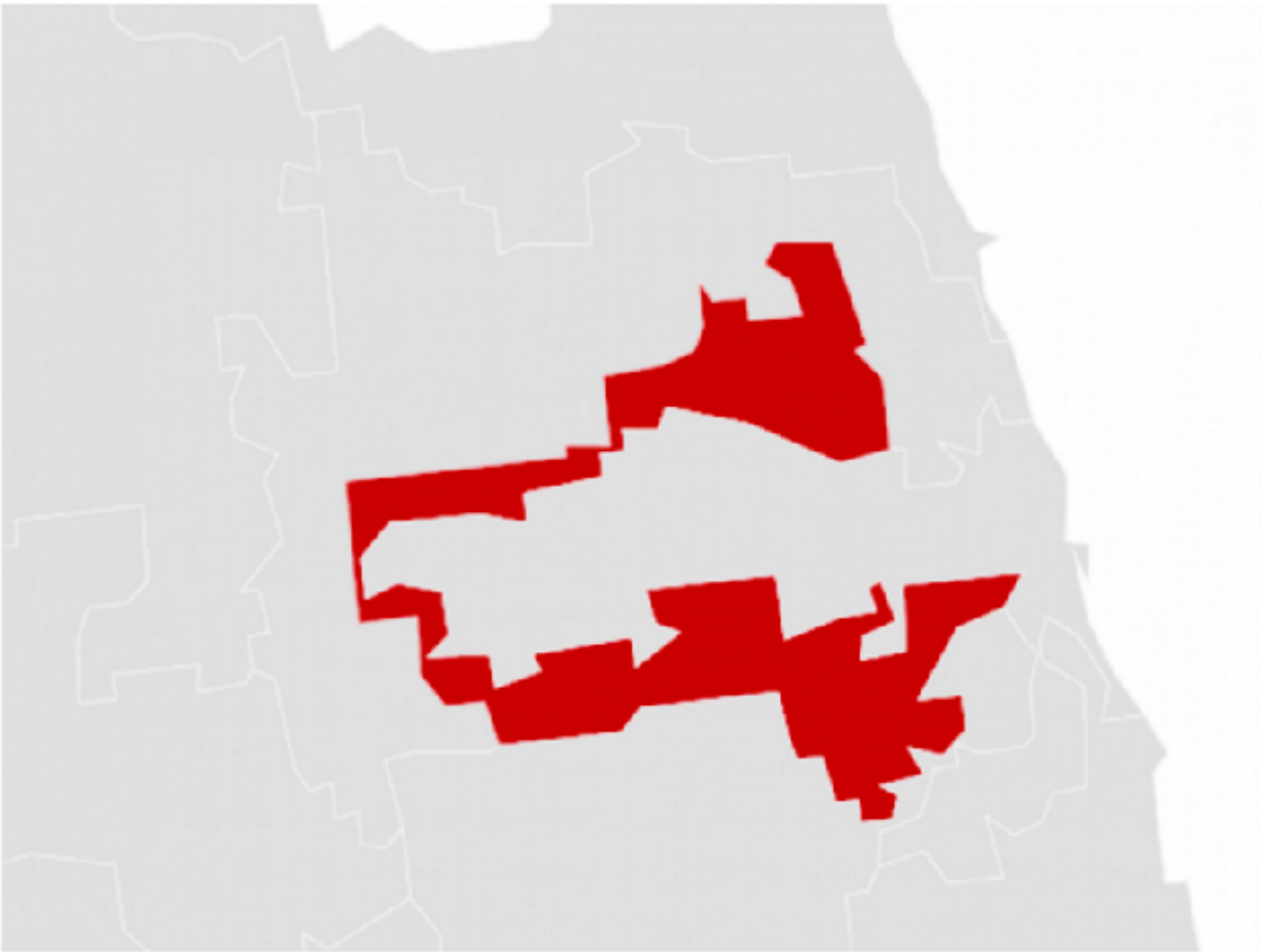
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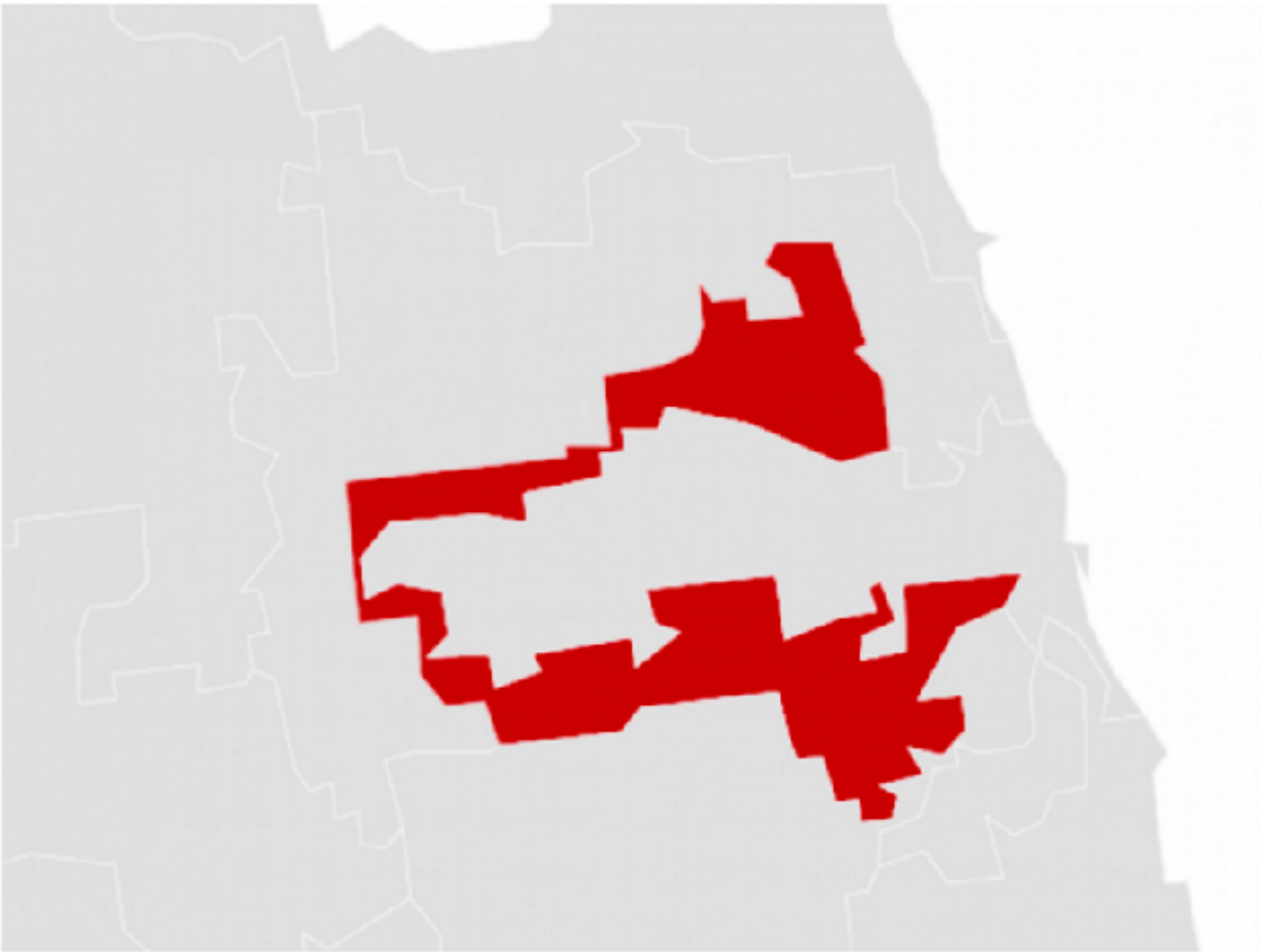
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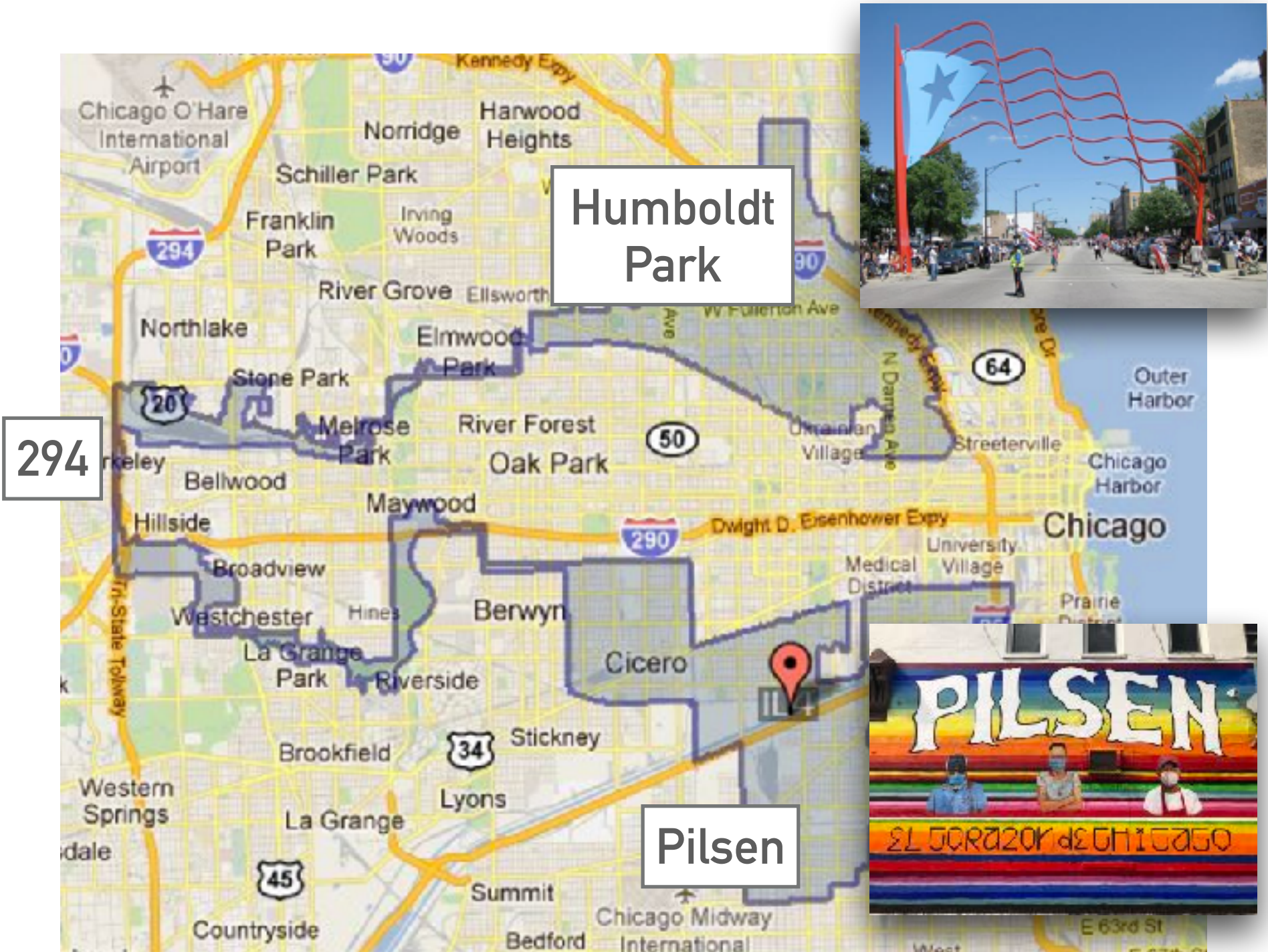
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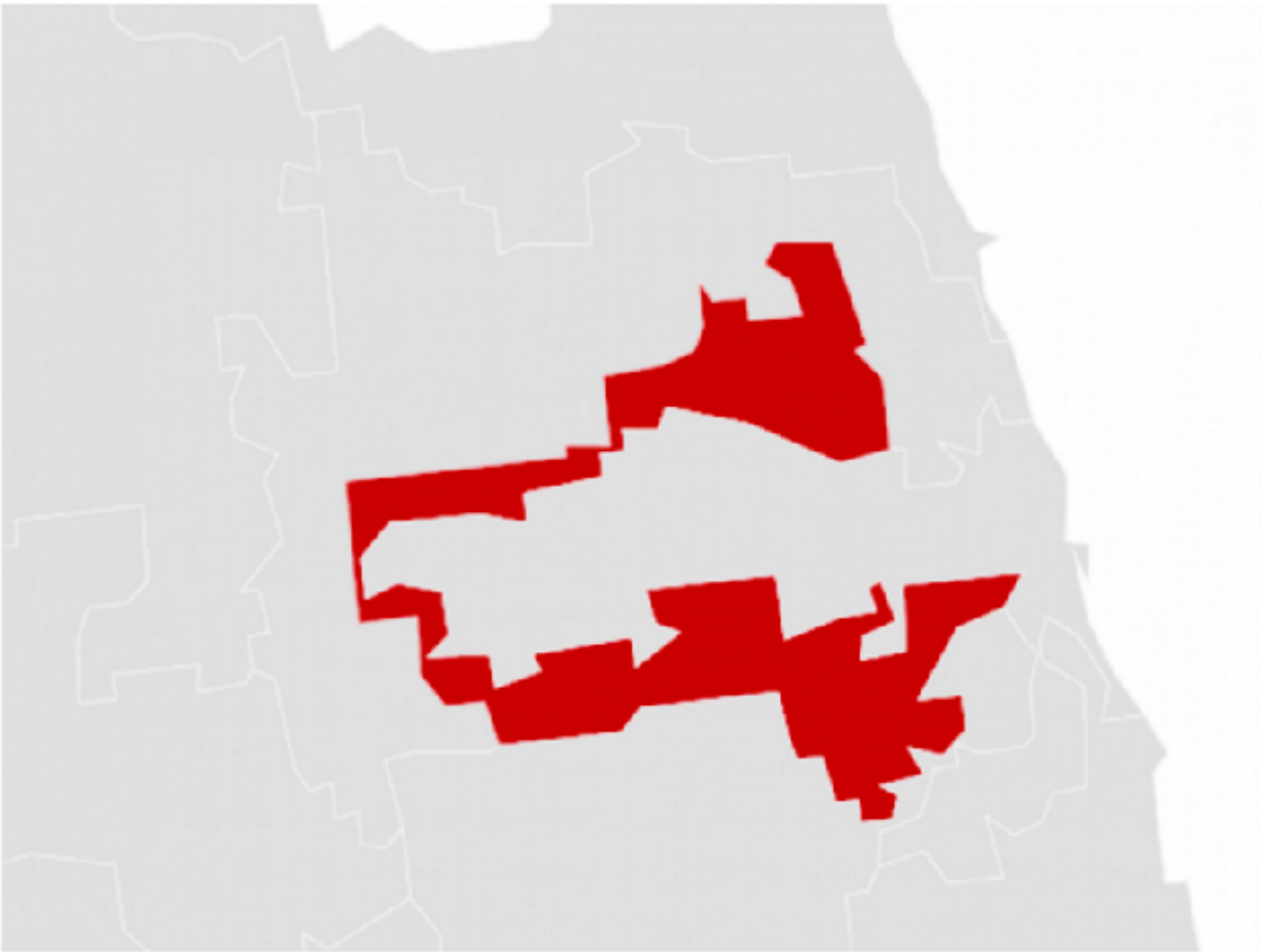
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Year	Democrat	Votes	Pct
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2008	Luis Gutiérrez (inc.)	112,529	80.6%
2010	Luis Gutiérrez (inc.)	63,273	77.4%
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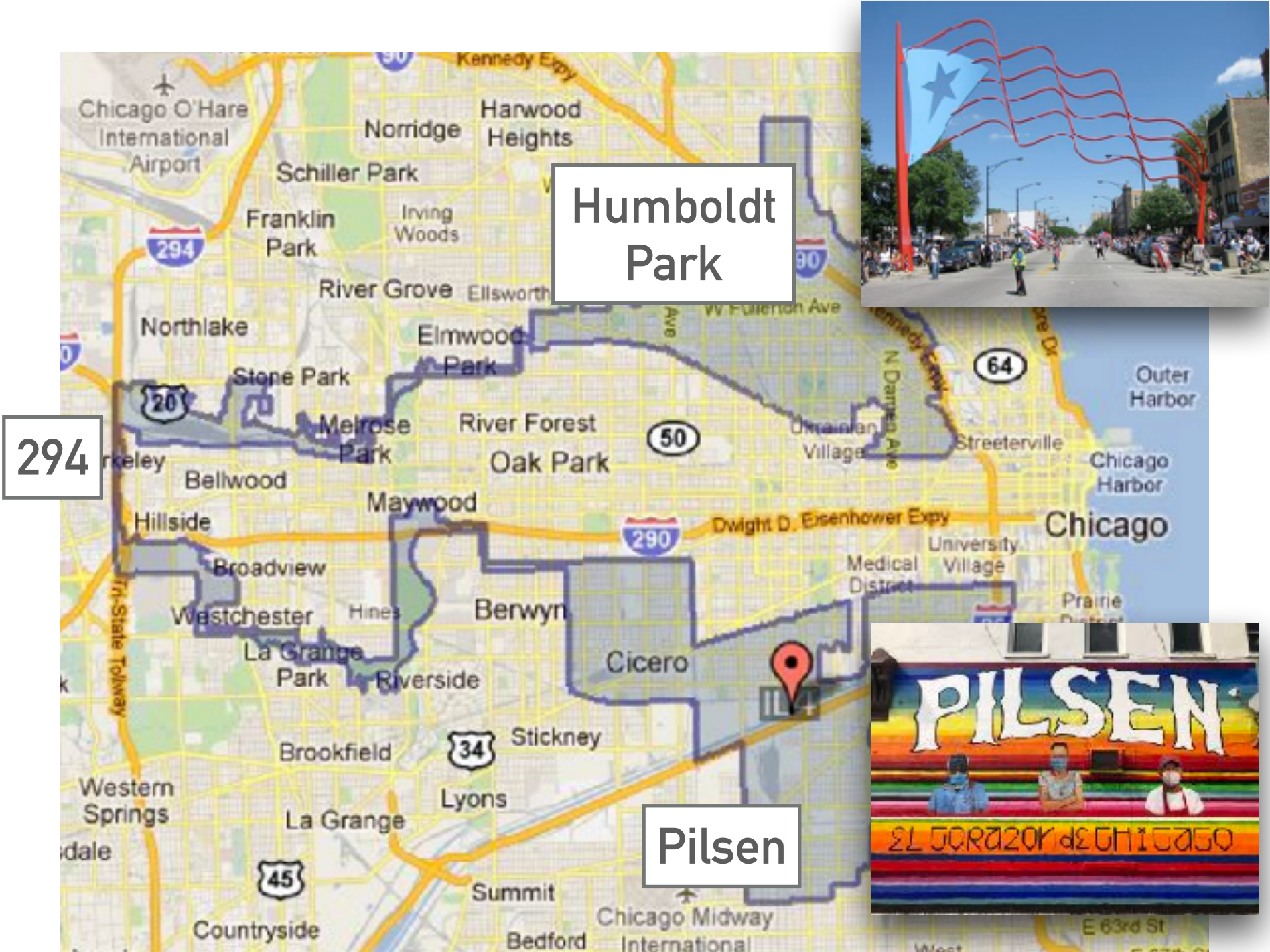


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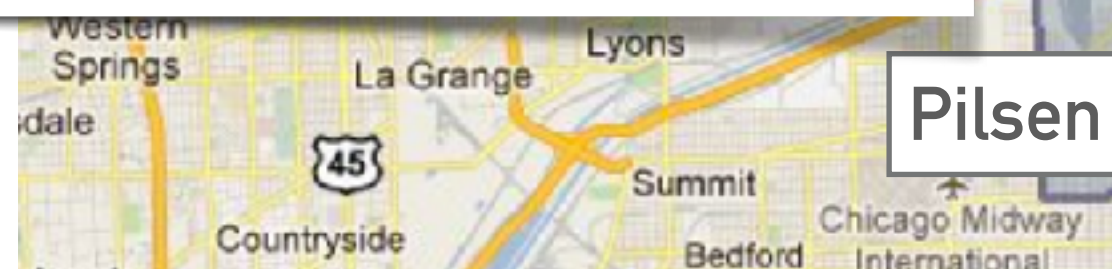
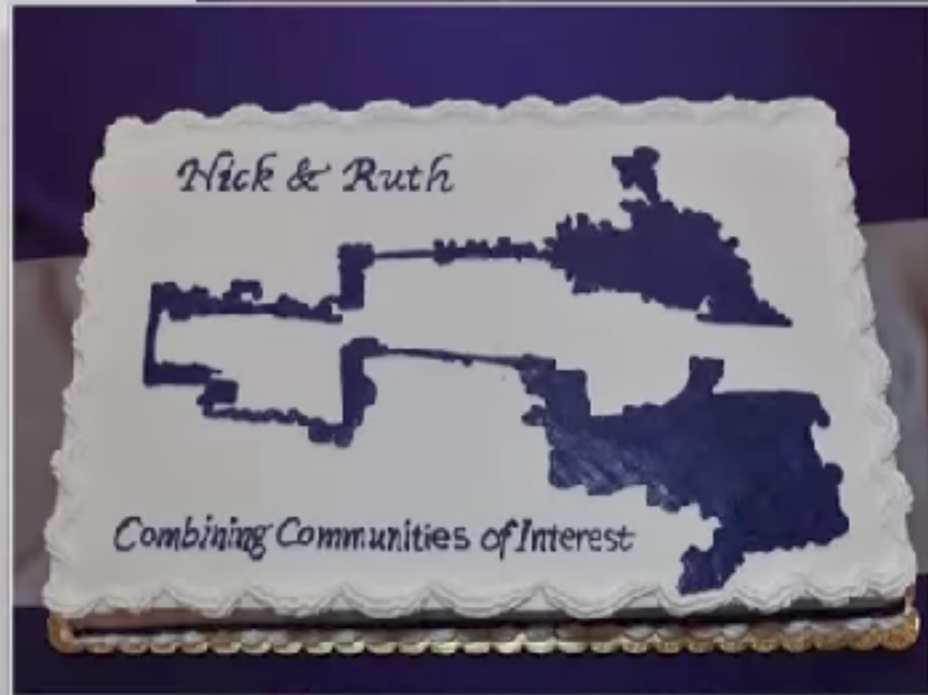


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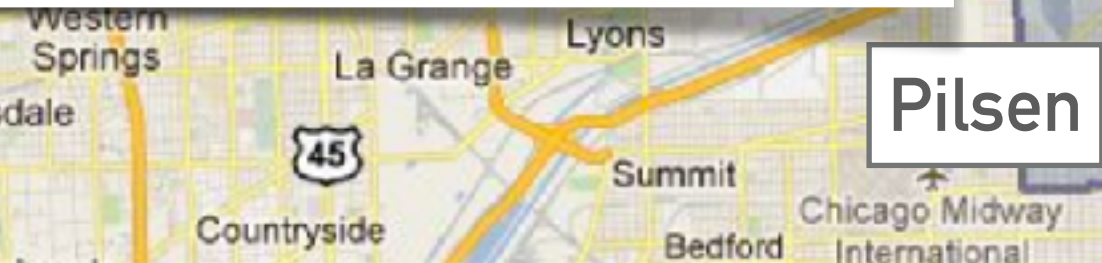
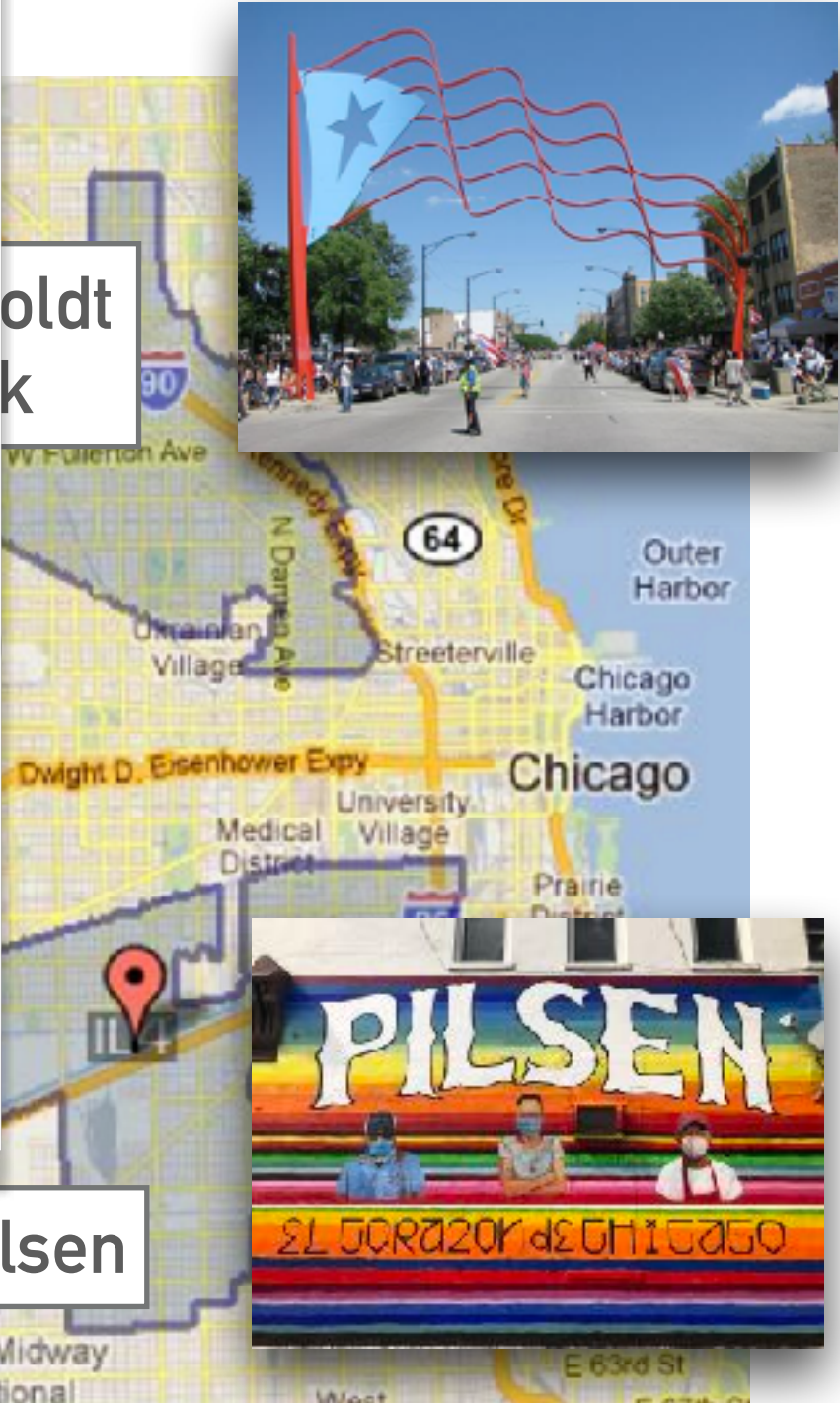
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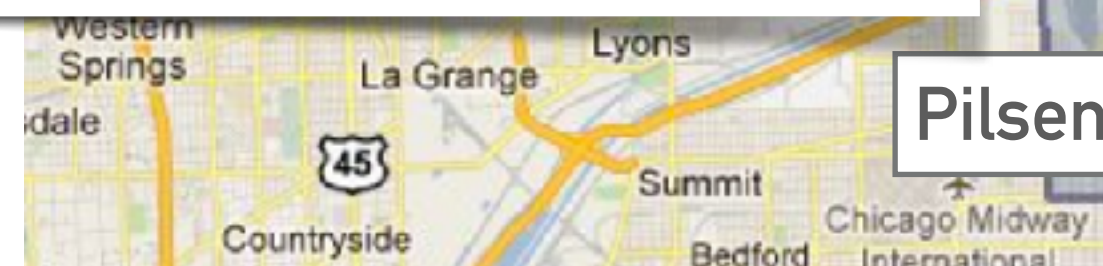
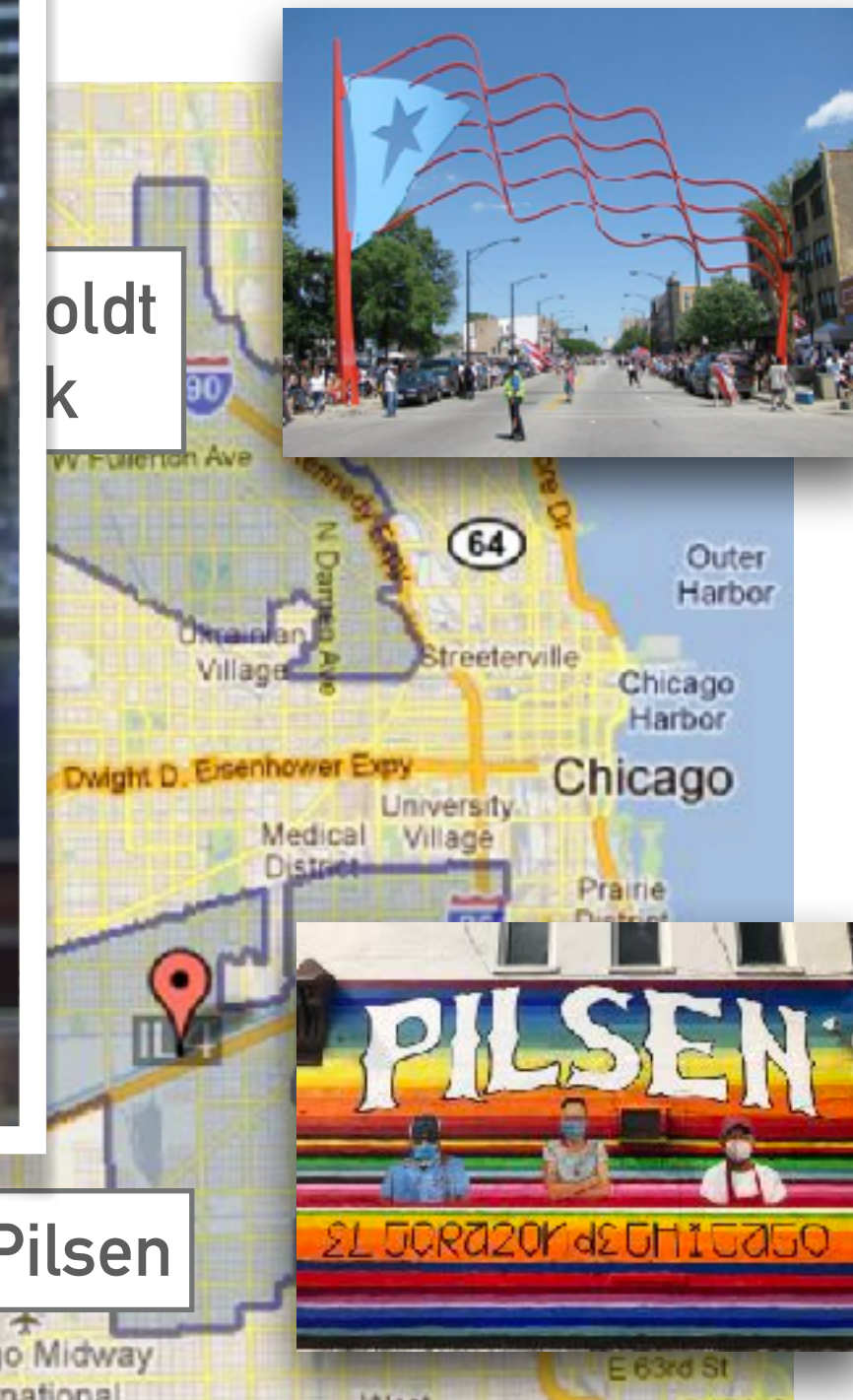
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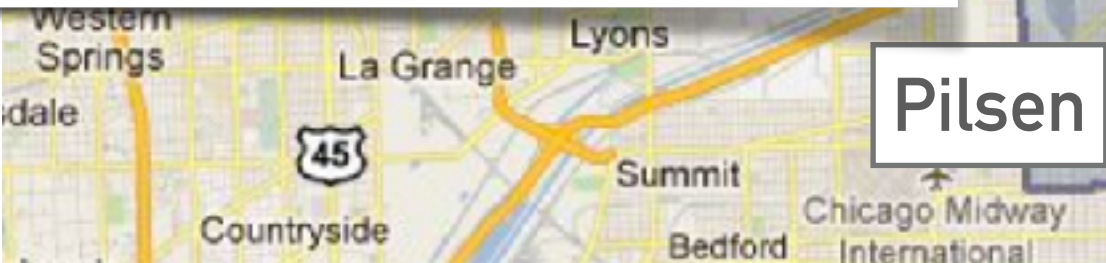
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dubious outcome



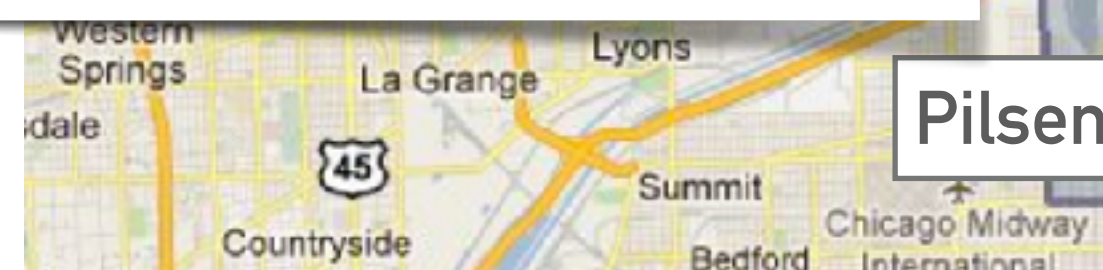
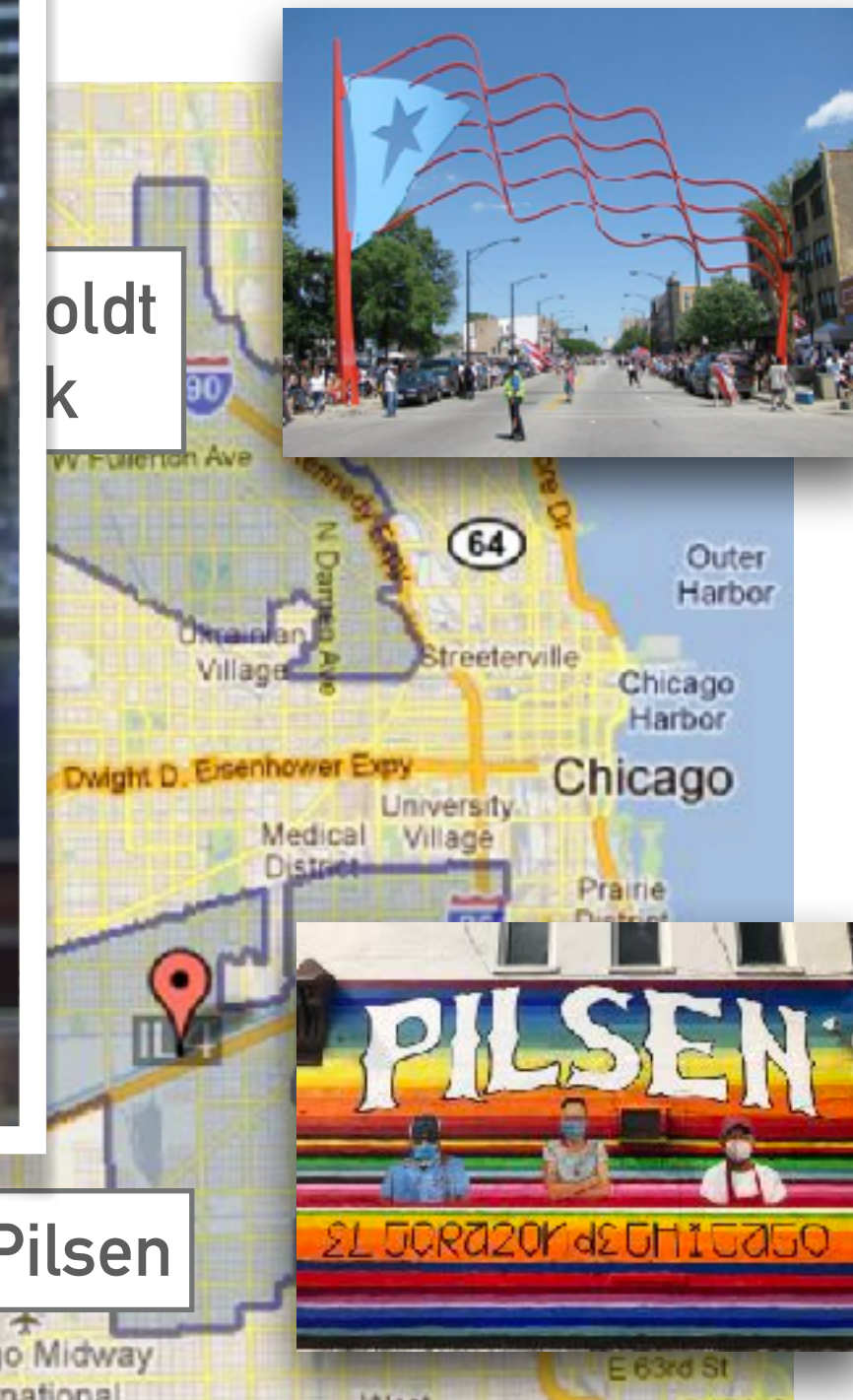
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???



Two important kinds of abuse:
partisan and **racial**
gerrymandering

Partisan gerrymandering



Partisan gerrymandering

- You can draw districts so that your “team” wins extra seats — win by a little, lose by a lot



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- The original Gerry-Mander was a salamander-shaped district approved by Governor Gerry of MA in 1812

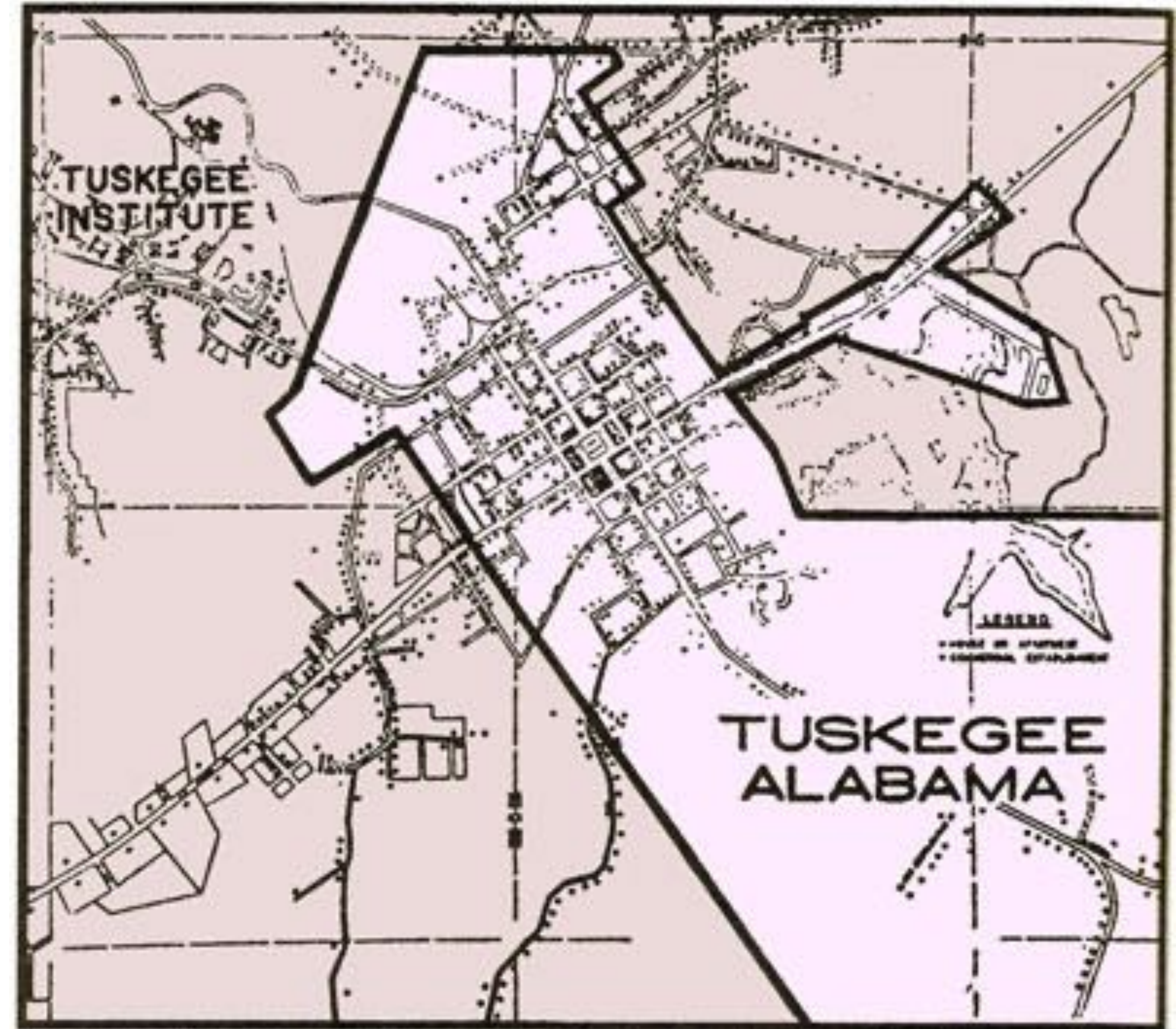


Racial gerrymandering

- Pretty much as soon as *Brown v Board of Ed* caused a spike in Black voter registration, many Southern states dug in with big gerrymanders

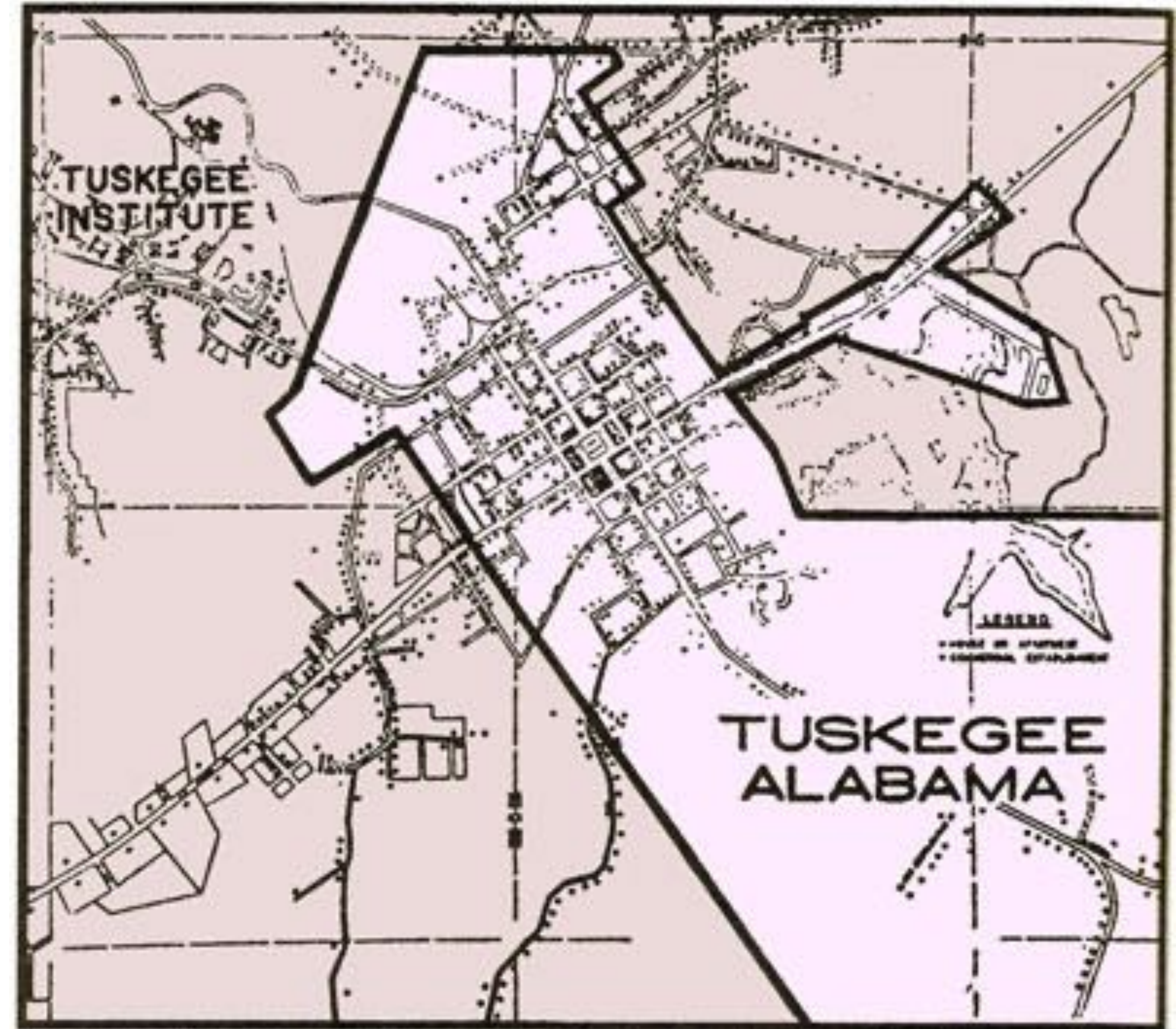
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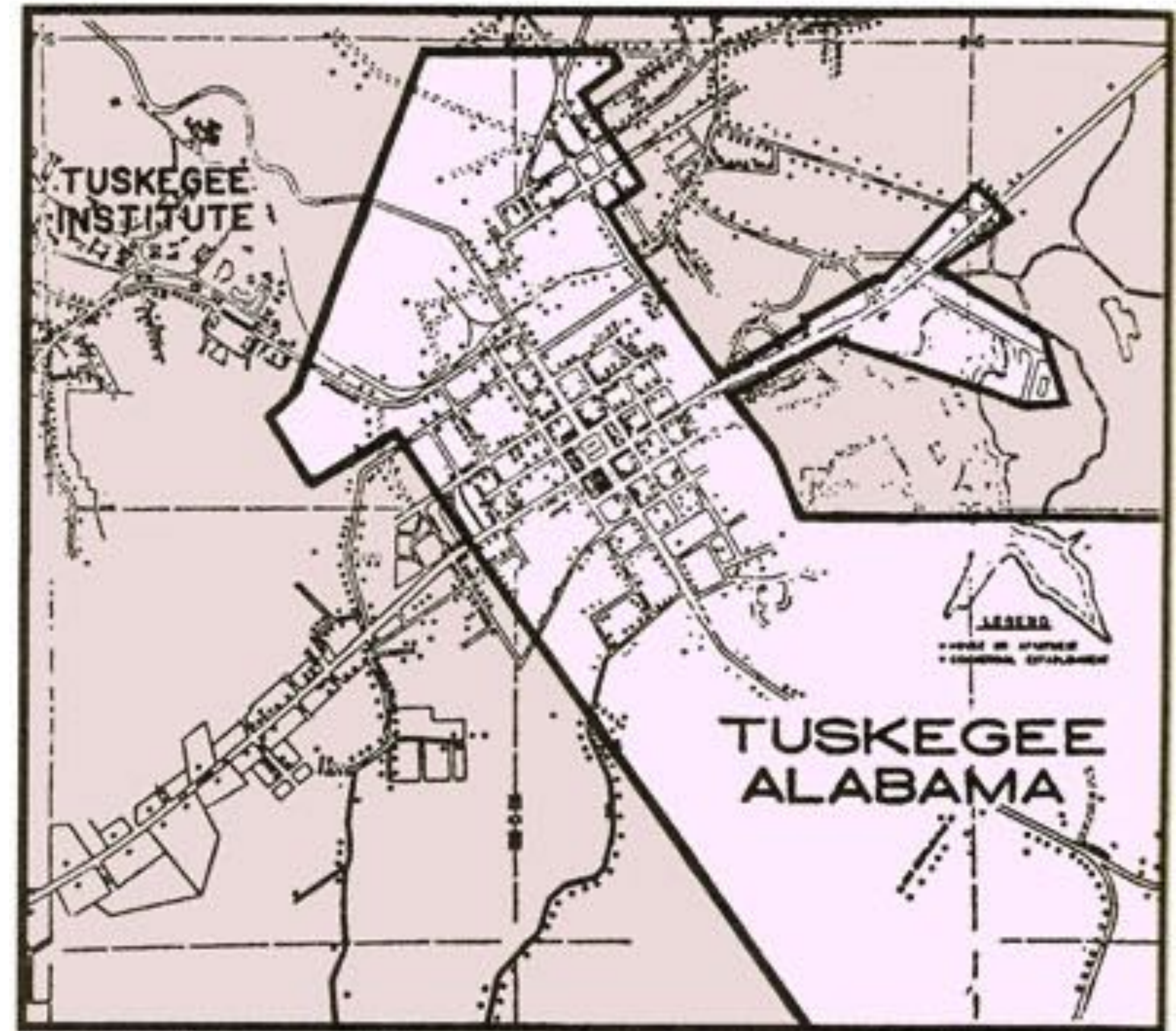
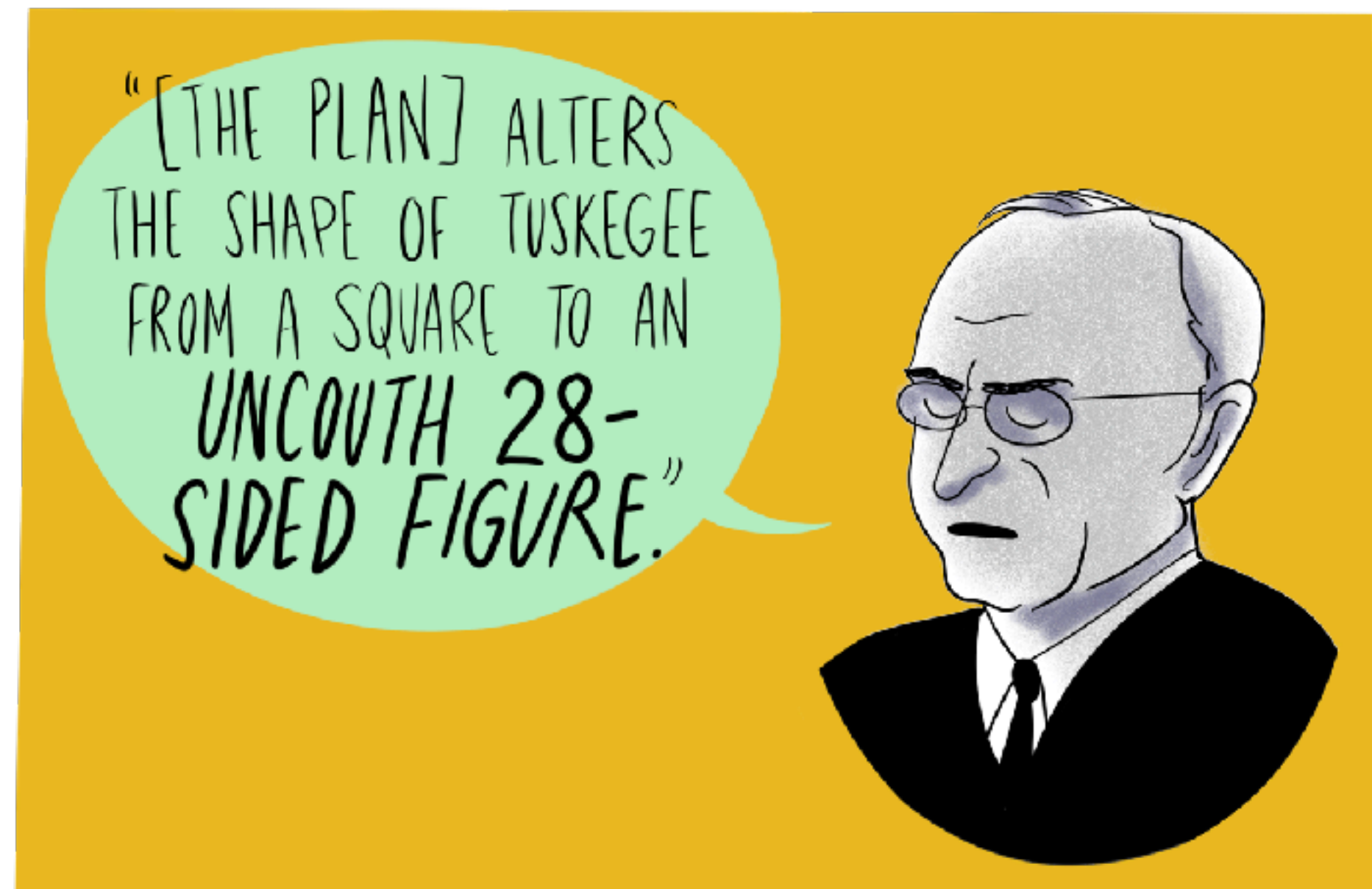


BEFORE: 80% Black

AFTER: 100% White

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Voting Rights Act of 1965



Voting Rights Act of 1965



landmark civil rights legislation blocks “denial or abridgement” of minority vote

Voting Rights Act of 1965



landmark civil rights legislation blocks “denial or abridgement” of minority vote
for many years, this meant drawing a handful of majority-Black districts on purpose

what does an expert witness do in a redistricting case?

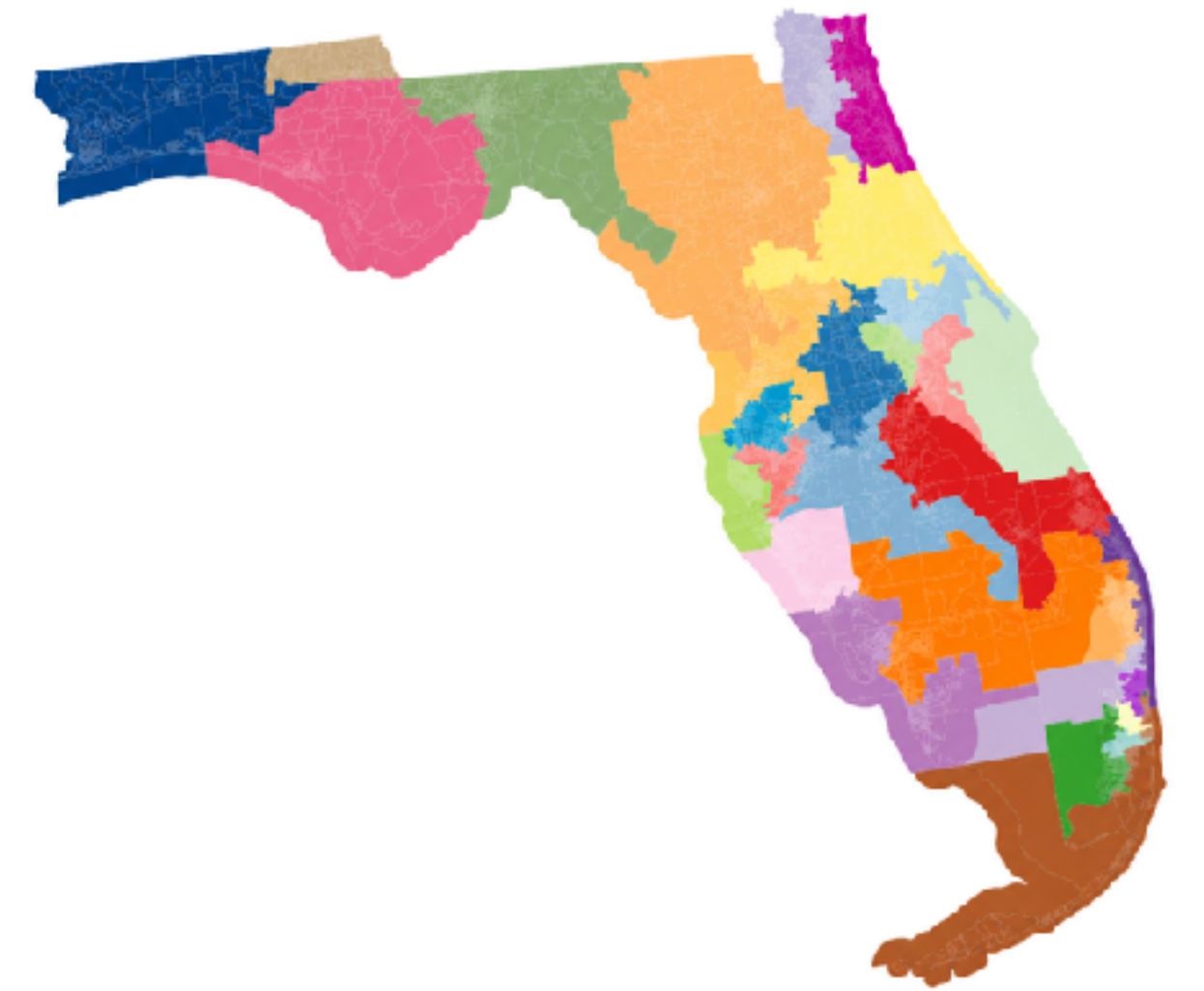
I've been an expert witness in court cases in Alabama, North Carolina, South Carolina, Pennsylvania, Wisconsin, Georgia, Texas, Florida, and Tennessee

Algorithmic redistricting

use computers to make zillions of “neutral” maps

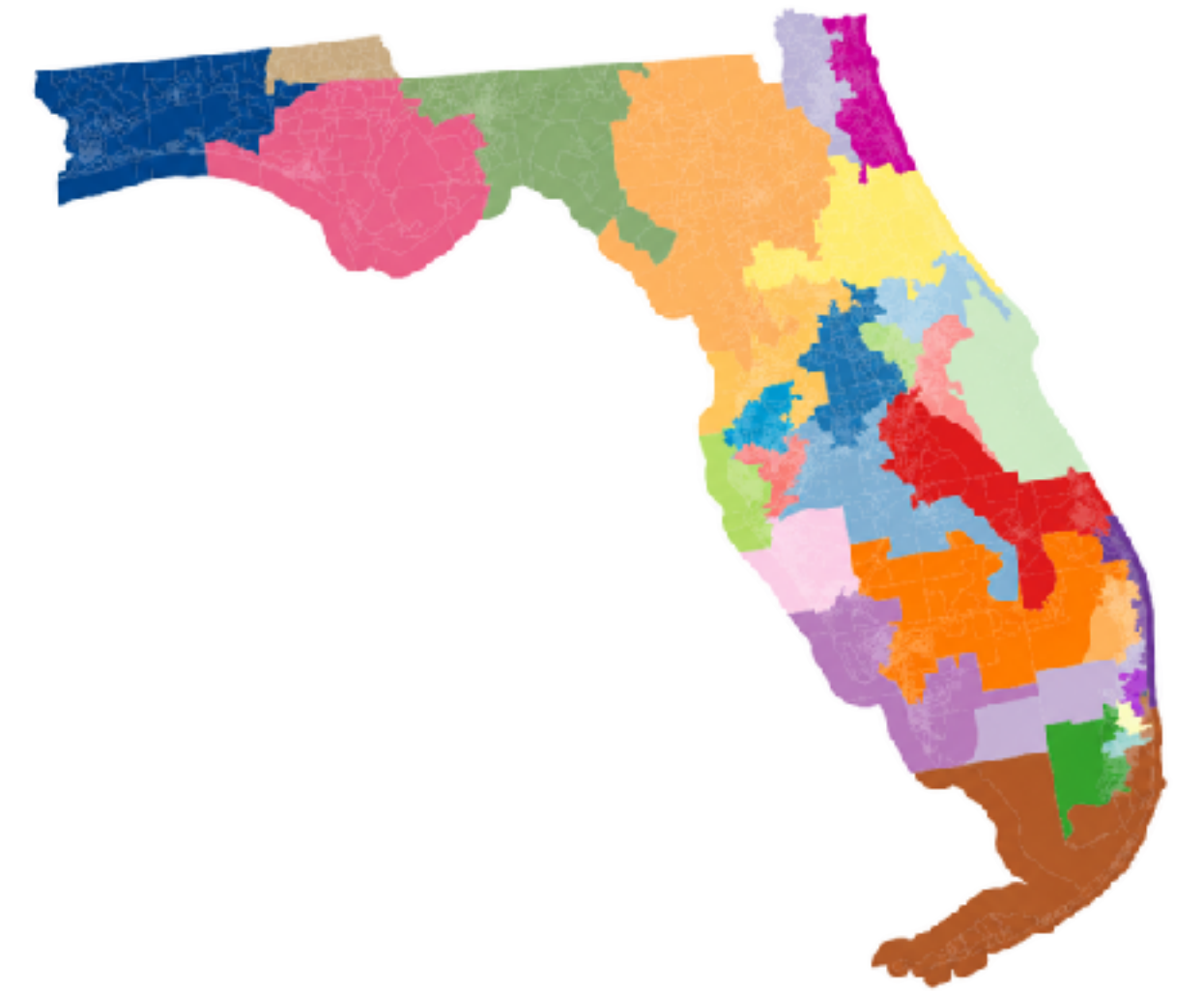
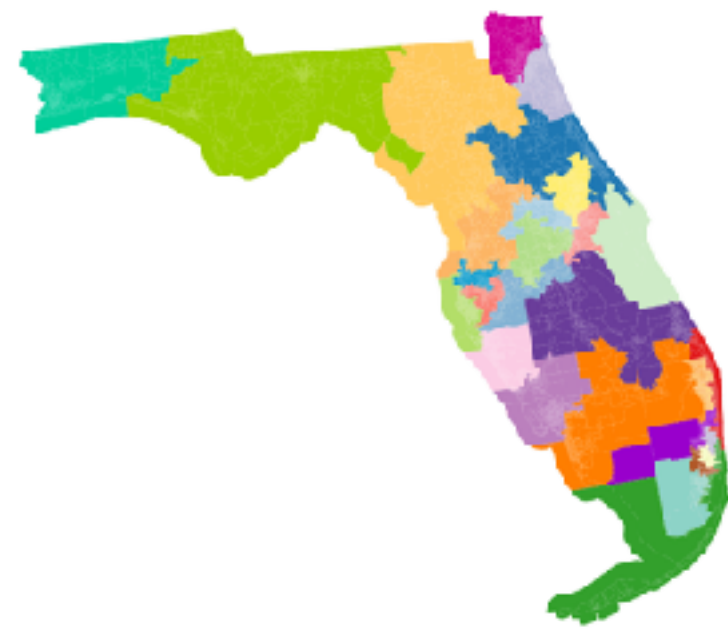
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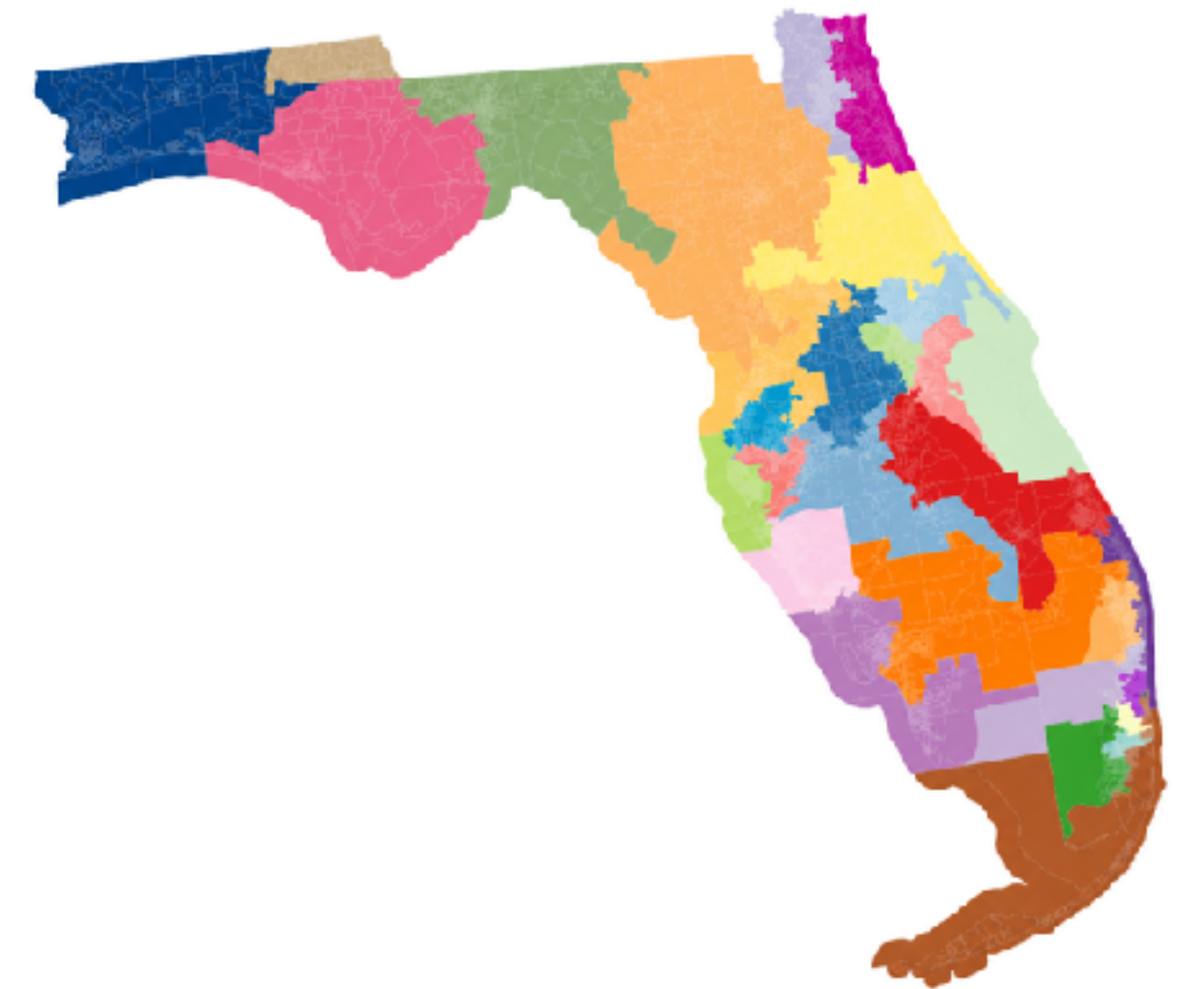
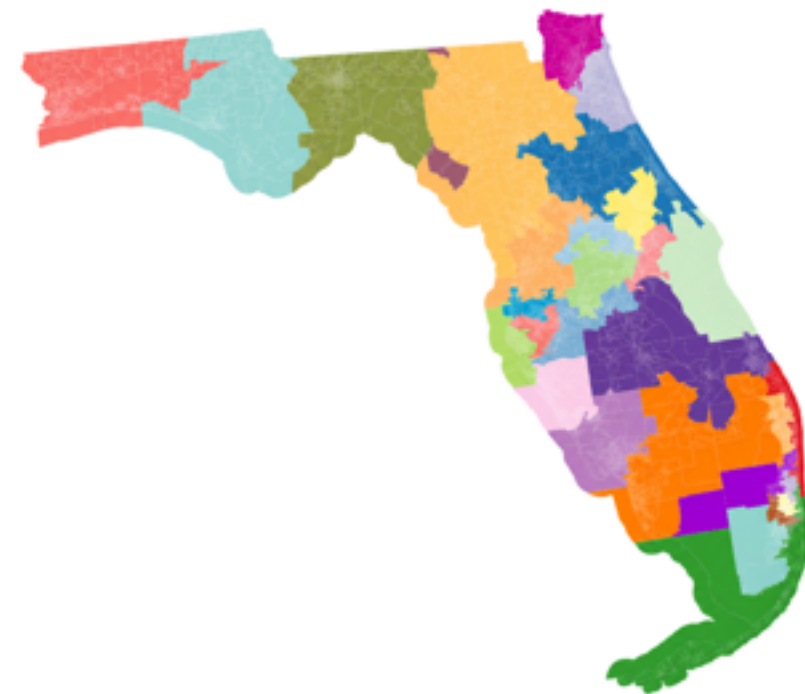
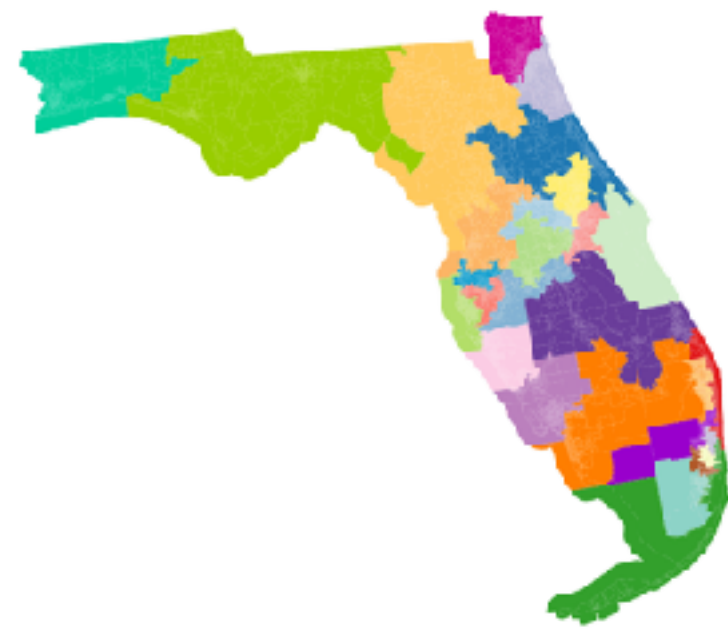
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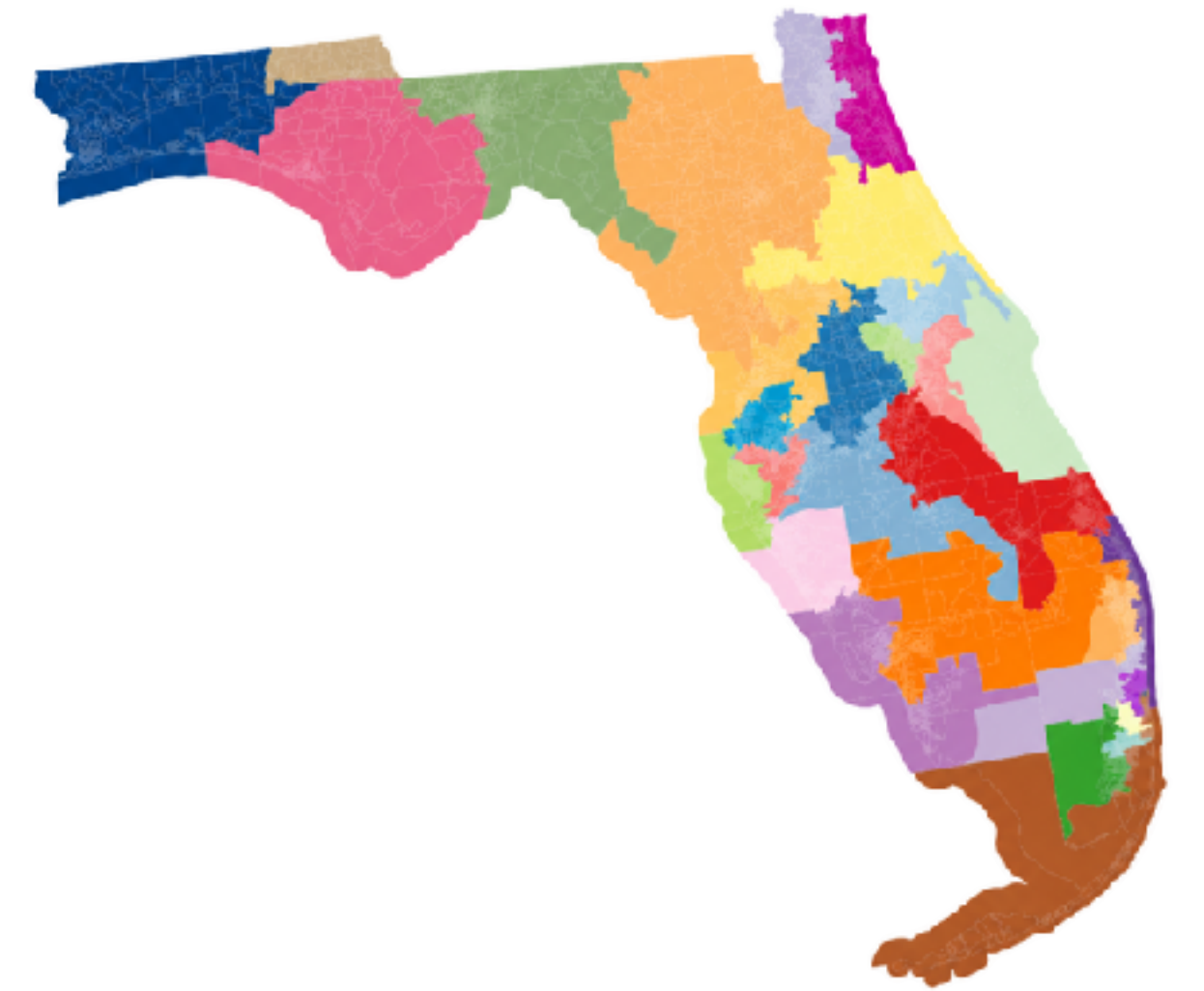
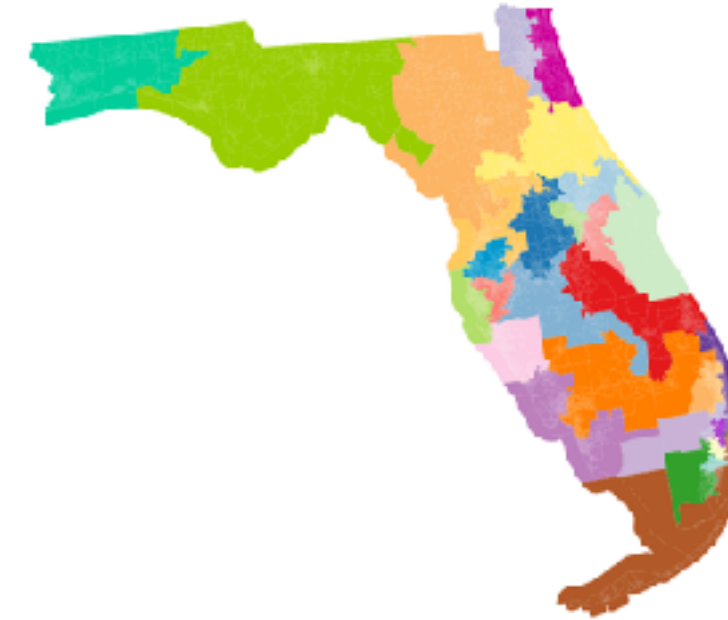
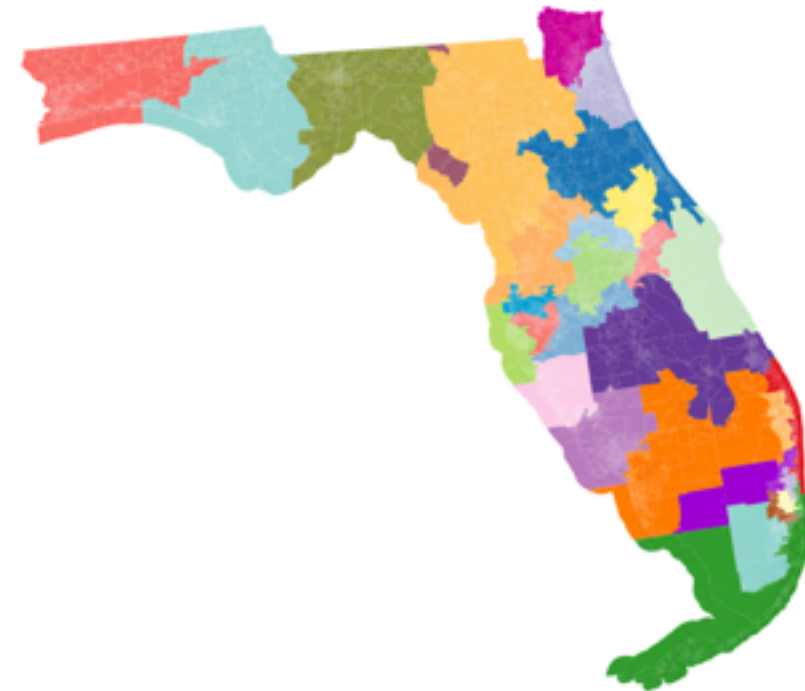
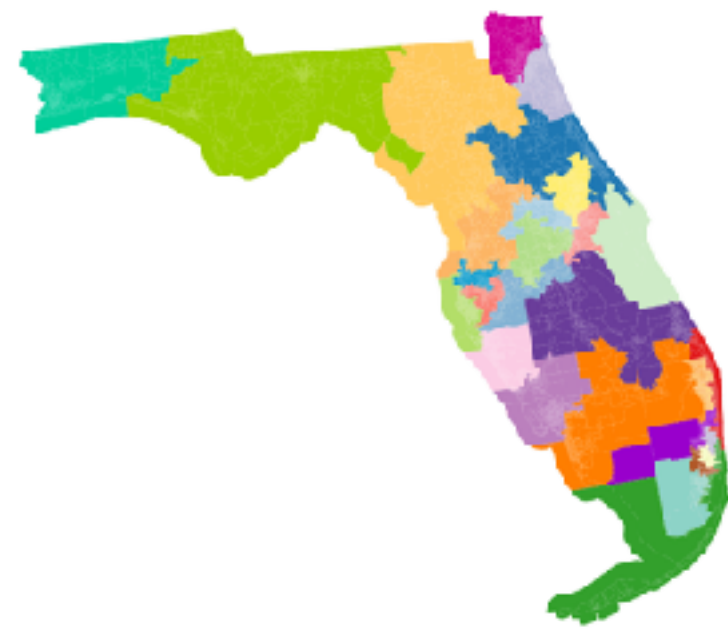
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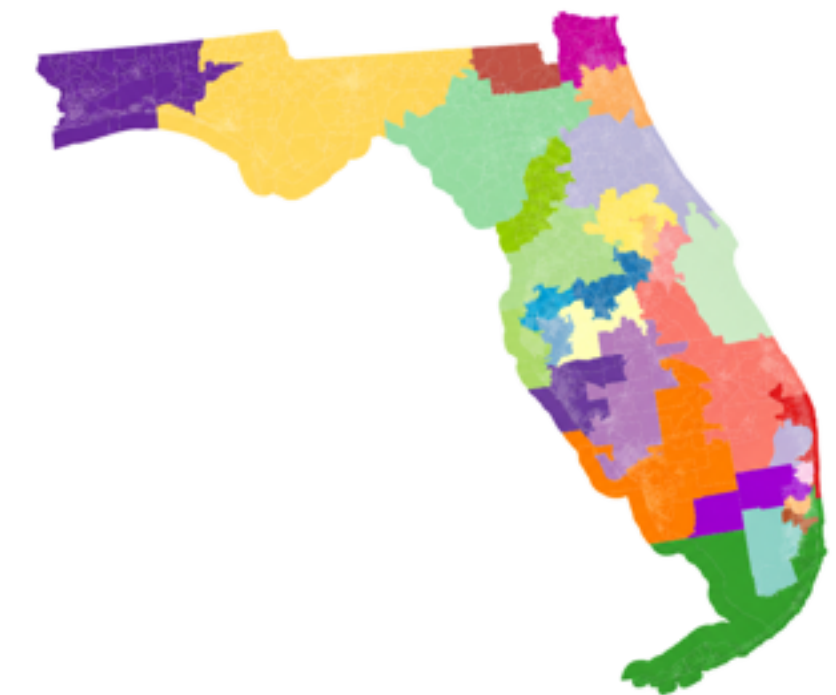
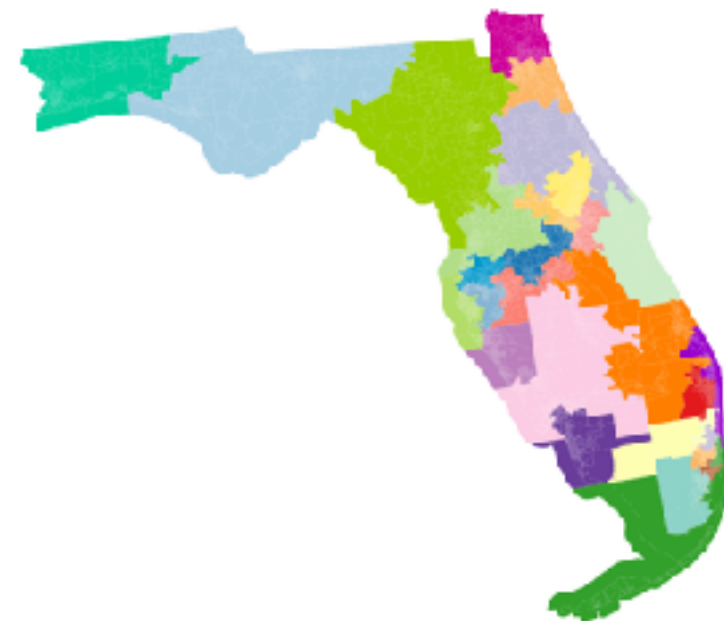
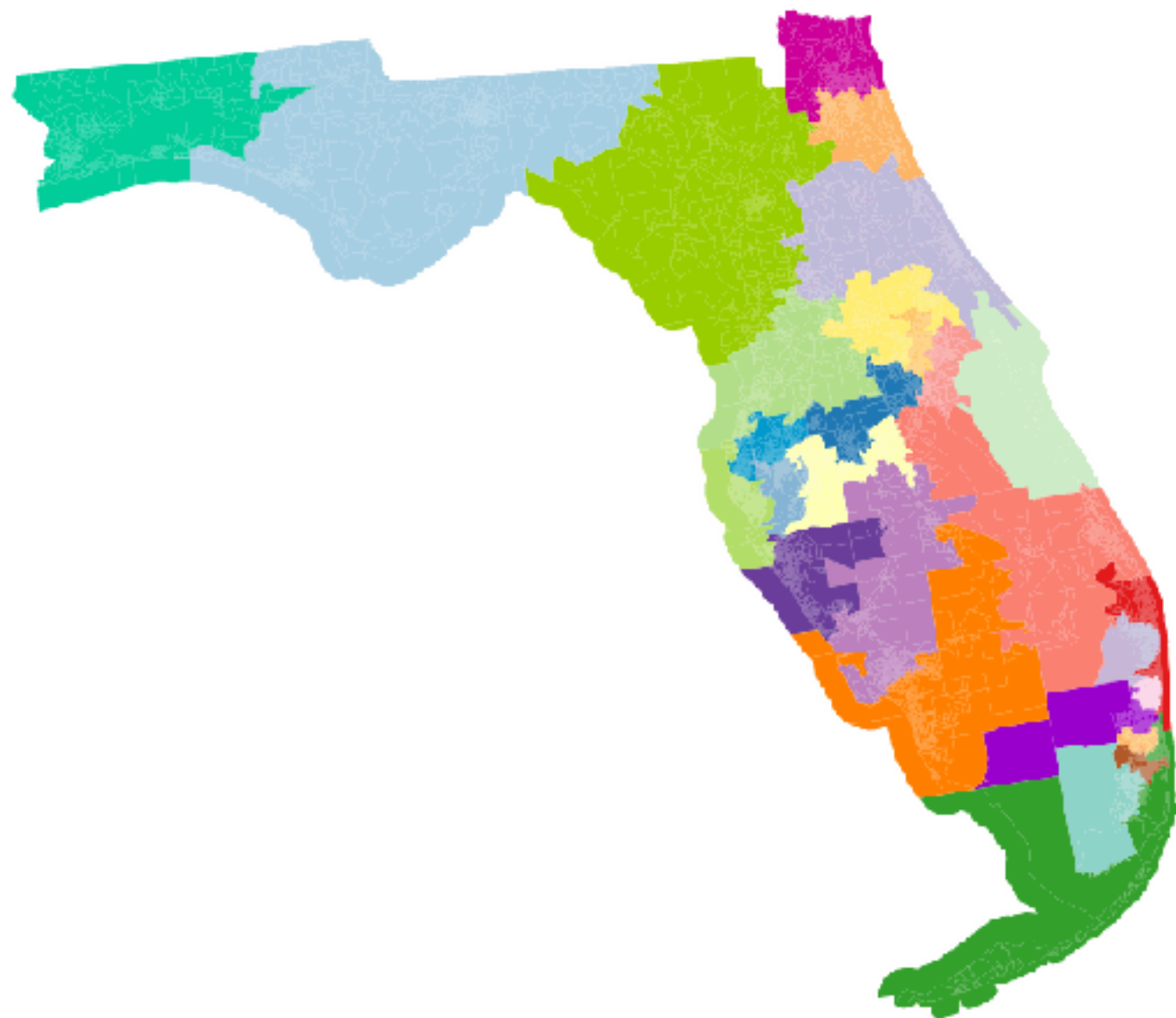
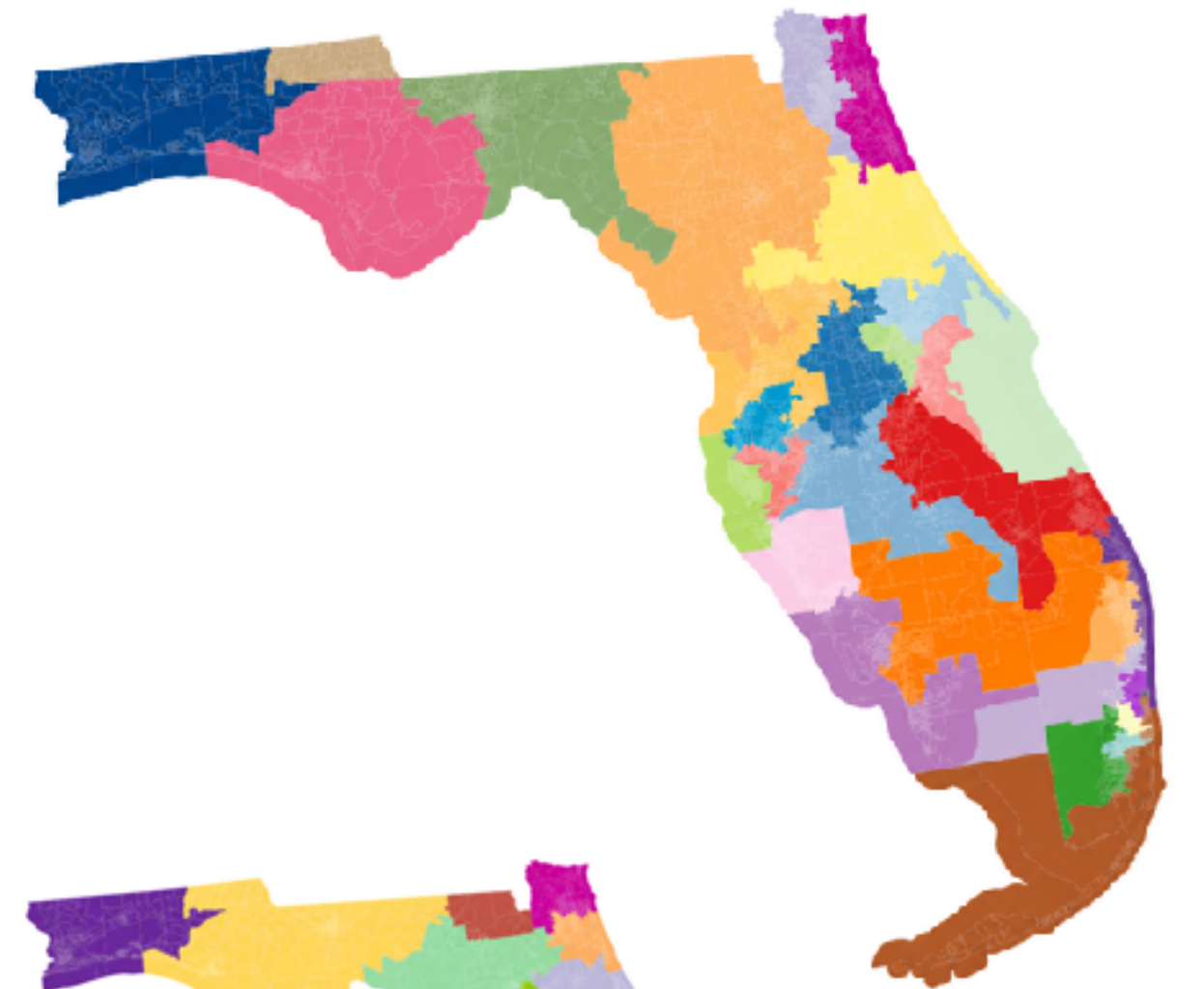
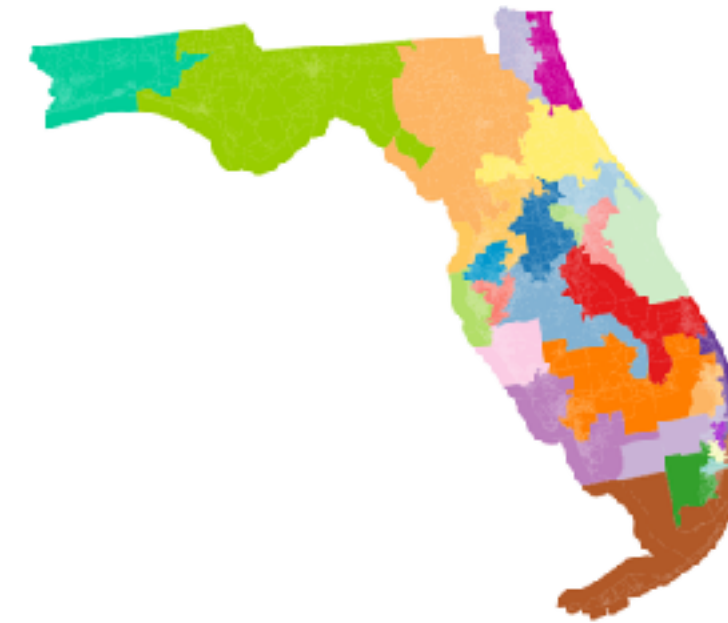
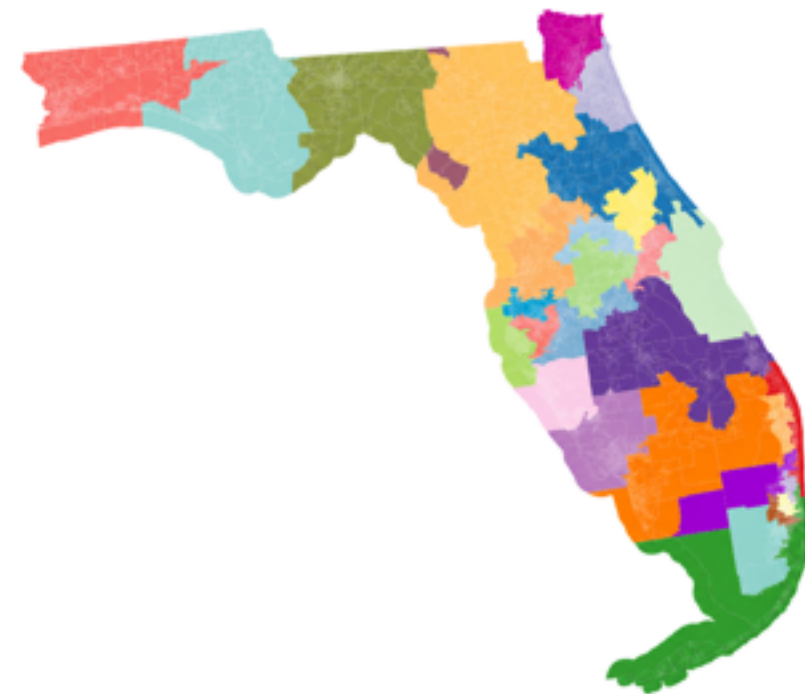
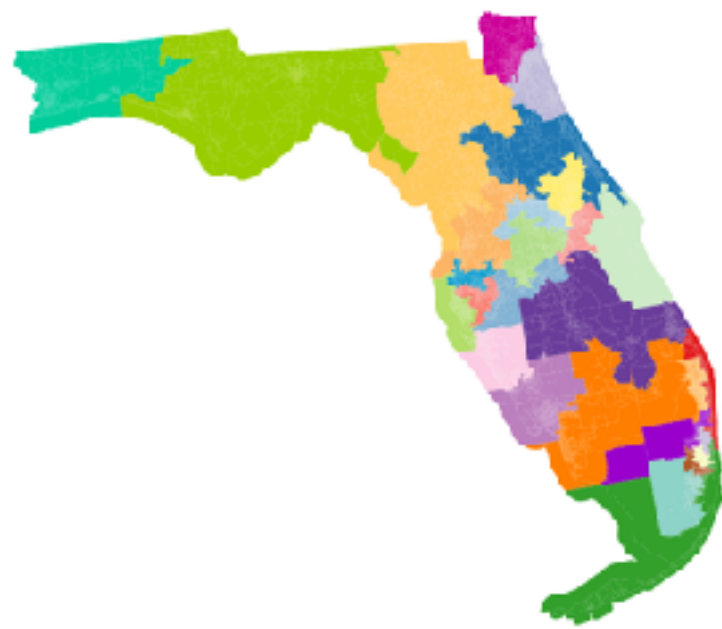
Algorithmic redistricting

use computers to make zillions of “neutral” maps



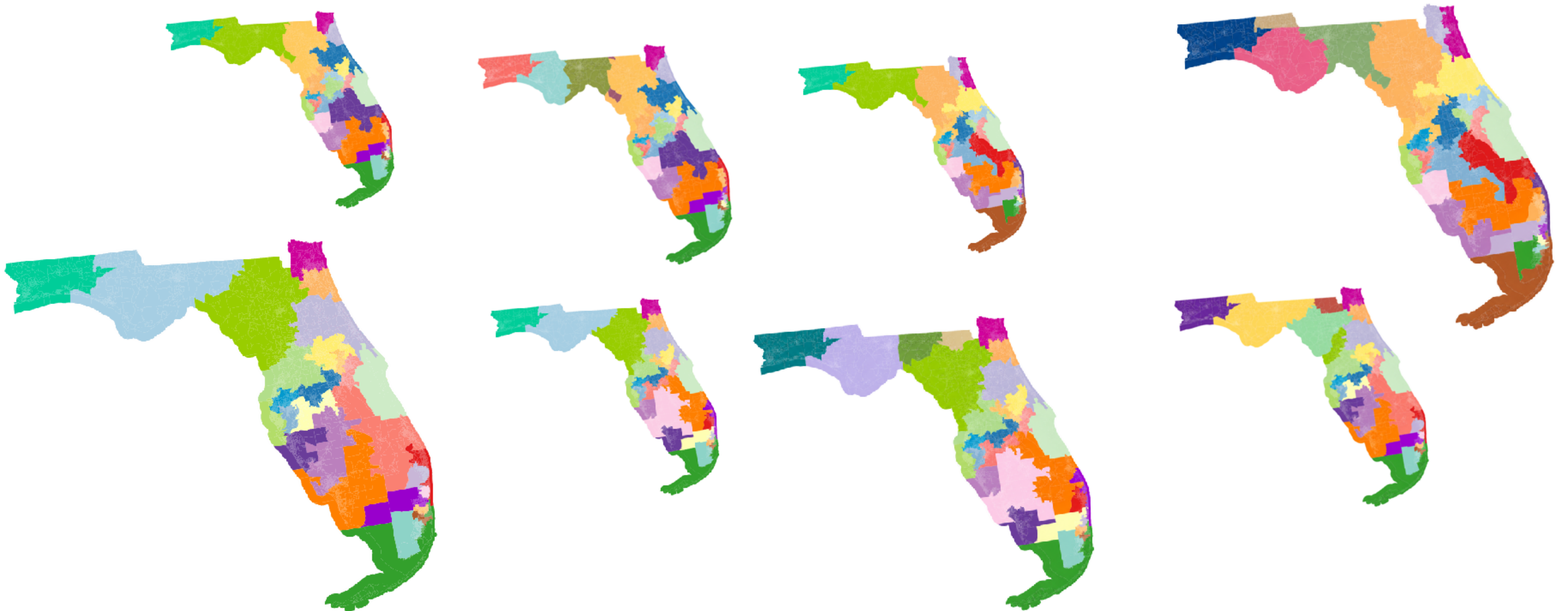
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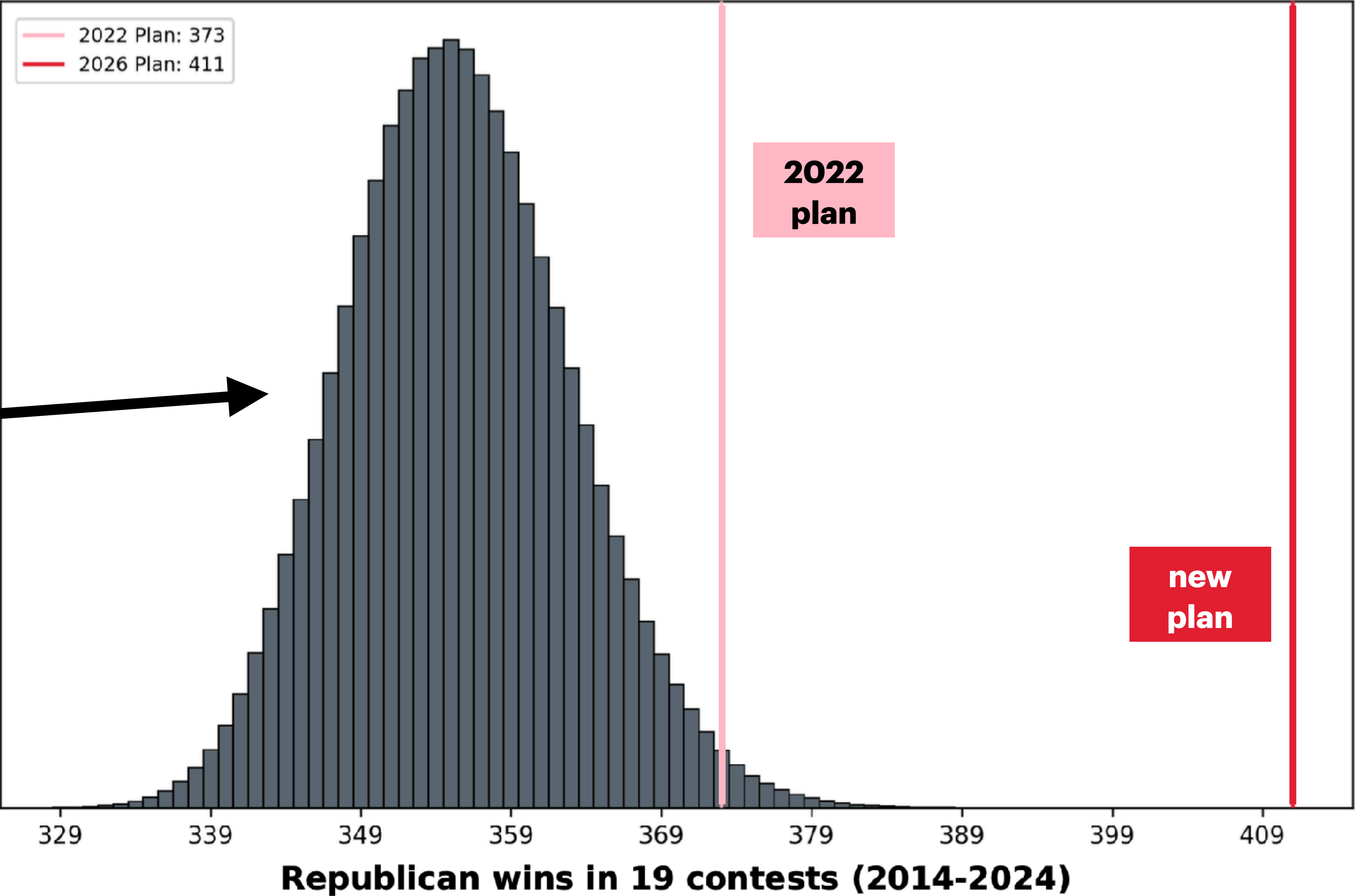


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30 million
maps

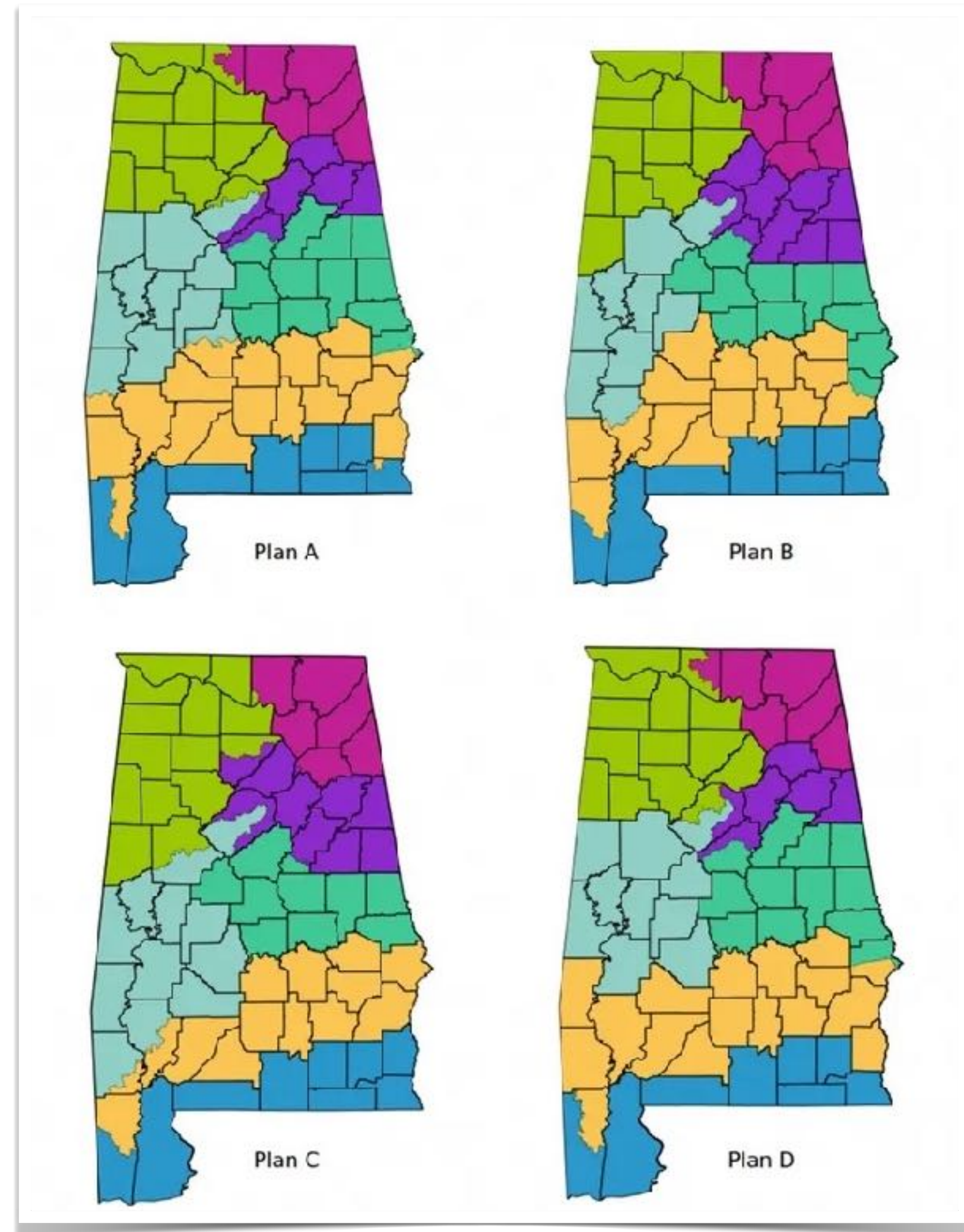


Designing maps that make
good demonstratives

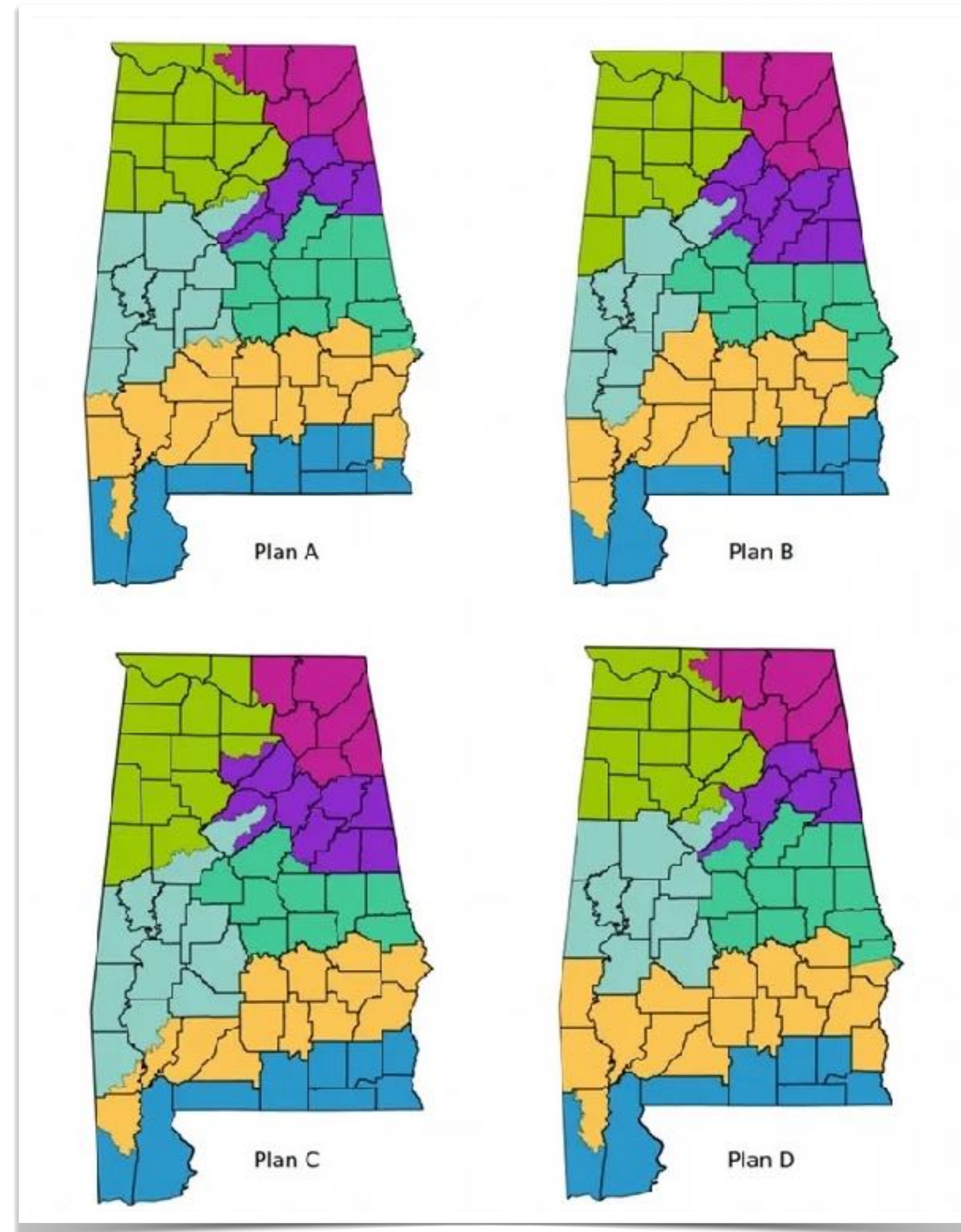
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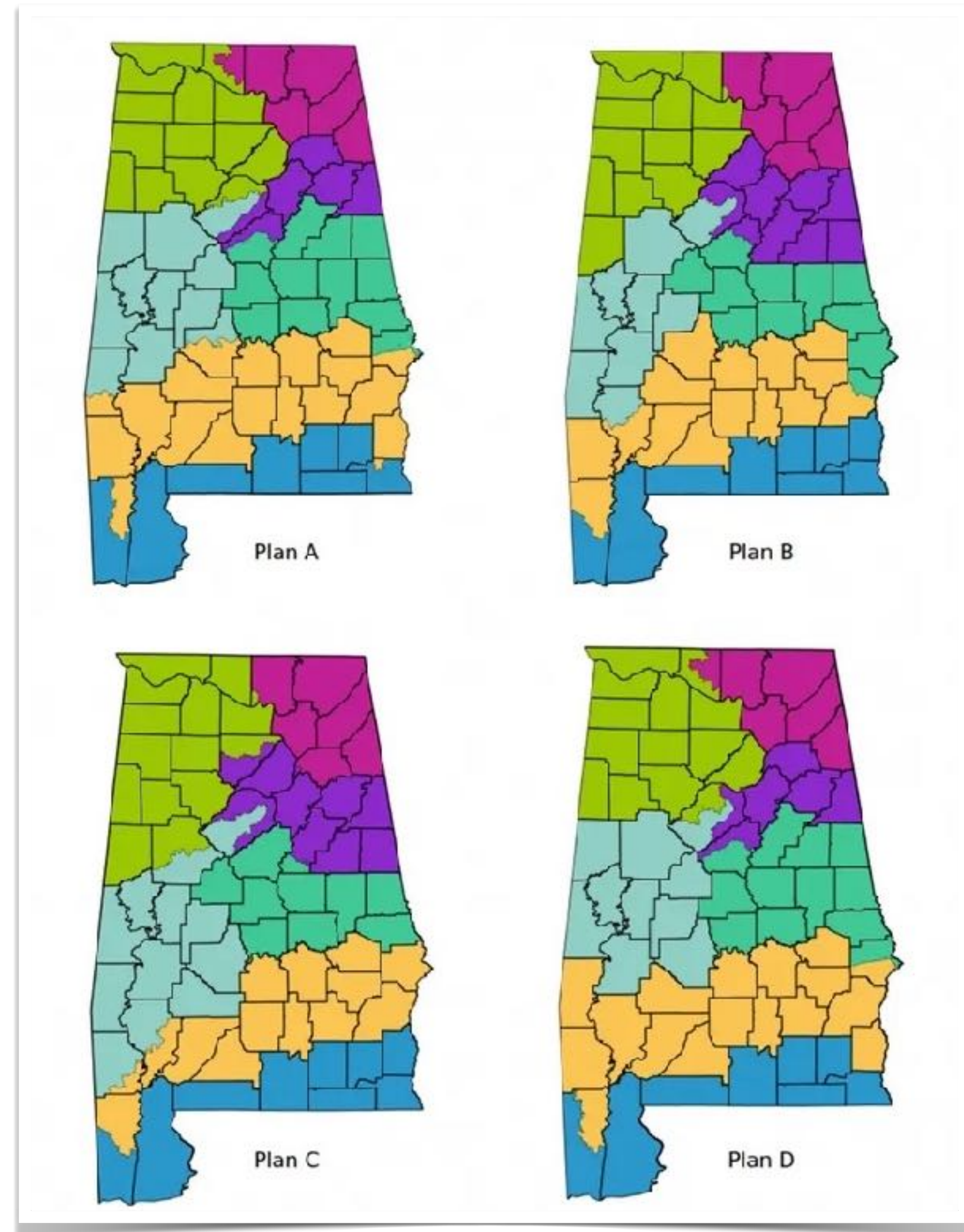
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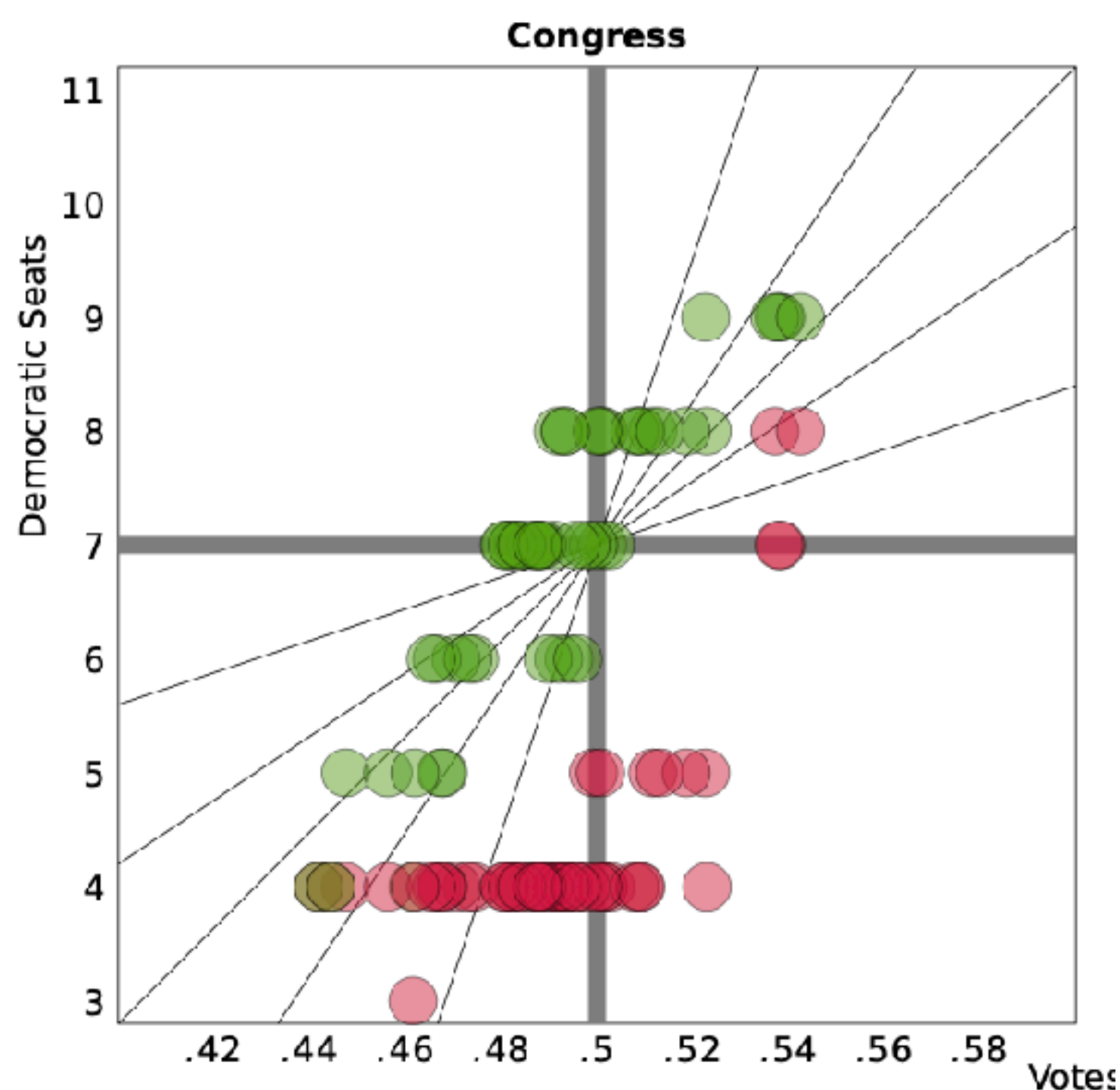
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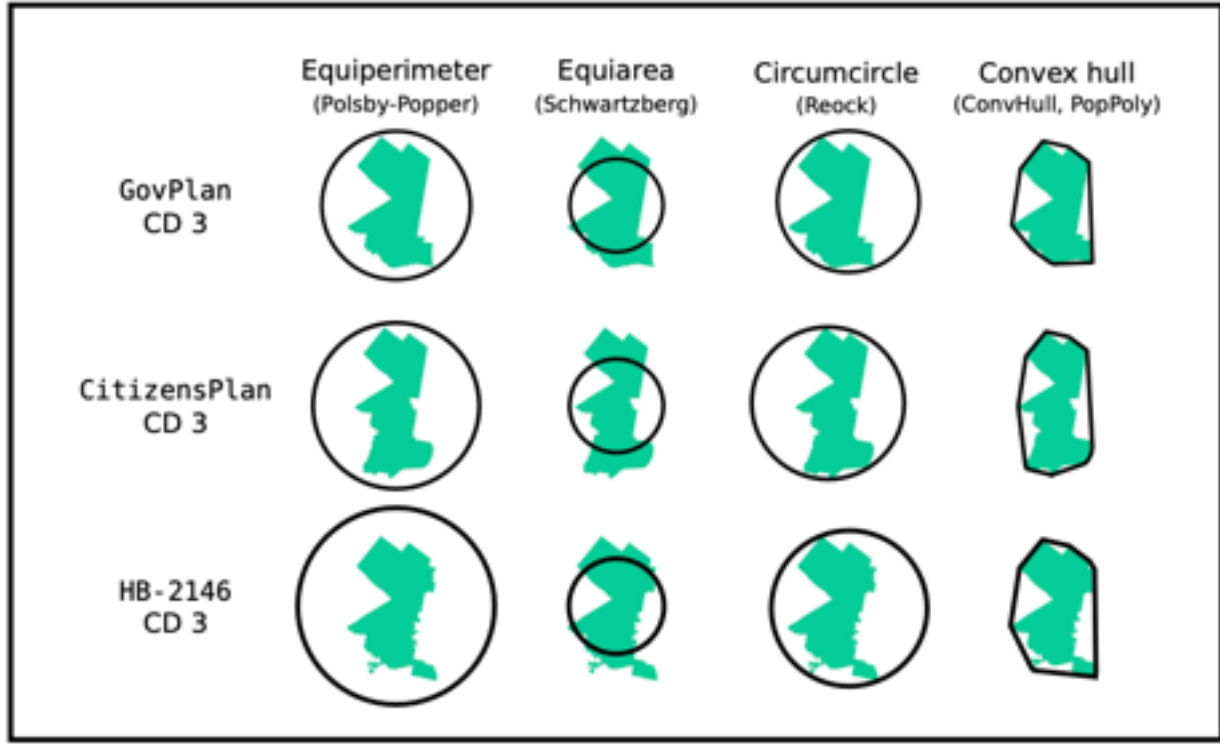
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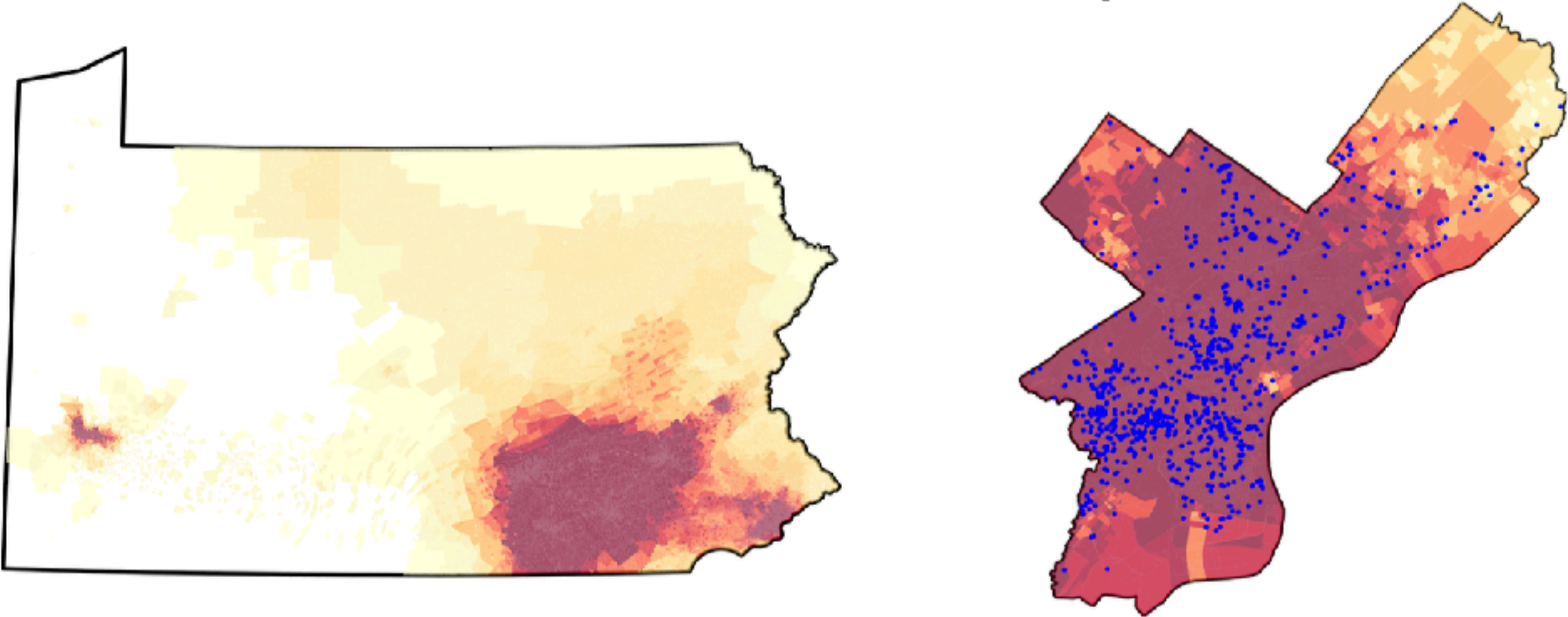
Statistics, metrics, visuals, and more



	NCLCV-Sen			SL-173 (old Senate plan)		
	All (52 contests)	Non-judicial (35 contests)	Up-ballot (14 contests)	All (52 contests)	Non-judicial (35 contests)	Up-ballot (14 contests)
Efficiency Gap	-0.020	-0.024	-0.017	-0.075	-0.068	-0.080
Simplified EG	-0.023	-0.028	-0.021	-0.076	-0.070	-0.081
Mean-median	-0.009	-0.012	-0.009	-0.036	-0.036	-0.037
Partisan Bias	-0.015	-0.023	-0.016	-0.072	-0.069	-0.08
Eguia County Skew	-0.040	-0.041	-0.030	-0.093	-0.083	-0.09
	SL-2 (new Senate plan)			Harper-Sen		
	All (52 contests)	Non-judicial (35 contests)	Up-ballot (14 contests)	All (52 contests)	Non-judicial (35 contests)	Up-ballot (14 contests)
Efficiency Gap	-0.045	-0.048	-0.046	-0.022	-0.023	-0.029
Simplified EG	-0.048	-0.051	-0.050	-0.027	-0.028	-0.034
Mean-median	-0.020	-0.022	-0.021	-0.003	-0.005	-0.003
Partisan Bias	-0.044	-0.045	-0.049	-0.013	-0.018	-0.002
Eguia County Skew	-0.065	-0.064	-0.059	-0.044	-0.041	-0.043



Overview of submitted COI maps



in my research, I became
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They tend to favor the rural party and they tend to lock out minority representation

One guy was persuaded by my research

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⁹The majority notes that this study used demographic data from the 2010 census, not the 2020 one. That is irrelevant, since the black population share in Alabama changed little (from 26.8% to 27.16%) between the two censuses. To think that this minor increase might have changed Dr. Duchin's results would be to entirely miss her point: that proportional representation for *any* minority, unless achieved "by design," is a statistical anomaly in almost all single-member-districting systems. Duchin & Spencer 764.

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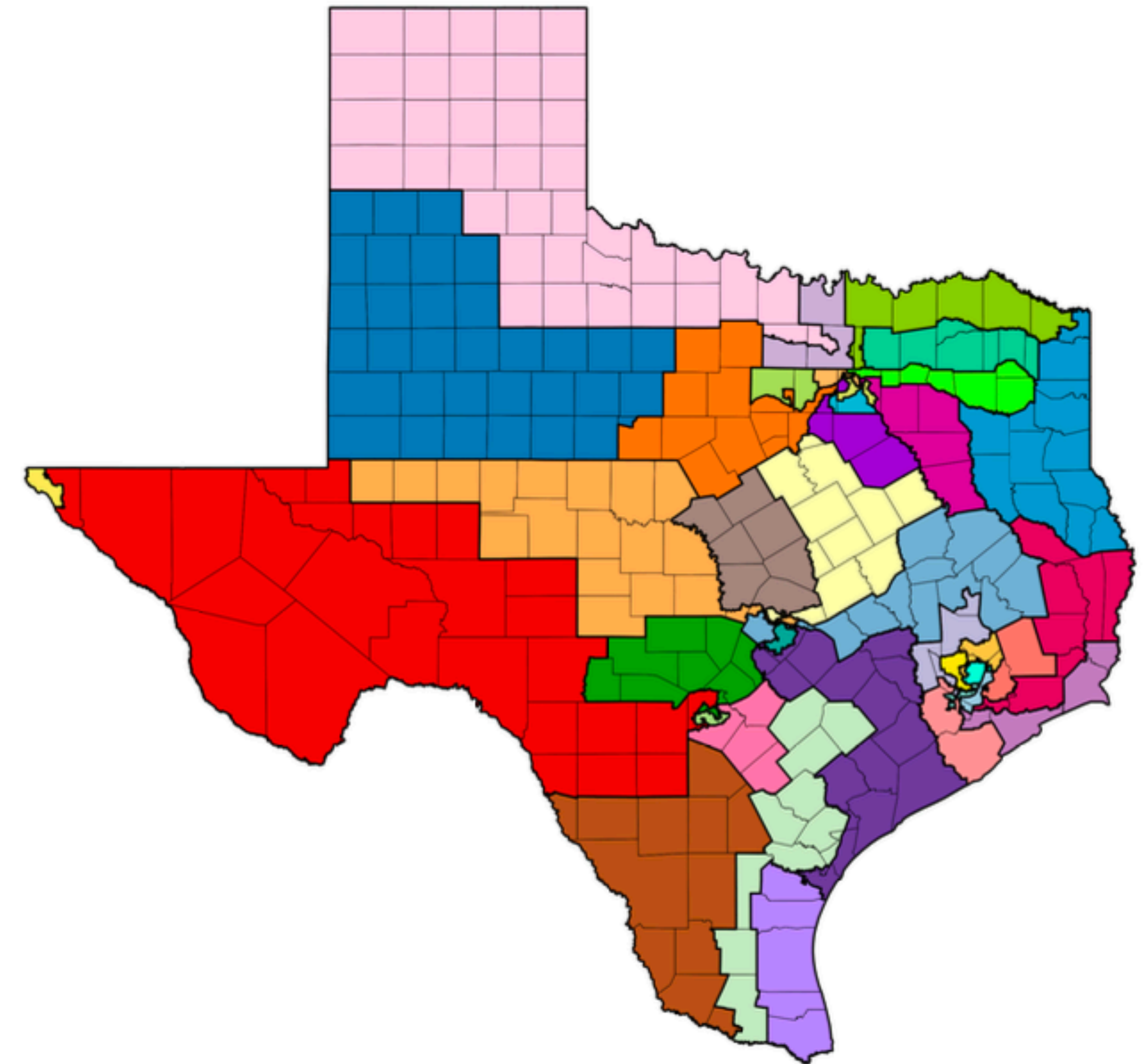
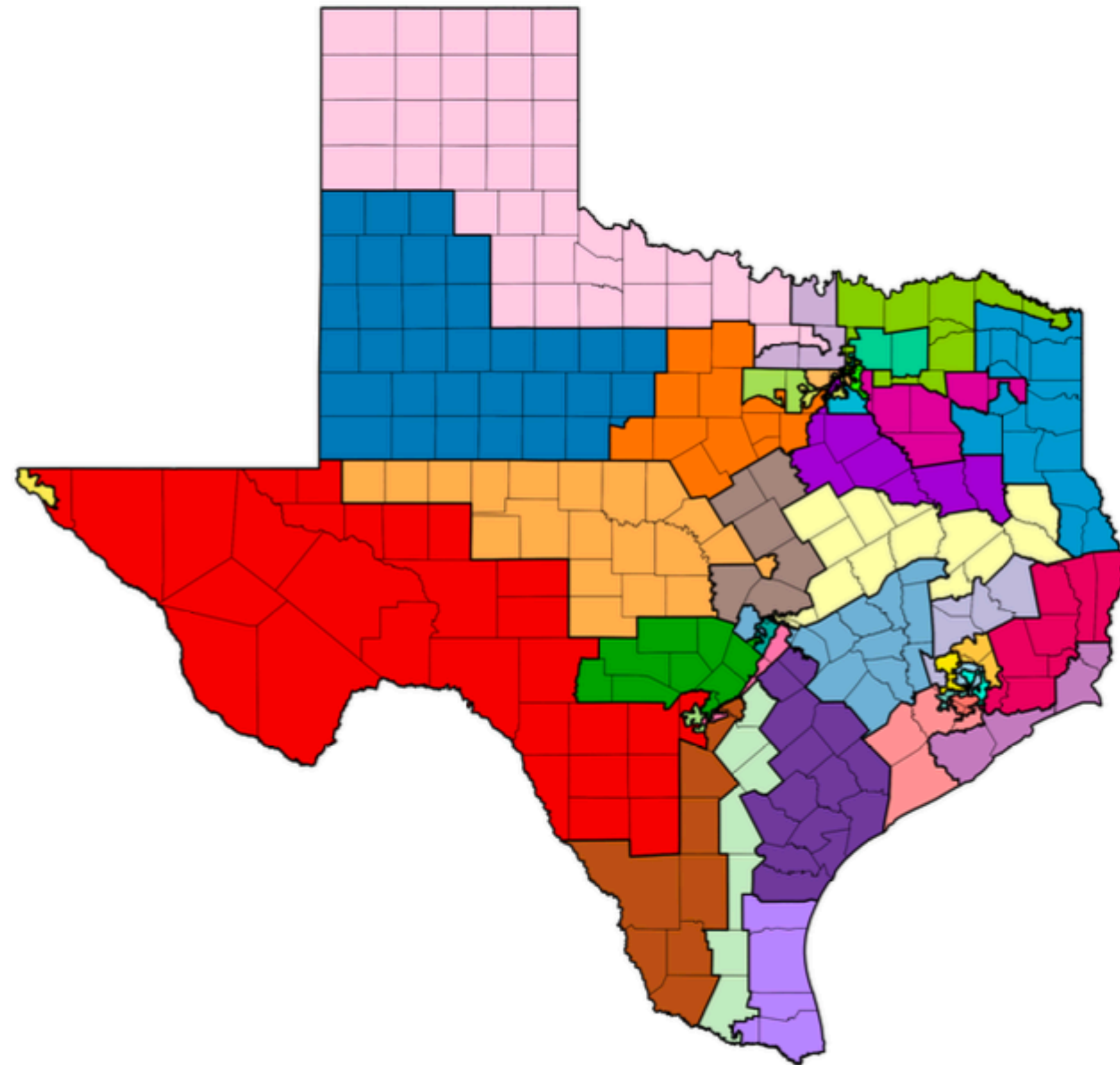


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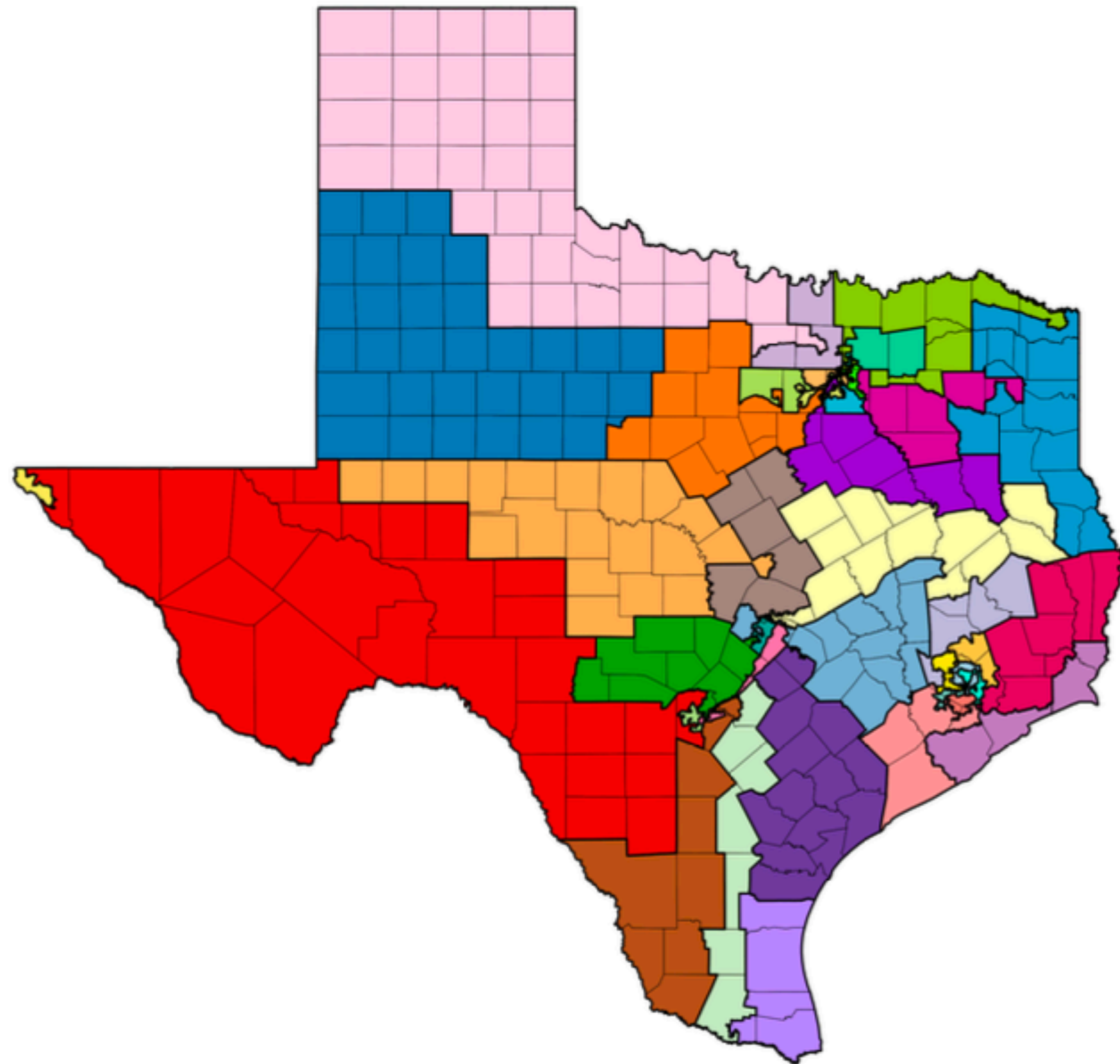
...but his **conclusion** was very different from mine

the **new** gerrymandering wars

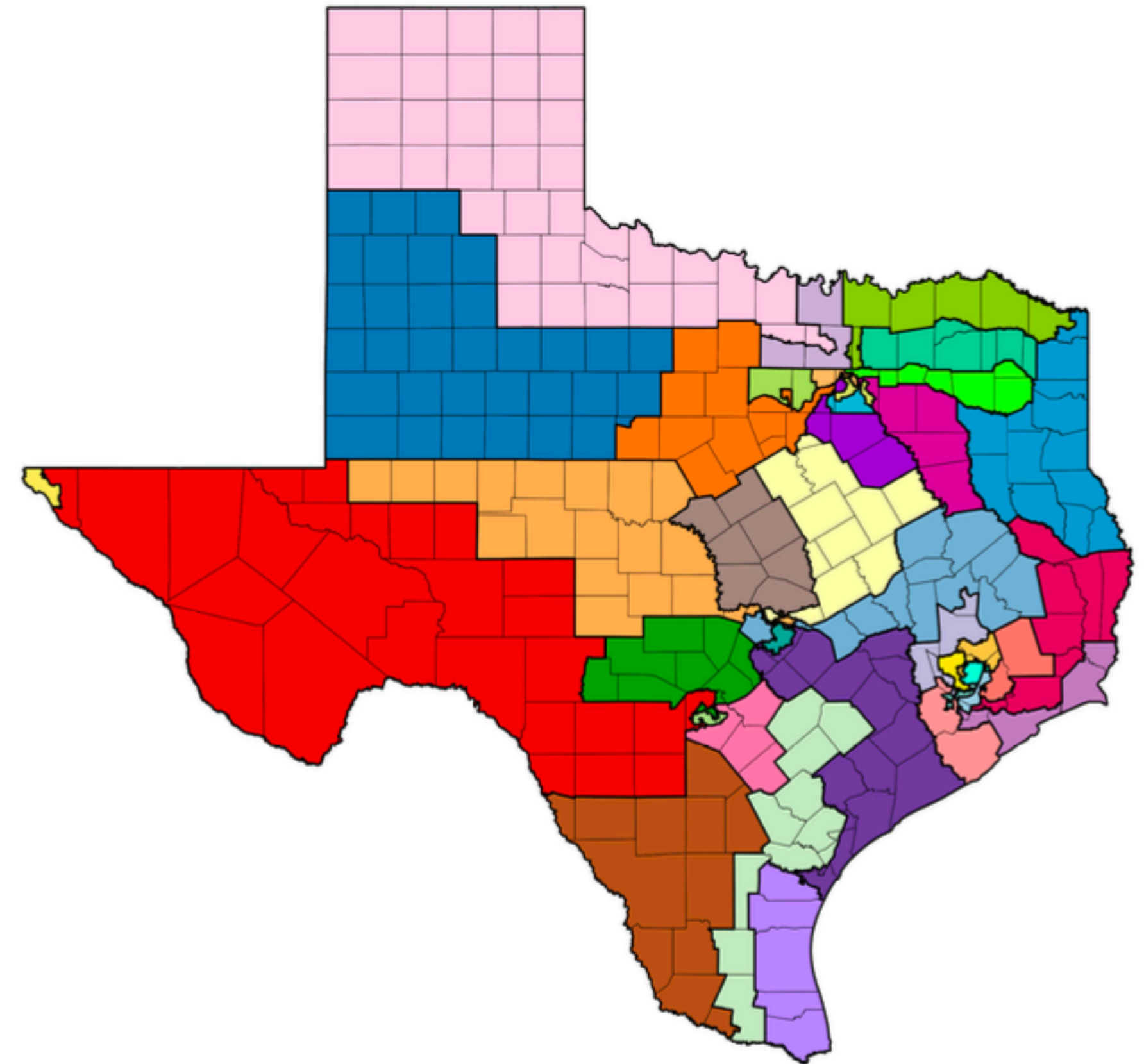
Texas went for Trump against Clinton, Biden, and Harris
— each time with about 57% of the two-party vote



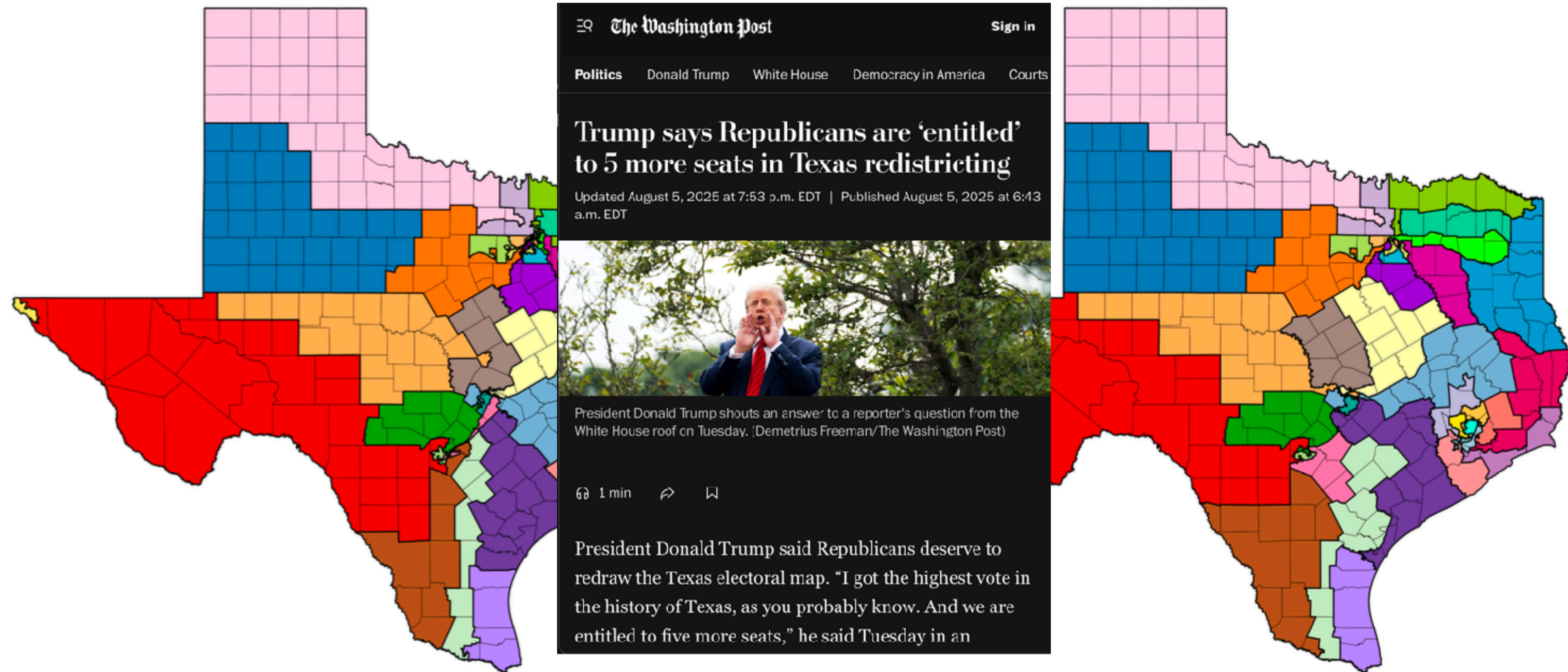
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2021 map tuned for 25/38 R seats
(66% of delegation)

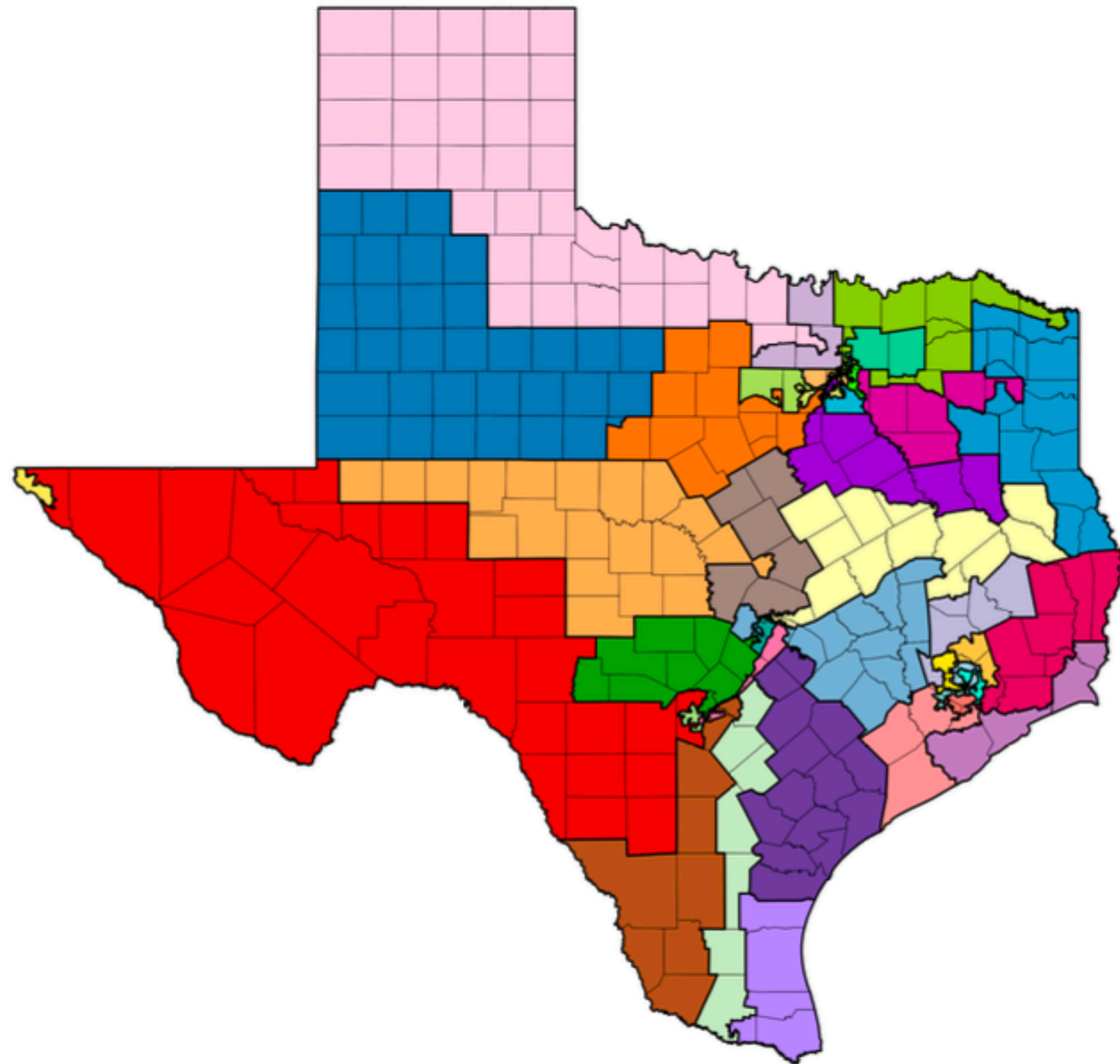


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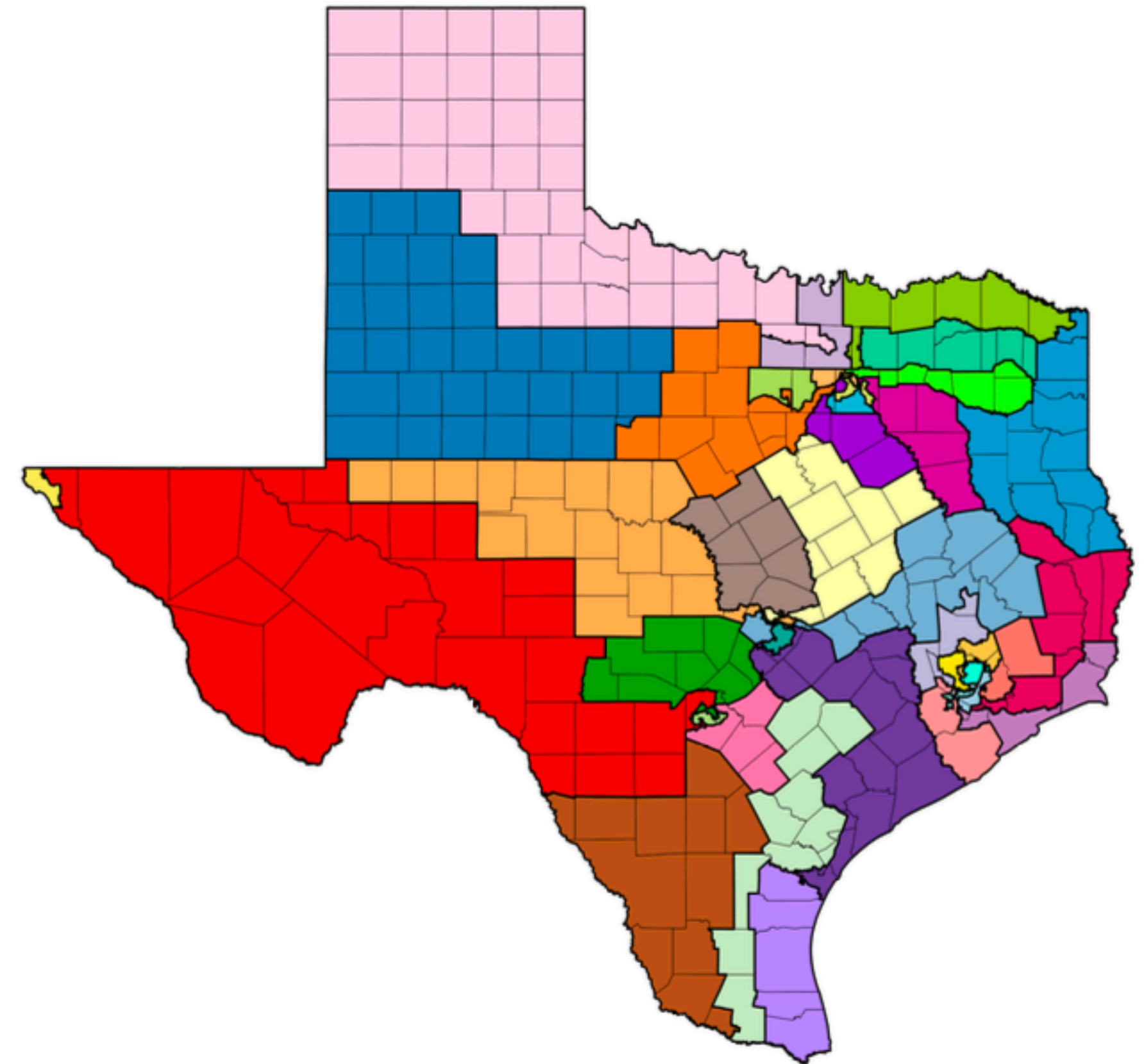


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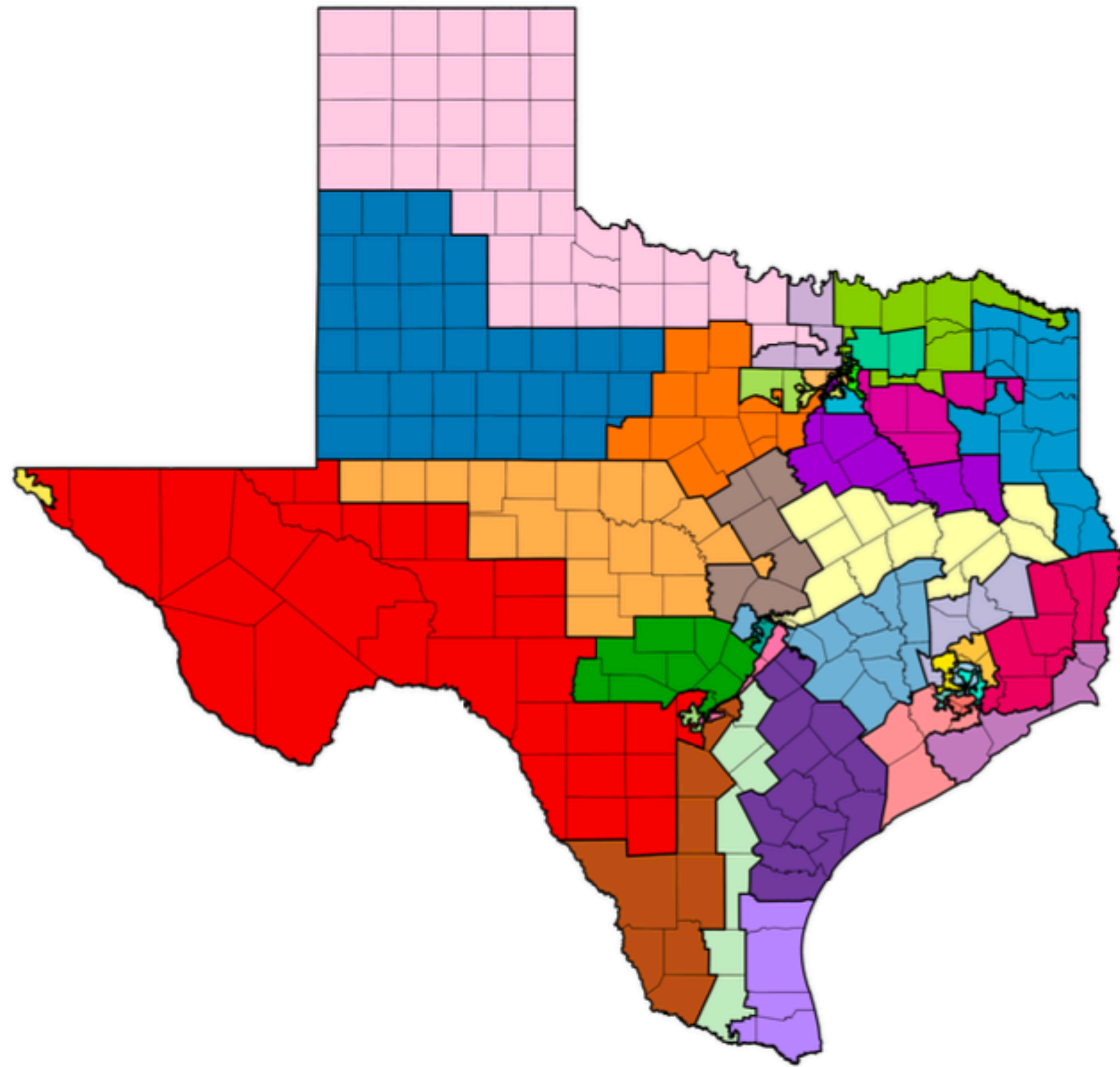
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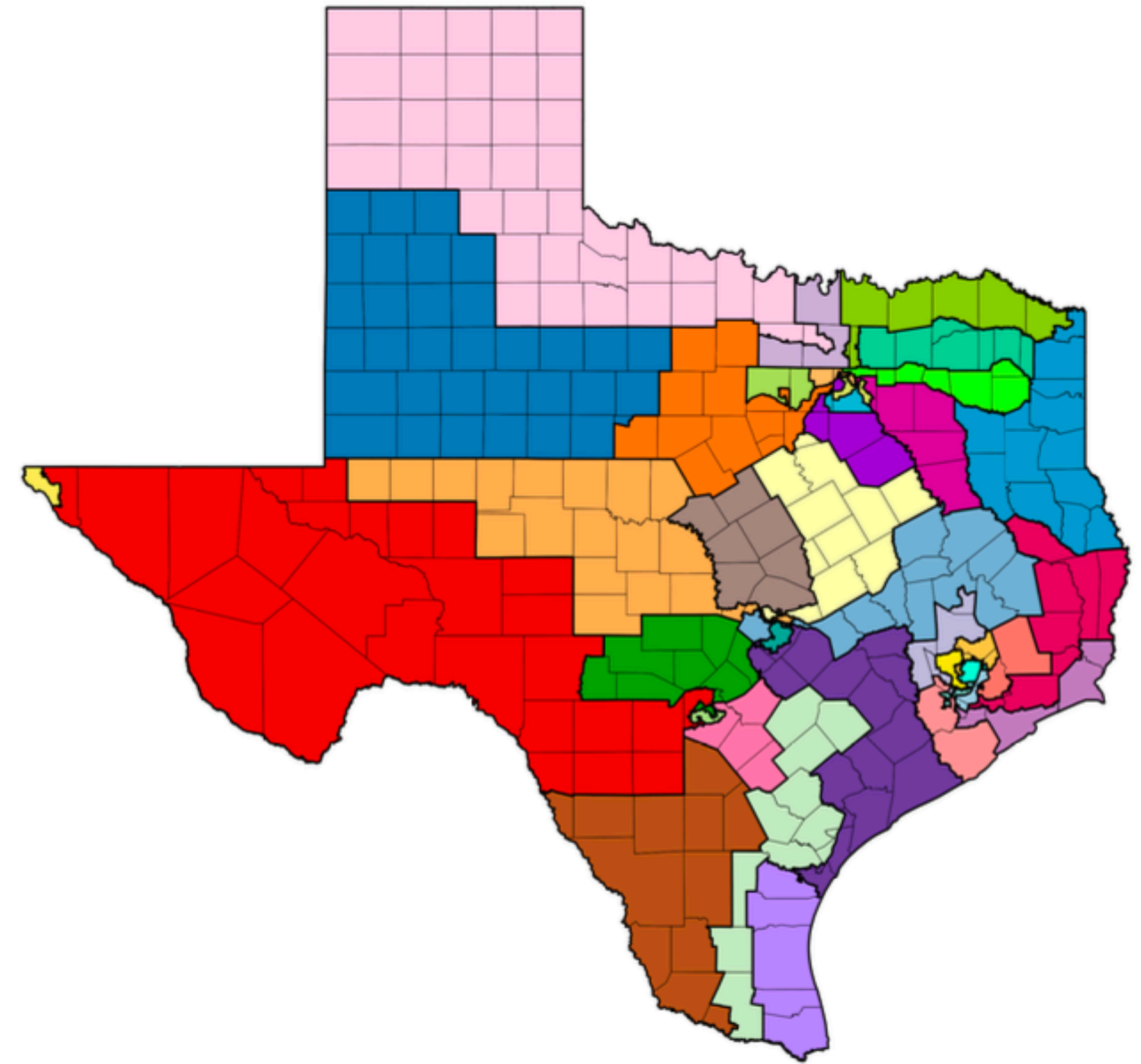
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2021 map tuned for 25/38 R seats
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2025 map tuned for 30/38 R seats
(79% of delegation)

Today's state of play

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TENNESSEE (+2R)

We'll just decline to give you a hearing till after the primary, then tell you it's too late

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- Organizers are thinking about **systems** and trying out new kinds of voting reform
- Researchers have a long time to get the **science** ready for the next time we face a reasonable Court



thank you!